# Beyond Horizons

## A Half Century of Air Force Space Leadership

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# INTRODUCTION
# The Dawn of the Space Age

> In making the decision as to whether or not to undertake construction of such a [space]craft now [1946], it is not inappropriate to view our present situation as similar to that in airplanes prior to the flight of the Wright brothers. We can see no more clearly all the utility and implications of spaceships than the Wright brothers could see fleets of B-29s bombing Japan and air transports circling the globe.[1]

In 1946, the authors of the first Air Force-sponsored Project Rand (Research and Development) study on the feasibility of artificial earth satellites aptly characterized the challenge and uncertainty surrounding the country's initial foray into the space age. Postwar skeptics dismissed proposed satellite and missile projects as excessively costly, technologically unsound, militarily unnecessary, or simply too "fantastic," while space advocates themselves remained hard pressed to convince opponents and stifle their own self-doubts. Space represented a "new ocean," a vast uncharted sea yet to be explored. The dawn of the space age brought many questions but offered few answers. Could satellites be successfully produced, launched, and orbited? If technically feasible, what military—or civilian scientific—functions should they perform? How should space functions be organized? What space policy would best integrate space into the national security agenda? What should be the Air Force role in space?

In view of the uncertainties involved, the period from the close of the Second World War to the launching of the first Sputnik in the fall of 1957 proved to be the conceptual phase of the nation's space program. Only by the mid-1950s, a full decade after the 1946 Rand study, could observers identify two sides of a national space policy that would characterize the American space program from the Eisenhower presidency to the present day. One side comprised a civilian satellite effort, termed Project Vanguard, designed to launch a scientific satellite by the end of 1958 as part of the International Geophysical Year. The other, an Air Force-led military initiative, sought to place into earth orbit a strategic reconnaissance satellite capable of providing vital intelligence about Soviet offensive forces.[2]

The Air Force played a central role during the formative era before Sputnik and afterward when the nation's leaders established space policy and organized to confront the Sputnik challenge. The National Space Act of 1958 created the civilian agency, the National Air and Space Administration (NASA), to operate the civilian space effort, while the Air Force and other military services and agencies jockeyed for position within the Defense Department and the overall national space program. Although the Air Force won the contest for military "supremacy" among the services, it seemed to many Air Force leaders that the policy of promoting the "peaceful uses of space" meant a diminished role for Air Force space interests and a threat to the nation's security. Nevertheless, by the end of the Eisenhower administration, the Air Force space program revealed the basic defense support mission characteristics it would retain for the remainder of the century.

## Arnold and von Kármán Form a Partnership

The Air Force space saga began with the partnership of General Henry H. "Hap" Arnold, Commanding General of the Army Air Forces (AAF), and the brilliant scientist, Dr. Theodore von Kármán, Director of the Guggenheim Aeronautical Laboratory at the California Institute of Technology (GALCIT). Together they provided the emerging Air Force with a strong research and development focus and championed Air Force interests in the new missile and satellite fields. Their legacy would endure.

Hap Arnold and Dr. von Kármán first met in 1935, when Arnold visited his friend, Dr. Robert Millikan, head of the California Institute of Technology (Cal Tech) in Pasadena, California, while serving as commander of the First Wing, General Headquarters Air Force, at neighboring March Field. The two men could hardly have appeared more different. Arnold radiated physical energy and heartiness from his large frame, while the short, slender intellectual Hungarian émigré exuded a quieter, less forceful presence. Yet the two men took to each other immediately. The Air Corps brigadier general's long-standing interest in aviation technology and association with the National Advisory Committee for Aeronautics (NACA) helped spark an immediate personal and professional friendship. Back in the First World

War Arnold had participated in primitive pilotless aircraft tests, and later served as a military representative to the NACA. For his part, renowned aerodynamicist von Kármán later recalled that while Arnold had no significant technical background or training, he possessed an appreciation for what science could contribute to aviation and the "vision" to persevere against long odds.[3]

After their first meeting, Arnold often visited Cal Tech to observe wind tunnel experiments and discuss with von Kármán various aeronautical and, especially, rocket propulsion initiatives Cal Tech had just undertaken. Von Kármán, who had established his reputation in structures and fluid dynamics as well as aerodynamics, showed the foresight to support a research project first proposed in 1936 by his bright graduate student, Frank Malina. Malina and his four-man team, known as the "suicide squad," had formed the GALCIT Rocket Research Group to develop both high-altitude sounding rockets and rocket-powered airplanes along the lines described by Austrian theorist, Dr. Eugen Saenger. With Cal Tech's move into rocketry, von Kármán's research placed him squarely at the center of the two areas of propulsion that would take the Air Force to "the fringes of space." One was the aerodynamic approach, represented by the NACA, which involved jet-propelled airbreathing "cruise" missiles; the other, astronautical approach, encompassed rocket-powered "ballistic" missiles.[4]

## NACA and the Rocketeers Lay the Groundwork

Since its founding in 1915, the National Advisory Committee for Aeronautics had served as the major government agency performing experiments in basic aviation technology and advanced flight research. During the 1920s and 1930s its research engineers worked closely with the Army, Navy, the Bureau of Standards, and the infant aircraft industry to improve aircraft design and performance. Relying primarily on wind tunnels at its Langley research laboratory in Virginia, its research led to use of retractable landing gear, engine cowlings, laminar flow airfoils, and low-winged all-metal monoplanes. It developed an outstanding reputation for its work in aerodynamics and with aerodynamic loads. Chartered to benefit both civil and military aviation, the NACA generally performed the research and left to the military services and industry the practical development of aircraft design and production. During the 1930s, the country's focus on Depression issues and budget retrenchment convinced the NACA to remain a small agency with interests primarily in aerodynamics. On the eve of World War II, Chairman Vannevar Bush's organization employed only 523 people and operated one research laboratory at Langley Field.[5]

Wartime, however, brought major expansion in the number of personnel, broader responsibilities in the area of structural materials and powerplants, and the addition of two new laboratories, one adjacent to Cleveland's municipal airport, and the other next door to the naval air station at Moffett Field forty miles southwest of San Francisco. During the war the NACA served as the "silent partner of US airpower," and solved a host of aeronautical problems. Alarmed by reports of German turbojet developments in 1940, for example, the NACA established a Special Committee on Jet Propulsion, and followed in 1944 with a Special Committee on Self-Propelled Guided Missiles. Although the NACA intended to work diligently with the Navy and Army Air Forces on these threats, the need to provide "quick fixes" throughout the conflict meant that basic research became secondary. At war's end the NACA proved eager to learn from the war by continuing its cooperative research efforts with the military. In an agreement signed between the NACA and the services in 1946, the parties agreed that "the effects of accelerated enemy research and development in preparation for war helped to create an opportunity for aggression which was promptly exploited. This lesson is the most expensive we ever had to learn. We must make certain that we do not forget it."[6]

The NACA's postwar vision embraced support of American supersonic flight probes by means of small solid-propellant sounding rockets, and the "X" series of high-altitude, rocket-propelled research aircraft. The first rocket-powered aircraft, Bell Laboratory's X-1, broke the sound barrier on 14 October 1947 with Captain Charles "Chuck" Yaeger at the controls. His historic flight became the first of many increasingly higher and faster experimental aircraft flights toward the fringes of space. The last, the single-place X-20A Dyna-Soar (named for dynamic soaring), would be the Air Force's best hope to launch a manned boost-glide rocket aircraft to the border of space. Although it did not become operational after initial development in the late 1950s, the Dyna-Soar served as a precursor of the Space Shuttle of the 1980s. Although the NACA expressed interest in rocket propulsion, its focus remained centered on aerodynamic experiments and manned flight within the earth's atmosphere. Space research seemed wholly outside its experience and interests.[7]

## Rocketeers Lead the Way

Spaceflight represented a challenge far more daunting than traditional aviation. Although future Air Force leaders would lay claim to spaceflight as a logical extension of Air Force operations in the atmosphere, aviation technology offered only limited solutions on the road to outer space. Although the technical advances that led from reciprocating to jet turbine engines powered aircraft higher into the upper atmosphere, the oxygen-dependent airplane remained confined to the atmosphere. Rockets, on the other hand, operate independent of the atmosphere by relying on their own internal propellants: fuel and oxidizer. In their flight through increasingly thinner atmosphere on the way to airless space, rockets become progressively more efficient. Although the post-World War II American rocket research airplanes could provide useful information on the guidance and control challenges facing vehicles in the upper atmosphere, their small rockets could never break the bonds of gravity, and they remained primarily aerodynamic vehicles. To operate either manned spacecraft or instrumented satellites in outer space, rockets needed sufficient thrust to boost their payloads into orbit where centrifugal force balanced the earth's gravitational field.[8]

The challenge of manned spaceflight had captivated the imaginations of dreamers for centuries. Yet their ideas remained only idle musings until technological progress in the late 19th century led serious enthusiasts to consider liquid-propellant rockets as "boosters" of spacecraft. Among the pioneers of liquid-propellant rocket research linked to visions of manned spaceflight, three men—Russian Konstantin Tsiolkovsky, German-Romanian Hermann Oberth, and American Robert Goddard—paved the way for the successful military and civilian space programs of the second half of the 20th century. While their research initially led to production of bombardment rockets for use by their respective military forces in the Second World War, they all remained committed to visions of spaceflight.[9]

The earliest of the space triumvirate, mathematics teacher Konstantin Eduardovich Tsiolkovsky, in 1895 published the first technical essays on artificial earth satellites. By the end of the century, he had worked out the theory of a liquid-fueled rocket dependent on kerosene to achieve sufficient exhaust velocity. For the next 20 years he immersed himself in theoretical studies but remained largely unknown to the world outside Russia. Yet, by the time of his death in 1935, his pioneering work had helped the Soviets establish a strong prewar rocket and jet-powered aircraft development program which led to the space program of the postwar era.

Although Hermann Oberth also taught mathematics and produced important theoretical studies on spaceflight, he assumed the role of publicist for rocketry and space exploration to enthusiastic European audiences after World War I. In 1923 he established his reputation in the new field of astronautics with the seminal publication, "The Rocket into Interplanetary Space," in which he described the technical requirements for propelling satellites into earth orbit. In 1927 he helped found the German Society for Space Flight, which became the most influential of the numerous rocket societies in Europe. By 1931, Oberth's work with the Society came to the attention of the German Army, which saw in sponsorship of the young rocket scientists a means of obtaining bombardment rockets for an army sorely constrained by the Versailles Treaty. Among the Society members who joined the Army project in 1932 was a 20-year old engineer named Wernher von Braun. After 1933, the Nazi regime expanded the Wehrmacht program and in 1937 began developing the Peenemuende experimental site on the Baltic coast under supervision of Captain Walter Dornberger. Although von Braun and his colleagues had now to focus on long-range rockets to help fuel Germany's military expansion, they continued to dream of manned spaceflight. During the Second World War, while the Luftwaffe produced the V-1 aerodynamic pulse-jet "cruise" missile, the Wehrmacht's Peenemünde rocketeers developed the far more impressive "big rocket," the V-2. Known as the A-4 to the rocket specialists, the V-2 measured 46 feet in length, weighed 34,000 pounds, and approached a range of 200 miles under 69,100 pounds of thrust produced by its liquid-propellant engine. To the Allies the V-2 presented a frightening weapon that could not be thwarted with any known defense. After the war Americans discovered that German plans had called for an intercontinental ballistic missile to strike New York by 1946. To the German rocketeers, however, the A-4 always represented the first rung on the ladder to space. After the war, the American Army's Operation Paperclip brought Dornberger, von Braun, and a host of other German rocket experts to the United States, where they joined the Army's rocket program—with their visions of spaceflight still alive.[10]

The German rocket specialists freely acknowledged their debt to American rocket pioneer, Robert H. Goddard. Unlike his Russian and German contemporaries, Goddard immediately moved beyond theoretical studies to practical experimentation. He always found applied research more exciting than theoretical studies. From his post as a physics professor at Clark University, Goddard began experimenting with powder rockets, and in 1914 received a patent for his liquid-propellant rocket engine. In 1920 the Smithsonian released his highly technical paper, "A Method of Reaching Extreme Altitudes," which described various rocket-propelled experiments that could be conducted as high as 50 miles in altitude. His paper also included a theoretical argument for rocketing a payload of flash powder to the moon, which drew public censure after a *New York Times* reporter ridiculed the idea in print. The experience left Goddard badly scarred and more than ever inclined to focus on private research. By 1926 he had built and tested the first liquid-propellant rocket, and in 1935 successfully launched a gyroscopic-stabilized rocket to an altitude of 7000 feet. Eventually, the prolific experimenter amassed an amazing 214 patents for his designs and devices. But Goddard preferred working alone and jealously guarded his work from other space enthusiasts like the intrepid members of the fledgling American Rocket Society.

In the 1930s Goddard moved his increasingly complex liquid propellant experiments from Massachusetts to the New Mexico desert, where he worked with his wife and various assistants supported by grants from the Guggenheim Foundation. Guggenheim officials quite naturally sought to bring Goddard and von Kármán's Cal Tech Rocket Research Project team together. Characteristically, Goddard proved reluctant, and von Kármán refused to collaborate without full disclosure of Goddard's research results.[11]

Despite the acknowledged importance of Goddard's work for future rocket development, active collaboration between von Kármán and Goddard might well have placed the postwar American rocket program on better technical footing and created more incentive for the Air Force to promote research in ballistic rather than aerodynamic missiles after the war. Cooperation between the two camps would certainly have helped the neophyte rocket group at Cal Tech, which had developed convincing theories about rocket flight but had no experimental data to work with. Moreover, as Malina recalled, in the 1930s most scientists generally considered rocket experiments a part of science fiction. With so little available practical data, Goddard's assistance would have been welcomed by von Kármán and the young rocketeers, who proceeded largely independently of Goddard.[12]

## Wartime Provides the Momentum—Arnold and von Kármán Establish the Foundation

Meanwhile, von Kármán and his Cal Tech rocket team continued their research into high-altitude sounding rockets and jet-assisted takeoff (JATO) devices by examining potential fuel types, rocket nozzle shapes, reaction principles, and thrust measurements. They managed to keep their experiments afloat with very little money until General Arnold came to the rescue in 1938. Late that year, Arnold, now chief of the Army Air Corps, helped convince the National Academy of Sciences to provide initial funding for the Cal Tech project. Shortly thereafter, in January 1939, the Air Corps assumed direction of the program, and in June awarded the researchers a $10,000 contract. Von Kármán explained that the program's label, "Air Corps Jet Propulsion Research, GALCIT #1," included the word "jet" rather than "rocket" because of wide-spread skepticism among his colleagues. As one of them told him, he was welcome to the "Buck Rogers" job.[13]

Malina wisely committed his team to explore both liquid- and solid-propellant rocket engine research. The team made rapid progress once they developed the first relatively long-duration, controlled-explosion solid-propellent engine. In August 1941, the Cal Tech engineers carried out their first flight tests in which Captain Homer Boushey, using four JATO canisters attached to his Ercoupe monoplane, rapidly climbed to an altitude of 20 feet. Malina was ecstatic. Continued test successes brought in a JATO contract from the Navy, and von Kármán and Malina in 1942 decided to capitalize on their growing project by forming a private company, Aerojet Engineering Company, to produce the jet canisters and work on other rocket-related contracts they expected to receive.[14]

In late 1943, after reviewing intelligence reports on German rocket development, von Kármán wrote a brief paper entitled, "Memorandum on the Possibilities of Long-Range Rocket Projectiles," in which he proposed that the AAF support development of a 10,000-pound air-breathing missile with a seventy-five-mile range as an extension of JATO research. When the AAF demurred, the Army Ordnance Department stepped in and offered von Kármán a far more challenging contract. The scientist readily agreed to the Army's project, which called for producing a 20,000-pound liquid-propellant rocket with a burn time of sixty seconds and a range of nearly forty miles. Organized under Frank Malina, the large project became known as ORDCIT (representing Ordnance, California Institute of Technology), until renamed the Jet Propulsion Laboratory (JPL) in November 1944. Their work would lead to the successful launching of the WAC Corporal series of liquid-propellant sounding rockets after the war. Meanwhile, as the Army's Ordnance Department focused primarily on rockets, the AAF's Air Materiel Command preferred to stress air-breathing missiles.[15]

During much of the war, von Kármán served as an aeronautical troubleshooter for Hap Arnold, the Commanding General of the AAF. By 1944 Arnold had become convinced that the next war, unlike the last, would demand far more technical competence. As Chief of the Army Air Forces, he said, his job was to

> project [himself] into the future; to get the best brains available, have them use as a background the latest scientific developments in the air arms...and determine what steps the United States should take to have the best Air Force in the world twenty years hence.[16]

In September of that year he called on von Kármán to lead a study group comprised of civilian and military experts to chart a course for the Air Force future. Arnold outlined his objectives for the group in a 7 November 1944 memorandum, "AAF Long Range Development Program." In order to place Air Force research and development programs on a "sound and continuing basis," he called for a plan whose farsighted thinking would provide a sound prescription for preparing Air Force research and development programs as well as congressional funding requests. Because "our country will not support a large standing Army" and "personnel casualties are distasteful, we will continue to fight mechanical rather than manpower wars." Given these constraints, he said, how can science be used to provide the Air Force with the best means to ensure the nation's security?[17]

With Arnold's strong support to overcome any bureaucratic impediments, von Kármán began work immediately, and by December had brought together a group of twenty-two renowned scientists and engineers. Calling itself the Army Air Forces Scientific Advisory Group, it would remain in place and continue as the Scientific Advisory Board after the Air Force became a separate service in September 1947. Following field trips to Europe and Russia to assess the current state of research, von Kármán's group on 22 August 1945 issued a preliminary report, *Where We Stand*, which explored the "fundamental realities" of future air power. The report argued that technological advances led by Germany during the war set the stage for an air force that must embrace supersonic flight, long-range guided missiles with highly destructive payloads, and jet propulsion to achieve air superiority. Von Kármán viewed government-supported research centers on the German model as a major element in the postwar national defense structure. *Where We Stand* raised crucial questions about the future of air power, and the Scientific Advisory Group intended to provide answers in its final report to General Arnold due at the end of the year.[18]

Meanwhile, while von Kármán and his team in late 1945 gathered additional field data and prepared their final report to the AAF chief, Arnold took additional steps to shape the future Air Force's scientific focus. Two of the most important involved the creation of Project Rand and a new Air Staff office to establish and direct the Army Air Forces' research and development agenda.

In September of 1945, Franklin Collbohm of the Douglas Aircraft Company proposed that the AAF establish a research project to provide it with long-range strategic planning based on ongoing scientific and technological advances. Collbohm's ideas had taken shape during his wartime association with Dr. Edward L. Bowles, who had served as General Arnold's special technical consultant. Late that month, Arnold and Bowles flew to California, where at Hamilton Field, north of San Francisco, they met with Collbohm and Donald Douglas, who strongly supported the proposal. At their meeting, Arnold decided to divert $10 million from the fiscal year 1946 procurement budget for Douglas Aircraft to organize a group of civilian scientists and engineers at Santa Monica, California, which would function independently of the company's existing research and engineering division. It would serve as a technical consultant group charged with operations analysis and long-range planning to examine future warfare and the best way the Air Force could perform its missions. Shortly thereafter, the Air Materiel Command (AMC) and Douglas Aircraft agreed to a three-year, $10 million contract for Project Rand to begin operating in May 1946.[19]

To provide an Air Staff focus for Project Rand and other research activities, General Arnold also created the office of Deputy Chief of Air Staff for Research and Development. The new position, which became effective 5 December 1945, drew criticism from the powerful Air Materiel Command, which heretofore tightly controlled the AAF procurement process from initial requirements to completed system. AMC favored rigid directives establishing specific AAF-determined goals for contractors without involving civilians in the planning process. Critics complained that research fell victim to production priorities at AMC. The new arrangement reflected Arnold's flexible approach to research and development whereby Rand would conduct broad investigations to see what could be accomplished and recommend courses of action and the new Air Staff office would provide central direction. AMC never reconciled itself to the new Air Staff position, while the Air Staff remained unwilling to assign it specific responsibility for the satellite and guided missiles programs. These would remain subjects of intra-Air Force organizational squabbles throughout the pre-Sputnik period. Nevertheless, initial prospects for achieving Arnold's goals appeared bright when he selected as his first Deputy Chief of Staff for Research and Development the hard-driving combat veteran, Major General Curtis E. LeMay.[20]

In November 1945, General Arnold became the first prominent military figure to address future warfare in terms of missile and satellite potential. In a report to Secretary of War Robert Patterson on 12 November, the air chief described the future importance of missiles and satellites as a means of preventing another Pearl Harbor-like surprise attack on the United States, and he outlined his vision for the nation's air arm. Strongly opposing shortsighted focus on present day forces, he cautioned that

> national safety would be endangered by an Air Force whose doctrines and techniques are tied solely to the equipment and processes of the moment. Present equipment is but a step in progress, and any Air Force which does not keep its doctrines ahead of its equipment, and its vision far into the future, can only delude the nation into a false sense of security.[21]

For Arnold, the forces of the future must never be sacrificed for the forces of the present. While the current state of technology convinced him to support manned aircraft, he envisioned a pilotless air force and supported developing intercontinental ballistic missiles (ICBMs) for the future Air Force. Profoundly affected by the German V-2 (A-4) rocket missile, he called for a similar weapon for the American arsenal, one "having greatly improved range and precision, and launched from great distances. [Such a weapon] is ideally suited to deliver atomic explosives, because effective defense against it would prove extremely difficult." In perhaps his most controversial prognostication, he proposed launching such "projectiles" from "true space ships, capable of operating outside the earth's atmosphere. The design of such a ship is all but practicable today; research will unquestionably bring it into being within the foreseeable future." Much of General Arnold's vision for his future Air Force received strong backing from Theodore von Kármán's monumental study on the state of air force technology—past, present, and future.[22]

After General Arnold suffered a massive heart attack in October 1945, von Kármán drove his group hard to conclude their work by the end of the year. In mid-December 1945 von Kármán's team produced the remarkable 33-volume report, *Toward New Horizons.* The first volume, "Science: The Key to Air Supremacy," set the tone by declaring that the Air Force should establish its policy, create new organizational alignments, and lay the "foundation of organized research" so that science would become an integral part of the Air Force. Von Kármán proceeded to discuss many specific means for providing technological training for service personnel and adequate research and development facilities, for disseminating scientific ideas at the staff and field levels, and promoting cooperation between the Air Force and science and industry. Regarding the latter, he noted that the Air Force preferred to sponsor research and development activities outside its own organizations, and this should be continued on a broader scale through extensive contacts with universities, research facilities, and scientists. As a means of providing continued scientific advice to Air Force leaders, he recommended that Arnold continue the Scientific Advisory Group as a permanent institution.[23]

*Toward New Horizons* expanded on issues discussed in von Kármán's "Science: The Key to Air Supremacy." The report's assessments of space issues are particularly interesting. Both jet and rocket propulsion received considerable attention, and von Kármán predicted the eventual operational success of ICBMs and declared the "satellite"…"a definite possibility." In his memoirs, von Kármán recounts that his group "examined the thrust capabilities of rockets and concluded that it was perfectly feasible to send up an artificial satellite, which would orbit the earth. We did not, however, give consideration to the military potential of such a satellite."[24] In fact, neither ICBMs nor satellites received more than passing mention because von Kármán and his colleagues believed that technological barriers would delay successful ballistic missiles for at least a decade. The report proceeded to emphasize what could be achieved within the atmosphere with jet propulsion. Indeed, von Kármán proposed that the Air Force implement a deliberate, step-by-step guided missile development program based on air-breathing missiles rather than ballistic rockets. The Air Force would accept von Kármán's argument and follow the air-breathing approach to missile development. Although von Kármán differed with other prominent scientists who dismissed the ICBM entirely, his recommendations served to chart an Air Force course that would delay development of the long-range ballistic missile.[25]

Nevertheless, *Toward New Horizons* proved to be a landmark because it established the importance of science and long-range forecasting in the Air Force. In staking out a role for military research, von Kármán differed fundamentally with colleagues like highly regarded Dr. Vannevar Bush, who believed that the military services should confine themselves to improving existing weapons and leave new scientific ideas to the civilian experts. *Toward New Horizons* helped ensure that the Air Force would reflect von Kármán's thinking. As his biographer aptly concludes, von Kármán's "detailed, highly technical blueprint set the agenda of research and development [in the Air Force] for decades to come."[26]

Arnold's and von Kármán's comments did not escape the attention of Dr. Bush, then the influential Director of the Office for Scientific Research and Development, and Chairman of the Joint Committee on New Weapons of the Joint Chiefs of Staff. Having sharply differed with von Kármán on military prerogatives in the research field, he turned his attention to the predictions of military officers on matters scientific. Appearing before a special Senate Committee on Atomic Energy in December 1945, Dr. Bush observed that

> We have plenty enough to think about that as [sic] very definite and very realistic-enough so that we don't need to step out into some of these borderlines, which seem to me more or less fantastic. Let me say this: There has been a great deal said about a 3,000-mile high angle rocket. In my opinion such a thing is impossible and will be impossible for many years. The people who have been writing these things that annoy me have been talking about a 3,000-mile high-angle rocket shot from one continent to another carrying an atomic bomb, and so directed as to be a precise weapon which would land on a certain target such as this city. I say technically I don't think anybody in the world knows how to do such a thing, and I feel confident it will not be done for a very long period of time to come. I think we can leave that out of our thinking. I wish the American public would leave that out of their thinking.[27]

When asked whether he was addressing his remarks to anyone in particular, he specifically identified General Arnold, whose report to Secretary of War Patterson had appeared the previous month.

Although Dr. von Kármán could characterize Vannevar Bush as "a good man… limited in vision,"[28] Bush and other prominent civilian scientists who expressed similar criticism had a major influence on the Air Force missile and space development programs. Their pessimism reflected current thinking in many postwar circles that contributed to stifling research by limiting it to the technical problems posed by ICBMs. In the postwar flush of victory and sense of American superiority, the American monopoly of seemingly scarce of fissionable uranium and the great weight of the first atomic bombs produced an air of complacency about the technological future. Atomic bombs of over five tons and relatively poor destructive capacity ("kill-radius") suggested that missiles could never be constructed with sufficient thrust and guidance accuracy to provide a credible operational weapon. Dr. Bush continued until his retirement in 1948 to manage research and development for the Defense Department, where he became known for his parsimonious funding of military programs that could not guarantee progress to his satisfaction.[29]

# CHAPTER 1
# Before Sputnik:
## The Air Force Enters the Space Age, 1945–1957

In the aftermath of World War II Air Force leaders laid the foundation for future operations in space by establishing a clear research and development focus for the new service. Commanding General of the Army Air Forces Henry H. "Hap" Arnold and his eminent scientific advisor Theodore von Kármán set the course through their policy statements, organizational decisions, and comprehensive analysis of Air Force scientific requirements for a technological future. Their legacy appeared endangered in the late 1940s when tight budgets and higher priorities confined space and long-range missile development to low level studies at best. Air Force leaders seemed intent on establishing Air Force responsibility for the as-yet-to-be-determined space mission, but unwilling to promote the development of satellites and booster missiles that would make possible such a mission.

By the early 1950s, however, change was in the air. New concerns about Soviet political activity and ICBM development compelled leaders to reexamine the country's defense posture. In doing so, missiles and satellites received new attention. Larger defense budget outlays and successful testing of thermonuclear devices offered the promise of a feasible ballistic missile and space booster. A number of government officials and Air Force officers who shared Arnold's legacy acted as catalysts for change by creating new organizational structures and promoting greater awareness of spaceflight opportunities. They faced strong opposition every step of the way. Yet on the eve of Sputnik, their considerable efforts helped bring the Air Force and the nation to threshold of space.

## Rand Proposes a World-Circling Spaceship

In a postwar America, with armed forces undergoing demobilization and reassertion of domestic priorities, Arnold and other Air Force innovators quickly realized that it was one thing to advocate an imaginative, liberally-funded research and development program for the Army Air Forces (AAF) and quite another to have it put into practice by a conservative military establishment. The Air Force's initial involvement with artificial earth satellites illustrates the difficulty of gaining approval for a system of the future rather than the present.

In early 1946, the AAF found itself about to be outmaneuvered by Naval officers who had been pursuing satellite feasibility studies since the end of the war. Captivated by a space study written in May 1945 by German space scientist Wernher von Braun, as well as the horde of captured V-2 rocket components, Dr. Harvey Hall of the Navy's Bureau of Aeronautics Electronics Division proposed a testing program to determine the feasibility of artificial satellites. Based on a current Naval hydrogen rocket motor development program, Commander Hall's team formed a Committee for Evaluation of the Feasibility of Space Rocketry and envisioned launching a liquid hydrogen-oxygen single-stage earth satellite to conduct scientific testing. Naval leaders agreed, and Hall called on four companies, including GALCIT, for technical assistance with fuels, electronic components and structural characteristics. Early in the new year, all four agreed that a satellite could be placed in earth orbit if the Navy proved willing to provide sufficient funding.[1]

Unable to gain the required Naval financial support, Commander Hall proposed a cooperative space venture to AAF representatives at a 7 March 1946 meeting of the War Department's Aeronautical Board. Although AAF Board members questioned the high costs involved, they expressed interest and promised to consult with Major General Curtis E. LeMay, Arnold's Deputy Chief of Staff for Research and Development, before the Board reconvened on 14 May. After discussions with General Carl A. Spaatz, who replaced General Arnold on 1 March as commanding general, LeMay informed Hall that the AAF could not support the Navy project but nevertheless would continue discussions on the subject. Already AAF leaders had decided that artificial earth satellite programs should be an AAF responsibility based on the argument that military satellites represented an extension of strategic air power. For the first time air leaders outlined the rationale for an Air Force space mission that would appear haphazardly over the next ten years, then surface prominently during the roles and missions debates after Sputnik.

To forestall the Navy's initiative in the spring of 1946 and help establish AAF primacy in the field, the service needed to demonstrate competence equal to the Navy's. In April LeMay turned to Project Rand for the necessary technical expertise. In just three weeks, the Rand team of sixteen experts completed their justly celebrated 250-page engineering analysis of a "World-Circling Spaceship."[2] Based on both the current state of technology and expected future engineering developments, the Rand team argued that "technology and experience have now reached the point where it is possible to design and construct craft which can penetrate the atmosphere and achieve sufficient velocity to become satellites of the earth." Indeed, the report predicted that the U.S. could launch a 500-pound satellite into a 300-mile orbit within five years at a cost of $150 million. Rand's analysts declared that even their most conservative engineers agreed, and they supported their prediction with a series of detailed studies in two chief areas.

One comprised technical feasibility studies dealing with such satellite-related issues as propulsion options, risks posed by potential meteor strikes, trajectory analyses, the important "re-entry" challenges posed by the intense heat objects would encounter returning through the earth's atmosphere, and, in contrast to the Navy's single-stage rocket, the use of a three-stage liquid hydrogen-oxygen rocket booster. The analysts argued that no technical challenge they investigated seemed overwhelming.

In a second area, noted radar expert Louis N. Ridenour examined a number of potential military satellite uses in a chapter titled "The Significance of a Satellite Vehicle." Focusing on defense support or "passive" military uses of satellites, he described satellites as nearly invulnerable observation platforms that could provide weather and bomb damage assessment data. He went on to describe the satellite as a communications relay station, in which satellites could be positioned at an altitude of approximately 25,000 miles so that "their rotational period would be the same as that of the earth." For the first time, a serious satellite proposal projected launching satellites into geosynchronous orbit for effective, worldwide communications.[3] Ridenour devoted most of his chapter to the satellites' scientific role in supplying important data unaffected by atmospheric conditions. As an aid to research, the satellite could facilitate the study of cosmic rays and provide precise gravitational measurements, as well as considerable astronomical data. Moreover, instrumented satellites could furnish important bio-astronomical information for medical scientists concerned with life in an acceleration-free environment.

Despite Ridenour's coverage of "passive" defensive military functions, he briefly raised the possibility of using satellites as offensive weapons. Given the onset of the missile age, he argued, satellites could provide both accurate guidance for missiles and serve as missiles themselves. He based his argument on the compatibility between missile and satellite technology as well as launch velocity requirements. As he explained,

> There is little difference in design and performance between an intercontinental rocket missile and a satellite. Thus a rocket missile with a free space trajectory of 6,000 miles requires a minimum energy of launching which corresponds to an initial velocity of 4.4 miles per second, while a satellite requires 5.4. Consequently, the development of a

> satellite will be directly applicable to the development of an intercontinental missile.[4]

In short, if you produce an ICBM, you also have a satellite launcher. In the future, however, the technical relationship between long-range missiles and satellites would remain largely unexploited. Even missile advocates normally argued that satellite development interfered with the greater need to accelerate missile programs. Later, closer examination would show that satellite technology could be applied to missile guidance systems and thereby contribute to missile development. The missions identified by Louis Ridenour would become a part of the Air Force and the national space program from the Eisenhower administration forward. Unfortunately, many of the Rand study's predictions and analyses would be forgotten in the years ahead. David Griggs, for example, in the report's introduction turned his vision to the future:

> Though the crystal ball is cloudy, two things seem clear: 1. A satellite vehicle with appropriate instrumentation can be expected to be one of the most potent scientific tools of the Twentieth Century. 2. The achievement of a satellite craft by the United States would inflame the imagination of mankind, and would probably produce repercussions in the world comparable to the explosion of the atomic bomb.[5]

Armed with the Rand study, General LeMay formally declined the Navy offer of a joint Navy-AAF program at the May meeting of the Aeronautical Board and staked out the AAF's claim to potential satellite operations. At the same time board members decided that the costs of developing and operating a satellite did not justify a major effort on a project of questionable military utility. The Board agreed to permit both services to continue their studies, with the jurisdictional assignment of satellite responsibility left unresolved.

The 1946 Rand report established the technical feasibility of orbiting a satellite but ruled out its likely use as an offensive weapon because available propulsion systems could not launch heavy atomic weapons into earth orbit. Given this restriction, the problem became establishing a credible role for an orbiting satellite that could justify the enormous cost and quiet the skeptics of "push-button warfare." During the next several years, Rand's satellite proposals would continue to founder on the criticism of cost and utility, while greater interest in developing guided missiles served to retard satellite progress further.

## The Air Force Shuns Ballistic Missiles

The analysts at Rand underscored the relationship between satellite and missile development. Not only would progress with satellites promote greater interest in missiles as boosters, but improvement in satellite technology could benefit missile development as well. Yet, the Air Force establishment, which focused on its bomber fleet, seemed unaware of the potential for mutual benefits, and later in the 1950s, when missiles promised additional strategic firepower for the nation's arsenal, critics of a forceful space program argued that satellites must not be allowed to interfere with missile development. The Rand analysts also might have noted that the missile-satellite relationship meant that any progress with satellites would depend on developments in the higher priority missile field. In the years after the Second World War, however, neither subject drew significant attention from the Truman administration and the defense establishment. As with satellite proposals, initial postwar interest in long-range guided missiles soon succumbed to an Air Force policy that relied on strategic bombers carrying air-breathing missiles, interservice conflicts over roles and missions, and administration-imposed budget ceilings that compelled Air Force planners to focus on present needs.[6]

General Arnold was not the only military leader impressed by the German V-2 achievements during the war. In the flush of victory, all the services sought to build on the wartime experience by conducting rocket and guided-missile experiments based either on aerodynamic, jet-propelled "cruise" missile principles, or the German V-2 short-range liquid-propellant ballistic rocket technology. Operation Paperclip brought nearly 130 leading German rocket scientists, a vast array of data, and approximately 100 dismantled V-2s to White Sands, New Mexico. There, under Project Hermes, the Army Ordnance Division conducted upper atmospheric research into airborne telemetry, flight control, and two stage rocket capability with representatives from the Air Force, the Air Force Cambridge Research Center, the General Electric Company, the Naval Research Laboratory, and a number of scientific institutions, universities, and government agencies. From 1946 to 1951, participants received valuable data from 66 V-2s that first carried various scientific instruments then, later, primates.[7]

By early 1949 the Army, which viewed rockets as extensions of artillery, had successfully used a V-2 as the mother vehicle to launch the Jet Propulsion Laboratory's WAC Corporal second-stage rocket to an altitude of 250 miles. As Frank Malina noted, "the WAC Corporal thus became the first man-made object to enter extraterrestrial space."[8] These early V-2-based Bumper-WAC experiments set the stage for the Army's future missile and space program involving the Redstone, Jupiter, and Juno boosters developed by the von Braun team under Army supervision after it moved in 1950 from Fort Bliss, Texas, to the Redstone Arsenal at Huntsville, Alabama. Postwar naval rocket research led by the Applied Physics Laboratory of Johns Hopkins University and the Naval Research Laboratory in Washington, D.C., produced two reliable and effective sounding rockets: the fin-stabilized Aerobee, a larger version of the WAC Corporal modified for production as a sounding rocket, which achieved a height of 80 miles; and the more sophisticated Viking, which would reach an altitude of 158 miles in May 1954. A modified Viking eventually would provide the booster for the four-stage Project Vanguard, the nation's first "civilian" space program.[9]

Despite General Arnold's interest in developing long-range missiles of the V-2 type, the Air Force followed the path charted by Theodore von Kármán, which stayed within the atmosphere and the initial Air Force domain. Short-range jet-propulsion weapons seemed to offer faster development and better range and payload capabilities. They also directly complemented the strategic bomber fleet, the nation's intercontinental strike force of the day. In October 1945 the Army Air Forces solicited proposals from seventeen aircraft companies for a ten-year research and development program for pilotless aircraft, and the fiscal year 1946 budget included an impressive twenty-six different projects. Yet only two involved missiles in the 5,000 mile range, and one of these consisted of a Northrop Aircraft supersonic turbojet vehicle. The other, a supersonic ballistic rocket design from the Consolidated Vultee Aircraft Corporation (Convair), would serve as the precursor of the future Atlas ICBM.[10]

If the Army Air Forces seemed devoted to shorter-range air-breathing missiles, it could not abandon long-range missile development to the Army or Navy. All three services jealously guarded their prerogatives and jockeyed fiercely with their rivals over roles and missions in the new postwar world. As it looked to a future as an independent service, the Army Air Forces proved particularly sensitive to new, unproved weapon fields such as rockets and missiles. While General LeMay in early 1946 staked out the AAF's claim to any prospective satellite mission, he also became embroiled with Army and Navy representatives over which service should be responsible for what types of missiles. Above all, the Army Air Forces took special interest in missiles it considered strategic.

Confusion and friction about missile development and operational control first emerged during the war in the competition within and among the services. A number of Army Air Forces offices asserted their "special" interests, while attempting to ward off the Army Ordnance Command and various elements in the War Department. A directive issued by Lieutenant General Joseph T. McNarney, Deputy Chief of Staff of the Army, on 2 October 1944, attempted to clarify the situation by assigning the AAF responsibility for "all guided or homing missiles launched from the ground which depend for sustenance primarily on the lift of aerodynamic forces." Although this ruling appeared to award the Army Ordnance Department (the Army Service Force) the ballistic missile mission, the AAF, which sought primary responsibility for all missile "programs," continued to complain of Army encroachment into the aerodynamic field.[11]

Conflict persisted into the postwar era as each of the services pursued its own guided missile program while keeping a wary eye on its competitors. In a revealing memorandum in September 1946 to AAF chief General Spaatz, General LeMay expressed his concerns about the Air Force maintaining its rightful "strategic role." Admitting that the "long-range future of the AAF lies in the field of guided missiles," he cautioned that the Army's success in controlling guided missiles might embolden its leaders to seek control not only of close support but strategic aircraft as well. After all, he noted, the "stated opinion" of the Army Ground Forces is that guided missiles are extensions of artillery. LeMay saw the possibility of the Air Force losing control of a weapon system that might replace manned aircraft in the future. Yet control of the weapon did not necessarily mean that it should be developed, at least at the present time. Meanwhile, the Navy entered the contest for preeminence. Given AAF aspirations and its strategic mission, Naval leaders joined their Army counterparts in arguing that each service should have the freedom to develop missiles in response to its particular needs.

On 7 October 1946 the War Department's Assistant Secretary of War (Air), W. Stuart Symington, attempted to settle the dispute by awarding the AAF responsibility for research and development activities pertaining to all guided missiles. The directive remained silent, however, on the important question of ultimate operational assignment. The issue lay dormant until after September 1947, when the establishment of an independent Air Force reopened the competition. A year later the Defense Department achieved a modicum of peace when the Air Force relinquished its responsibility for conducting research and development work for the Army. In return, the Air Force received authority to develop strategic missiles, while the Army became responsible for tactical missiles. Meanwhile, the Air Force continued its pathbreaking ballistic missile defense studies, Projects WIZARD and THUMPER. Although the latter was cancelled in March 1948, Project WIZARD continued until early 1958, when then Secretary of Defense Neil H. McElroy reacted to persistent feuding over ballistic missile defense responsibilities by awarding the Army the mission of strategic defense and merging WIZARD with the Army's NIKE-ZEUS anti-ballistic missile system.[12]

The problem of interservice rivalry over missiles received little help from the defense committees most responsible for providing direction. With passage of the National Security Act of 1947, the Research and Development Board replaced the Joint Research and Development Board. Dr. Vannevar Bush continued as chairman until his retirement in October 1948. Neither he nor those active on subordinate committees, like the Committee on Guided Missiles, possessed the authority needed to provide the firm direction. Too often they allowed the complex committee system to work to their disadvantage and avoid decisive action.

Throughout the conflict over roles and missions, the Air Force demonstrated more interest in gaining and preserving its prerogatives than moving ahead with a strong missile research and development program. Paradoxically, as the Air Force's commitment to develop an ICBM diminished, its determination to be designated sole authority responsible for long-range missiles increased. Even with long-range cruise missiles, for which Air Force leaders sought exclusive control based on the service's strategic mission, it normally chose not to implement programs leading to operational missiles. Efforts to garner exclusive control of missiles would continue. In

September 1948, for example, the Defense Department awarded the Air Force operational control of surface-to-surface pilotless aircraft as well as strategic missiles. Two years later, in a very important March 1950 decision, the Air Force received official responsibility for developing long-range strategic missiles and short range tactical missiles. Later, near the end of the Truman administration, the Air Force successfully defeated the Army's bid to develop the Redstone rocket's range beyond 200 miles. The strategic mission would remain with the Air Force.[13]

In the late 1940s Air Force leaders signaled their attitude about research and development when forced to respond to the Truman administration's drastic economy drive that began in late 1946. Compelled to choose between supporting the forces of the present and those of the future, the Air Staff ignored the admonitions of General Arnold and Dr. von Kármán by focusing on manned aircraft to the detriment of guided missiles. As a result, Air Force research and development programs for missiles suffered drastically in the late 1940s.

Budget figures help tell the story of decline. Air Force leaders needed to show a firm commitment to research in terms of policy advocacy and budget allocation. As expressed by General Benjamin W. Chidlaw, commander of Air Materiel Command, "Many people have given lip-service to the magic phrase 'Research and Development.' Very few of us have really fought for it—and made sacrifices for it."[14] Without such a commitment, the Truman economy drive was bound to seriously erode research and development funding and projects. The fiscal year 1946 Army Air Forces budget allocated $28.8 million for research and development, with half earmarked to support the twenty-six guided missile programs sponsored in 1946. The initial fiscal year 1947 budget reflected the importance of research with a grant of $75.7 million, $29 million of which was dedicated to guided missiles research. Then the budget ax fell. Under pressure from the Defense Department, in December 1946 the Air Force cut the missile budget by $5 million and eleven missile projects. Additional funding cuts in May led planners to eliminate five more programs.[15]

Faced with drastic reductions in the guided missile program, the Air Materiel Command decided to protect those programs promising the earliest tactical operational availability, and in June 1947 General Hoyt S. Vandenberg approved the AMC recommendations. This criterion effectively eliminated the only long range guided missile project, the MX-774, and the Air Force terminated the Convair contract on 1 July. That same month the Air Staff established development priorities for managing the smaller budgets they expected in the future. The subsonic bomber and air-to-air and air-to-surface missiles received top priority. In the belief that long-range surface-to-surface missiles would be prohibitively expensive and require ten years to develop, build and launch, long-range ballistic missiles stood at fourth priority.[16]

By 1947 the pressure to downgrade the development priority of long-range missiles proved overwhelming. In the growing Cold War conflict the administration increasingly looked to strategic bombers, supported by cruise missiles, and the atomic bomb as the country's main line of retaliatory defense. Moreover, manned aircraft remained the heart of the Air Force, and advocates of a new, if potentially revolutionary, weapon and "push-button" warfare found themselves outmatched in competition for funding. Critics focused on the technological challenges of missile development. The budget slashers argued that putting scarce funds into a research program that might not be realized for a decade or possibly never could not be justified in light of current priorities. Therefore one must continue with a cautious step-by-step approach to any long-range missile program. Missile advocates found themselves victims of a circular argument: missiles seemed too challenging technologically, but no funds could be spent on solving the technological dilemmas; so the problems would go unresolved and the missile would remain "impossible." To questions about the logic of budgeting for missile programs, the answer always seemed to be the dogmatic response: "the time is not right" for an expanded program.

The Air Force's devotion to aerodynamic missiles like the intercontinental Navaho, with its combination ramjet-booster rocket propulsion, and the subsonic Snark and Matador missiles also must be seriously questioned. Planners consistently offered the rationale that aerodynamic research benefited ballistic missile research. This proved correct to a point, as shown by the transfer of the Navaho Rocketdyne engines for use in the Redstone and Atlas systems. Yet cruise missile guidance systems offered little commonality, while aerodynamic vehicles could provide no help with the ICBM's high-speed reentry from space into the upper atmosphere. Sadly, Air Force scientists never reexamined the assumptions so forcefully established in the 1945 von Kármán reports. Moreover, not one of the reports called for research and development to achieve strategic reconnaissance.[17] Although the Air Staff reassessed guided missile priorities in 1948 and 1949, it elected not to change them. Fortunately, Convair decided to use its own funds to continue the MX-774 project under imaginative Karel Bossart. Bossart's team persevered with their innovative experiments involving swiveling engines, internal fuel storage and tank design, and various means of separating the nose-cone warhead as a solution to the formidable reentry problem. All would prove important in designing the Atlas ICBM in the early 1950s. Meanwhile, in the late 1940s the outlook for the long-range guided missile project appeared bleak.

## Ballistic Missiles Receive New Life

The first serious signs of a change in attitude toward research and development in general and guided missiles in particular appeared in 1949. Faced with growing criticism, General Vandenberg requested the Scientific Advisory Board to examine the state of Air Force research and development. It appointed a special committee chaired by widely respected Louis N. Ridenour. Throughout the summer of 1949 he and his committee examined research and development programs, then on 21 September submitted a highly critical report. The committee determined that "existing organizations, personnel policies, and budgetary practices do not allow the Air Force to secure the full and effective use of the scientific and technical resources of the nation." Its major recommendations included ensuring better assignment and promotion opportunities for technical officers and reorienting budget priorities because "if war is not imminent, then the Air Force of the future is far more important than the force-in-being and should, if necessary, be supported at its expense." The Ridenour Report is best remembered, however, for its organizational recommendations: the creation of a deputy chief of staff for research and development on the Air Staff, and a new major Air Force command for research and development.[18] This was von Kármán's wish, too.

Because of expected opposition within the Air Force to a "civilian" report that called for radical change, sympathetic officers like Major General Donald L. Putt, the Director of Research and Development in the office of the Deputy Chief of Staff for Materiel, helped create a parallel, senior-level military group that would undertake a review similar to the Ridenour study and thereby promote broader acceptance for its recommendations. Their efforts produced the Anderson Committee, named for its chairman, Air University's General O. A. Anderson, which conducted extensive interviews throughout the Air Force before issuing its report on 18 November 1949. The Anderson Report strongly supported the Ridenour Committee's findings, and used the effective argument that failing to implement the recommendations might easily lead the Army and Navy to "take over responsibilities abdicated by the USAF."

The powerful arguments for change convinced General Vandenberg to promptly implement the organizational recommendations. On 23 January 1950, the Air Force created the Office of the Deputy Chief of Staff, Development, and the Air Research and Development Command (ARDC) with headquarters at the Sun Building in Baltimore, Maryland. Yet it would take the "personal salesmanship" of Lieutenant General James H. ("Jimmy") Doolittle, acting as special assistant to General Vandenberg a year later in the spring of 1951, to end Air Materiel Command's foot-dragging. In late March General Vandenberg ordered the immediate transfer of AMC's Engineering Division and other designated responsibilities and functions to the new command, and reassignment of ARDC directly to Air Force headquarters rather than AMC. If the new arrangement divided responsibility for weapons acquisition between the two commands, it nevertheless served to highlight the importance of the research and development function in contrast to the heretofore production-oriented Air Materiel Command. Significantly, the Air Staff assigned the guided missiles program to the new command.[19]

While the Air Force made organizational changes in the early 1950s, events on the international scene contributed to major reassessments of the country's defensive posture. News that the Soviet Union had successfully detonated an atomic bomb in

August 1949, communism's triumph in China, and alarming reports of Soviet progress in missile development led to calls for increased military preparedness both in and outside the administration. The outbreak of the Korean War in June 1950 served to heighten the growing sense of national weakness. In the summer of 1950, for example, Under Secretary of the Air Force for Research and Development John A. McCone submitted reports on America's vulnerability to Soviet attack to Secretary of the Air Force Thomas K. Finletter, advocating a "Manhattan-type" program for missiles under the "most capable man who can be drafted." In late August 1950 President Truman responded to calls for action by appointing T. K. Keller, chairman of the Chrysler Corporation, "Director of Guided Missiles." Unfortunately, Keller approached his job as missile "czar" on a part-time basis, and focused largely on cruise-type missiles and the Army's tactical Redstone missile. Convair's low-priority Atlas ballistic missile project received little attention. Nevertheless, the McCone reports contributed to the movement for action on guided missiles.[20]

Other efforts to enhance defense proved more significant. President Truman early in 1950 authorized immediate development of the hydrogen or thermonuclear bomb, while after the outbreak of the Korean War, Congress authorized a 70-group Air Force and nearly doubled the administration's defense budget request from $14.4 to $25 billion. Armed with its new wealth, the Air Force reconsidered Convair's long-range rocket proposal. The company's presentations led to a contract in January 1951 for project MX-1593, whereby Convair would examine both the ballistic approach and the "glide" vehicles which use rocket power to reach the outer atmosphere then use their wings to glide through the atmosphere to their targets. The boost-glide approach reflected continued Air Force interest in the postwar "X"-series of high-altitude rocket-powered aircraft.[21]

Convair's six-month contract to conduct a "study and test program" for two types of missile propulsion hardly represented a ringing endorsement of the ICBM concept. Nevertheless, by late summer 1951 the Convair engineers had selected the ballistic-type rocket largely because it represented a weapon considered unstoppable for the foreseeable future, while they believed the formidable technical problems solvable by the early 1960s. ARDC, which had responsibility for the guided missiles program, agreed that the missile deserved greater support. Convincing Air Force headquarters to award it sufficient funding and project priority, however, proved next to impossible. In the fall of 1951, the Air Staff's Research and Development Directorate rejected ARDC's request for increased funding and directed a slowed-down five-year test program before considering further commitments. Convair continued to lobby Air Force headquarters in late 1951 and early 1952, while ARDC's new commander, General Putt, in a letter to his former office of Research and Development, argued that the ballistic missile project should be approved immediately because of its total "invulnerability to all presently known countermeasures and because of the relative simplicity of the entire weapons system." Putt also warned that the Soviets appeared to be pursuing development of such a weapon. In the spring of 1952 Air Force headquarters referred the ARDC request to the Guided Missiles Committee of the Defense Department's Research and Development Board. The Committee authorized only continued studies and component testing, not the complete Atlas system.[22]

Despite growing evidence to the contrary, skeptics on the Air Staff and in the Defense Department continued to view the intercontinental ballistic missile as a weapon system too complex and likely impossible ever to reach the operational stage. Much of the criticism focused on the old issue of warhead weight. Yet by 1950 the Atomic Energy Commission affirmed the existence of a sufficient number of atomic weapons small enough to be carried in guided missiles. Moreover, President Truman noted that in early 1950 his military service chiefs proceeded with elaborate plans to use the H-bomb on the assumption that the tests he had just authorized would be successful. Test results at Eniwetok in November 1952 proved the feasibility of thermonuclear technology and confirmed their optimism. Based on the test results, ARDC petitioned the Air Staff to reassess the overly restrictive weight and accuracy parameters for the Atlas. In response, a Scientific Advisory Board ad hoc committee chaired by Dr. Clark Millikan reviewed the technical issues. Although the Millikan Committee concluded that anticipated warhead yields called for reducing accuracy and guidance requirements, it saw no need to accelerate the program. Rather it recommended a "step-wise" project that would guarantee "a review of the project at appropriate intervals." A sense of urgency remained absent.[23]

At the end of the Truman presidency strategic bombers and cruise missiles represented the key elements in the nation's offensive arsenal, while the ICBM project moved painfully forward as a cautious, low-funded, phased study and test program that reflected the traditional skepticism of the Air Staff. Given the fate of ballistic missile development over the course of the Truman years, satellite proposals could be expected to garner even less support. Most decision-makers remained blissfully unaware that missile propulsion, guidance, and reentry technologies could be useful for early stages of space exploration, while the response to guided missiles suggests that such knowledge would have had little bearing on satellite developments.

## The Air Force Studies Satellites

In postwar America satellite development followed a pattern similar to that of guided missiles. Initial interest faded under budget austerity, and serious government action only began to appear in the early 1950s. Although critics of ICBMs could stress their technological challenges, the German experience of World War II had demonstrated their potential military worth. Satellites, however, not only suffered from association with the "fantastic," but left many unconvinced of their military utility.

Back in 1946, while service jurisdiction over satellites remained undecided following the May meeting of the Aeronautical Board, Rand continued with its remarkable series of satellite studies. In February 1947, the "think tank" produced a second, multi-volume study that expanded on the initial 1946 report.[24] Led by James E. Lipp, head of Project Rand's Missiles Division, it provided detailed specifications for a reconnaissance satellite comprised of a three-stage rocket booster with a gross weight of 82,000 pounds, orbiting at 350 miles, and costing $82 million. Accompanying documents covering a variety of technical subjects from "Flight Mechanics of a Satellite Rocket" to "Communication and Observation Problems of a Satellite" offered contractors guidance for their own design work. The Rand analysis also identified for further development various component areas such as guidance control, orbital control, communications equipment and procedures, and reliable auxiliary power sources. Solar power and miniaturized electronic equipment had yet to be developed.

Two reference papers provided particularly insightful comments on the potential importance of reconnaissance satellites. In one, Yale astronomer Lyman Spitzer, Jr., addressed tactical satellite support of naval operations and the vulnerability of satellites to attack. Most interesting, he proposed satellites as communications relay stations and the application of astronomical telescopic principles to space reconnaissance. His work would contribute later to experiments using long-focal-length panoramic camera systems for surveillance purposes.

James Lipp's "The Time Factor in the Satellite Program" proved especially significant in light of future developments. He described the importance of satellites for scientific research, for military operations, for encouraging development of long-range rockets, and for providing the nation psychological and political benefits. Among his observations, he discussed polar orbits for recurring surveillance, geostationary orbits to compensate for the earth's rotation, and the use of television equipment and special telescopes for transmitting electro-optical images to ground stations. Several of Lipp's perceptive political and psychological assessments would prove hauntingly accurate. Noting that other nations would likely pursue satellite development, he argued that satellite feasibility had been proven, and "the decision to carry through a satellite development is a matter of timing, depending upon whether this country can afford to wait an appreciable length of time before launching definite activity."

Fully aware of the danger in waiting too long to develop satellites, he echoed the warning of David Griggs the year before by declaring: "The psychological effect of a satellite will in less dramatic fashion parallel that of the atom bomb. It will make possible an unspoken threat to every other nation that we can send a guided missile to any spot on earth." The importance of orbiting satellites outweighed the expense, he argued, and a "satellite development program should be put in motion at the earliest time."

Air Force leaders did not share James Lipp's sense of urgency. Six months passed before they requested Air Materiel Command to evaluate the Rand reports. In its late December 1947 evaluation, AMC officers offered a judgment that became commonplace in the years ahead. While they affirmed the technical feasibility of the reconnaissance satellite, they questioned both the high costs and lack of clear military utility. Constrained by "scarce funds and limited component scientific talent," the Air Force should not risk supporting a satellite development program when guided missiles deserved research funding priority. Characteristically, the Air Staff called for more studies on requirements and desired design specifications. In view of the severe missile program cuts in the fiscal year 1947 and fiscal year 1948 budgets, satellite advocates had no reason for optimism. With the only ICBM research program eliminated in July 1947, satellite studies represented the most proponents could expect and the least skeptical Air Staff planners needed to offer. Even so, during the next three years defenders of satellite utility studies needed to work hard to protect the "fantastic" elements from the budget ax.

Even though Air Force leaders proved unwilling to promote satellite development, they were not averse to campaigning for "exclusive rights in space." In January 1948, Chief of Staff Vandenberg became the first service chief to issue a policy statement on space interest when he declared that

> The USAF, as the service dealing primarily with air weapons—especially strategic—has logical responsibility for the satellite. Research and Development will be pursued as rapidly as progress in the guided missiles art justifies and requirements dictate. To this end, the program will be continually studied with a view to keeping an optimum design abreast of the art, to determine the military worth of the vehicle—considering its utility and probable cost—to insure development in critical components, if indicated, and to recommend initiation of the development phases of the project at the proper time."[25]

Although Vandenberg's statement might be faulted for its lack of clarity, clearly, once again, progress in the satellite field would depend on advances in missile technology without recognition that satellite technology might benefit the missile engineers. At the same time, funding would remain a major determinant. With sufficient money available, the Air Force, like the other services, would likely pursue a new mission to increase its share of the budget.

General Vandenberg's declaration appeared at a most opportune time for Air Force interests because Defense Department officials had decided once again to address the organizational squabble over roles and missions. Since September 1947 responsibility for satellite issues in the Defense Department belonged to the Research and Development Board's Committee on Guided Missiles. In December 1947, the latter formed a Technical Evaluation Group to assess satellite feasibility. Two months following Vandenberg's policy dictum, the Committee issued a report that verified the technical feasibility of satellites, but proceeded to assert that "neither the Navy nor the USAF has as yet established either a military or a scientific utility commensurate with the presently expected cost of a satellite vehicle." In hindsight it seems difficult to appreciate the question about military use, especially after the comprehensive, technical Rand report of 1947. At the same time, the satellite represented a "passive" weapon system that seldom elicited the interest of planners worried about supporting conventional strategic weapons. After all, they argued, what could the satellite do that aircraft could not, and at lower cost? Several years of analysis and promotion seemed to be required to establish military satellite utility. Significantly, the Committee recommended continuing with utility studies at Rand and allowing the research agency permission to consult with industry on system and component designs for a reconnaissance satellite.[26]

## Satellites Receive New Life

While Air Force leaders might have been disappointed that the committee did not endorse Vandenberg's policy statement, at least the Rand studies continued to receive Defense Department funding. The Navy attempted to join the Air Force as joint sponsor of the Rand project but failed to overcome the opposition of LeMay and other Air Force leaders. By the end of 1948, the Navy had "suspended" its satellite work. The Army, meanwhile, would not reenter the satellite arena until its Redstone rocket team proposed Project Orbiter in 1954. This left the Air Force alone on the satellite field, such as it was. Based on the findings of the Technological Capabilities Committee, Rand proceeded to develop a satellite project with component analyses for "eventual construction and operation of a satellite vehicle." Rand's research and study subcontracts would be subject to AMC's approval and the availability of funds. The key question involved utility. Rand's 1947 study had shown the serious complications associated with designing a recoverable space vehicle. This drew their attention in the years ahead almost entirely to instrumented satellites rather than manned spaceships. The issue for instrumented satellites then became what equipment would be necessary and what military purposes would they serve?

Rand analysts addressed these questions in several 1949 studies, including one entitled "Utility of a Satellite Vehicle for Reconnaissance," and in a study conference in 1949 it sponsored on the military usefulness of satellites. The conference produced an unusually convincing argument for developing a reconnaissance satellite. Noting that technology did not yet permit satellites to operate as destructive weapons, conferees emphasized the passive satellite roles of communications and reconnaissance—especially as political and psychological weapons designed to alter Soviet political behavior. After establishing a list of eight basic satellite characteristics, the analysts assessed the possible functions such a satellite would likely perform. They concluded that as a surveillance instrument it could serve as a major element of political strategy. As a vehicle capable of penetrating the secrets behind the Iron Curtain, it could provide intelligence that might be used in various ways to modify Soviet actions. As the conferees concluded, "no other weapon or technique known today offers comparable promise as an instrument for influencing Soviet political behavior." The study group recommended that Rand impress the Air Force with the surveillance potential of such a satellite.[27]

This Rand study, too, produced few immediate results. As one more study, however, it helped foster growing awareness of reconnaissance satellite capabilities and helped lay the groundwork for passive surveillance applications when Rand commenced its component studies and designs in 1950 after the Air Force received authority to develop booster rockets. Advocates hoped the new concern with Soviet missile advances and the Korean War would generate increased interest in strategic satellites as it seemed to do for missiles.

In late November 1950 Rand recommended the Air Force authorize extension of Rand's research into specific areas of the reconnaissance satellite mission. With Air Force approval, Rand investigators produced two reports in April 1951: "Utility of a Satellite Vehicle for Reconnaissance" and "Inquiring into the Feasibility of Weather Reconnaissance from a Satellite Vehicle." The reconnaissance portion drew the most attention from the Air Force. Based on detailed analysis, it advocated "pioneer reconnaissance," or extensive coverage using television with a resolution of between 40 and 200 feet, in a 1,000-pound payload with a space vehicle weight of 74,000 pounds. With improvements in television technology, the researchers expected to achieve the 40-foot dimension in the near future. They hoped this would permit satellites to conduct all military reconnaissance and finally satisfy the skeptics.

The newly activated Air Research and Development Command enthusiastically supported the Rand findings and authorized Rand to recommend measures needed to begin development work in the reconnaissance program. Eventually this research would lead to the milestone Project Feed Back report of 1954. Rand began in 1951 by subcontracting key subsystems such as orbital sensing and control to North American Aviation, and optical systems, television cameras, and recording equipment to the Radio Corporation of America (RCA). In November 1951 the Air Force contracted with the Atomic Energy Commission to study small nuclear reactors as satellite power sources. By June 1952 the Commission reported encouraging results from preliminary testing, and Rand moved forward with its Feed Back research, which focused on designing and evaluating satellite components.[28]

The findings of the Air Force Beacon Hill Study reflected the state of these efforts at the close of the Truman era. In early 1952, the Air Staff authorized a study group, chaired by Eastman Kodak's Carl Overhage, and consisting of fifteen prominent reconnaissance specialists, including Polaroid's Edwin Land, Louis Ridenour, and Lieutenant Colonel Richard Leghorn, USAFR, considered by Rand one of the few "integrative" thinkers concerned with so-called pre–D-Day reconnaissance. The report called for various improvements to obtain strategic intelligence, and specified refinements to sensors lofted in high-altitude aircraft and balloons, sounding rockets, as well as long-range air-breathing missiles like the Navaho. The group also recognized the need for high-level approval for any overflight of foreign territory, an issue that would dominate political space policy debates during the Eisenhower administration. Although the Study addressed important issues, Rand officials referred to the Beacon Hill Report as "Reconnaissance without Satellites," and considered it a setback for reconnaissance satellites. Not a single Beacon Hill briefing or study addressed either weather reconnaissance or electro-optical reconnaissance, important applications Rand had been considering for years.[29]

On the eve of the Eisenhower administration, satellite advocates had cause for both hope and dismay. The Air Force-sponsored Rand studies had identified a mission, strategic reconnaissance, and produced increased technical justification for developing a military satellite. Feed Back research involving several hundred scientists and engineers seemed well underway by the end of 1952 and promised at long last to set the stage for satellite development. Renewed Air Force interest in the Convair long-range ballistic missile also indicated that large satellite booster rockets might soon be available. Yet the Rand reconnaissance proposal remained a planning project, and the ICBM program moved forward at a very leisurely pace. At the beginning of 1953, it remained to be seen how strongly the new Eisenhower regime would support both satellites and missiles.

Reviewing the course of missile and satellite development in the Truman years, clearly both satellites and missiles fell victim to skepticism about their practical, military use and to economic retrenchment that grew unabated through the 1940s. In a sense, General Arnold's retirement in March 1946 left no one of his stature in either the Air Force or the defense establishment willing to challenge national policy that favored strengthening the forces in being at the expense of future capabilities. Nor did Air Force leaders in the late 1940s question seriously the service's gradualist approach to guided missile development or the priority accorded aerodynamic, cruise missiles rather than long-range ballistic missiles. By the 1950s, however, heightened security concerns and technological change offered the prospect of breaking with the past and accelerating the satellite and missile programs.

## Eisenhower Faces the Threat of Surprise Attack

President Dwight D. Eisenhower took office in January 1953 determined to implement a "New Look" defense policy that stressed strategic nuclear striking power at the expense of conventional forces.[30] In order to do this and roll back the Truman administration's Korean War budget from nearly $45 billion to $35 billion, he charged his Defense Department to end waste and duplication throughout the services. Missile and space programs could be expected to absorb their share of Defense Department cutbacks. Indeed, in early 1953 the administration expressed no particular interest in accelerating either program. Yet in the space of just four years, the regime would come to preside over a costly expansion of both military missile and satellite programs and a civilian satellite project that together represented the birth of the American space program. These events have left their mark on the nation ever since.

The rapid growth in space activities under Eisenhower, however, became lost in the wake of the Sputnik launches of October 1957. Critics contended that the administration had allowed the nation to be humiliated and endangered by failing to appreciate the political and psychological importance of being first into space and the demonstration of Soviet leadership in large operational boosters and ICBM technology. The public sensed a directionless program.

In fact, on the road to a national space policy, Eisenhower and his advisors followed a far more sophisticated, secretive, and complex path than many at the time appreciated. Early in the administration, they decided to follow what amounted to a dual space program that focused on launching a civilian scientific satellite to establish the principle of unimpeded overflight in space for the military satellites to follow. The administration had no intention of "racing" the Soviets in space affairs and gambled that the low priority and modestly funded civilian satellite venture could be completed in time for launch of the International Geophysical Year (IGY). Meanwhile, the major defense effort would be devoted to developing ICBMs for the "New Look" doctrine of "massive retaliation" as soon as possible. Given these priorities, the military reconnaissance satellite momentarily represented the odd man out in the space program.

The Eisenhower space program remains an impressive achievement, if not entirely preplanned. Early in the administration, three developments served to propel the nation to the threshold of space. One involved the President's determination to take all possible measures to forestall another "Pearl Harbor." Another concerned the technological "thermonuclear breakthrough" that solved much of the ICBM payload weight dilemma. Finally, several determined government officials risked violating bureaucratic routine to energize the decision-making process. Throughout the period, the Air Force remained divided between reform minded individuals who favored accelerated growth of missile and space programs, and more conservative officials who preferred a cautious, step-by-step approach leading to commitment well into the future. Although the reform group proved victorious, their members had to bypass traditional Air Force bureaucratic structure and procedures to achieve their goals.

Like General Arnold, World War II veteran General Eisenhower could never forget Pearl Harbor. As president, his scientific advisor, James Killian, remarked that Eisenhower remained "haunted"..."throughout his presidency" by the threat of surprise nuclear attack on the United States.[31] To avoid this horror, intelligence data on Soviet military capabilities became essential. Yet, neither news of Soviet advances in long-range bombers like the TU-4, or reports on Soviet long-range missile progress could be verified. At the same time, the development of a thermonuclear device and its testing in both the United States and the Soviet Union raised alarms about a potentially devastating surprise attack. A number of Rand studies in 1952 and 1953 heightened awareness by describing the vulnerability of strategic air bases to attack. The Rand assessments complimented the Central Intelligence Agency's (CIA) national intelligence estimates that forecasted imminent Soviet atomic weapons production and delivery capabilities.[32]

But reports remained confusing or contradictory, and the administration quickly realized that current intelligence methods could not provide meaningful data. Pre-hostilities intelligence information became increasingly essential, and all parties realized that aerial reconnaissance offered the most effective means to solve the dilemma. The near-term answer became the U-2 high-altitude reconnaissance plane, while the long-term solution would prove to be the military reconnaissance satellite. Meanwhile, the best potential satellite boosters also represented the best weapons to prevent surprise nuclear attack.

## Trevor Gardner Energizes the Missile Program

While Eisenhower and his advisors worried about intelligence data, Trevor Gardner, the "technologically evangelical" Assistant Secretary of the Air Force for Research and Development, made it his mission in public life to convince the government that the nation must pursue a crash program to develop an operational Air Force ICBM or face nuclear disaster.[33] Ironically, he assumed his office with the mandate to implement the expected economy program in the Defense Department by ending waste and duplication in the Air Force missile program. Assistant Air Force Secretary Gardner was to have a profound influence on the nation's missile program, but he and his allies felt compelled to go outside established Air Force and Defense Department structures to carry out their goals.

In April 1953 Gardner called for review of all Air Force missile programs. He instinctively rebelled against ARDC's cautious approach and the Air Staff's persistent delaying tactics. Their reasoning reflected the dilemma of the self-fulfilling prophecy: missiles represented too costly an investment for an "impossible" system. But no development money meant that the problems would continue unsolved and the missile remain "impossible." Gardner, who had heard reports of the "thermonuclear breakthrough," knew that, now, accuracy and guidance performance requirements could be relaxed and the missile no longer need be considered "impossible."[34]

Fortunately, Gardner found willing allies to accelerate missile development among middle echelon ARDC and Air Staff officers, as well as the Convair group promoting Atlas. At the same time, the Joint Chiefs of Staff, as part of its military posture review for the incoming administration, called for a broad-based reexamination of the entire Defense Department missile picture. Gardner received the assignment to review the country's missile programs based on Secretary of Defense Charles Wilson's drive to eliminate waste and duplication among the services.

At this point Gardner decided to bypass the Air Force bureaucracy and appoint a full-time group of experts on whom he would rely for advice. Late in the fall of 1953 he convened the Strategic Missiles Evaluation Committee (SMEC) under the chairmanship of renowned Princeton Institute for Advanced Study mathematician and activist John von Neumann. This group, which came to be referred to as the von Neumann Committee, comprised an impressive assemblage of scientists and engineers, all of whom had been handpicked by Gardner for their "progressive" views on ICBM requirements as well as their technical brilliance. Trevor Gardner charged von Neumann's committee to determine the measures necessary to accelerate development of the Atlas missile.[35]

While von Neumann committee members deliberated, a Rand Corporation group directed by Bruno W. Augenstein neared completion of a similar study on mounting thermonuclear weapons atop ICBMs. Responding to Air Force direction to investigate aerodynamic systems, Rand analysts had produced a number of reports on missiles in the early 1950s that favored ramjets and boost-glide rockets over ballistic missiles. When nuclear weapons were made smaller, Rand concluded that ICBMs represented the optimum surprise-attack weapon, which heightened the challenge to produce pre-hostilities strategic intelligence. At the same time, an accelerated ICBM program would mean having space boosters available at lower costs. Rand evaluators worked closely with the von Neumann team, and Augenstein briefed von Neumann Committee members personally in December 1953 on his findings. To no one's surprise, the two groups reached similar conclusions in their final reports, which appeared two days apart in early February 1954. These reports would help convince President Eisenhower to convene that spring the Surprise Attack Panel or, as it was soon renamed, the Technological Capabilities Panel, chaired by James Killian.[36]

The von Neumann report confirmed the Rand analysis by calling for a drastic revision of the Atlas ICBM program in light of Soviet missile progress and newly available thermonuclear technology. Referring to the recent Operation Castle tests in the spring of 1953, von Neumann predicted the advent of thermonuclear warheads weighing only 1,500 pounds with a yield of one megaton. This meant that performance criteria for the Atlas could be reduced, making its development more feasible within the state of the art.[37]

Critical of Convair's management practices and design, which envisioned an enormous five-engine rocket to boost the earlier, heavier warhead, the committee recommended a thorough study of various alternate design approaches and the establishment of a new development-management agency in the Air Force authorized to provide overall technical direction. Committee members considered this agency more important than all the technical guidance, warhead weight, and reentry problems yet to be solved. Finally, panel members urgently recommended the project be given high priority and substantial funding. The von Neumann report would stimulate the revision necessary to develop the large boosters required for military reconnaissance satellites.[38]

Armed with the findings of the Rand and von Neumann Committee studies, Gardner set off to win support throughout the Air Force hierarchy to expedite an expanded ballistic missile development effort. After gaining approval from Chief of Staff General Nathan Twining and Secretary of the Air Force Harold Talbott, Gardner could successfully counter any disapproval from key air staff agencies and Air Research and Development Command. The traditional Air Force bureaucracy did not favor this civilian-sponsored initiative that proposed creating a separate development-management agency that would bypass established administrative channels. In the end, the Air Staff supported the Gardner-engineered initiative, perhaps because disapproval might result in appointment of a new missile "czar" completely outside the Air Force framework. If not all that the Gardner group desired, the results nevertheless proved "revolutionary." In April 1954 the Air Staff proceeded to create a new Air Force headquarters position, an Assistant Chief of Staff for Guided Missiles, with responsibility for coordinating all Air Force guided missile activities. The following month, Air Force leaders took a more significant step by directing ARDC to form a West Coast project office at Inglewood, California. Organized as the Western Development Division (WDD), the latter represented the central von Neumann committee recommendation, and Gardner insured that the new organization's chief would be his ally, Brigadier General Bernard Schriever. Shortly after the Western Development Division began functioning in August, General Schriever arranged for the Air Force to contract with the Ramo-Wooldridge Corporation as full-time technical consultant to his command. Schriever proved to be a splendid choice to head a crash ICBM program. A young disciple of Hap Arnold, whom he considered "one of the most farsighted persons" he had ever known, he had joined Trevor Gardner's reform group in early 1953 while serving on the Air Staff as Assistant for Development Planning in the office of Deputy Chief of Staff for Development. He used his intelligence, patience, and superb negotiating skills with military, government and private industry leaders to become an effective advocate for missile and space systems causes.

In order to produce an operational missile by the end of the decade, Schriever's command adopted a number of managerial innovations that would become common practice for the Air Force in future years. Help again came from the von Neumann committee, which had been reconstituted in April 1954 as the Atlas Scientific Advisory Committee. Together with Ramo-Wooldridge, the committee convinced Convair and the Air Force to design a smaller missile capable of carrying the lighter, powerful hydrogen warhead. Given the time constraints, the planners chose to develop a "light-weight" three-engine rocket with a thin metal air frame skin housing the liquid-fuel and oxidizer tanks made rigid through overpressure. The crash program called for special management techniques, too. In the summer of 1954 the ICBM committee recommended that the Western Development Division award alternate subsystem contracts, whereby each Atlas component would be "backed up" by an alternate relying on different technology. This more costly parallel development approach meshed effectively with the new "concurrent" procedures pioneered by Schriever and his staff. Under concurrency, all measures necessary to construct and deploy the system would be completed simultaneously. Still skeptical of Convair's capabilities, however, Air Force officials applied the parallel development concept on a larger scale by producing at the same time a second, more-sophisticated "back-up" ICBM, the Titan. Designers configured the new Titan as a two-stage liquid-propellant missile, with a more advanced guidance system, and rigid frame to permit underground deployment. Parallel development allowed Atlas and Titan program managers to replace subsystems in case of failure or technological breakthrough, while advanced designs could be pursued without risk to the overall ICBM program. It served as an effective risk mitigation approach that proved its worth when the Air Force launched both Atlas and Titan missiles successfully by the end of the decade.[39]

General Schriever could hardly have expected such future success when he surveyed the state of his command in the spring of 1954. Indeed, he faced a major battle within the Air Force to retain control of his project. Despite his relatively independent status under ARDC with responsibility for system planning, technical direction, and budgeting, the Air Materiel Command continued to control the major funding areas of system production and procurement. To do the job assigned, General Schriever believed he needed authority over all aspects of missile acquisition, from design, research and development, through production. The Air Staff, however, refused to compromise on this issue, and AMC maintained its production prerogative by establishing a Special Aircraft Project Office at Western Development Division to handle ICBM procurement. According to General Schriever, initial friction soon gave way to a reasonable "partnership" arrangement after the general established good rapport with the AMC officers. This far from optimum division of system management responsibilities would continue until the creation of Air Force Systems Command during the organizational reform of 1961.[40]

Managerial problems with the Air Materiel Command proved only the tip of the iceberg. Even though the Secretary of Defense had declared Atlas of "critical importance" in early 1955, the bureaucratic labyrinth at the Air Staff and the Defense Department continued to cause bottlenecks and delays because of the multiple program review levels. Once again Trevor Gardner—encouraged by General Schriever's active support—decided to bypass the Air Force bureaucracy by going directly to Congress. Meetings with Senators Clinton Anderson and Henry M. Jackson, the two most influential members of the Joint Committee on Atomic Energy, and congressional visits to Schriever's suburban Los Angeles headquarters, convinced the congressmen to support streamlined management procedures to eliminate the bureaucratic obstacles. At the same time, additional reports of new Soviet long-range bombers and missile tests picked up by radars in Turkey raised fears that the United States might be falling behind in the ICBM race.[41]

The congressmen wrote President Eisenhower in late June 1955 about their concerns and recommended immediate action on the Atlas program to avoid funding delays, overcome interference from major Air Force commands, and bypass the multiple review levels. By fall the President had designated the Atlas ICBM the "highest national priority" weapon system. Still, procedures remained unchanged, prompting Trevor Gardner again to seize the initiative by directing Hyde Gillette, Air Force Deputy for Budget and Program Management, to form a committee to devise new, more effective procedures for the missile program. In October 1955 the Gillette Committee's recommendations led to the establishment of a ballistic missiles committee at both Air Staff and Defense Department levels to function as the sole reviewing authorities for Western Development Division programs. Gone were the various separate offices that Schriever had to consult individually. Now he submitted a yearly development plan to a single committee, made up of representatives from the offices concerned with the ICBM program. Although not entirely able to overcome all Air Staff skeptics and AMC opponents, the Gillette procedures removed many bureaucratic bottlenecks, and the ICBM program moved ahead rapidly.[42]

By 1955 the momentous procedural and organizational decisions for ICBM development proved to have a major impact on the military space "program" as well. Gardner and Schriever, given their focus on missile requirements, could not be expected to devote their energies to lower-priority satellite activities. In fact, they viewed the military satellite space program as a competitor for personnel, funds, and contractors. Nevertheless, the relationship between satellites and missiles had become better understood as rockets with sufficient thrust soon would be available to launch the heavier satellites preferred by the Air Force. If the Western Development Division were to gain responsibility for the Air Forces' advanced reconnaissance satellite project, advocates hoped that the Gillette procedures would benefit satellite development as they promised to do for the ICBM.[43]

## The Air Force Commits to the First Military Satellite

While Secretary Gardner and General Schriever worked on missile issues in the spring of 1954, the military satellite project also cleared major hurdles. Now, with ICBMs representing a practical option, Rand's studies on satellite systems received new life, as the Eisenhower administration sought solutions to the intelligence dilemma of providing accurate data on Soviet offensive capabilities.

Rand studies had proceeded on the assumption that the Atlas ICBM would provide the space booster required to launch a reconnaissance satellite. Rand also assumed that spaced-based sensing systems offered the best means of quickly relaying important intelligence data to ground stations. By the spring of 1953, Rand satellite studies of the previous two years—now referred to as Project Feed Back—began to draw a wider audience in view of new high-level interest in Soviet missile advances. Promising results from Atomic Energy Commission tests on nuclear power for satellites encouraged the Air Force in May to direct further study of the matter and to have ARDC begin "active direction" of the reconnaissance satellite program advanced by Feed Back. In the fall of 1953 Rand officials discussed satellite issues with a number of important government officials and military officers and, based on realistic near-term operational feasibility, recommended the Air Force issue a design contract within a year leading to full system development. By year's end ARDC had published a management "Satellite Component Study," and assigned it weapon system [WS] number 117L. Project Feed Back would place the satellite on the sure path of development.[44]

Authored by analysts James E. Lipp and Robert M. Salter, Jr., Rand's Feed Back report appeared in March 1954 in the midst of deliberations about the optimum ICBM organization.[45] It drew together findings from the previous two years' intense study of reconnaissance satellites. The "milestone" Feed Back study proposed an electro-optical reconnaissance satellite with a television-type imaging system projected to achieve a resolution of 144 feet from an altitude of 300 miles. The report readily admitted that this resolution could not deliver the accurate intelligence required and encouraged the Air Force to foster a competition among industrial firms to develop a higher resolution system based on long-focal-length, panoramic camera technology. It also discussed newly analyzed operational issues dealing with subsystems, cost projections, likely international political reactions, and a host of additional engineering requirements. With this "blueprint" in hand, Rand encouraged the Air Force to proceed on a full-scale basis with this "vital strategic interest" by implementing a seven-year development program budgeted between $165 and $330 million. In the next few years, while Air Force scientists and project officers worked to develop techniques for safe reentry of space payloads through the atmosphere, Rand engineers would stress two types of non-recoverable reconnaissance systems: one relied on television technology and "immediate" data transmission to ground stations; the other used tape storage of sensed data that would be transmitted at a later time.

After some initial hesitation, the Air Force agreed to pursue the Feed Back recommendations further, and in May 1954 directed Air Research and Development Command to review the military applications of the Rand satellite concept. Meanwhile, Rand and ARDC met with various Air Force, Defense Department, and industry leaders to "sell" the Rand proposal. At the same time ARDC proceeded with analyses of intelligence processing options, solar-electrical energy converters, auxiliary power sources, and guidance and control mechanisms. Following approval from the Defense Department's Coordinating Committee on Guided Missiles, the command issued a system requirement on 27 November 1954. With this decision, the Air Force in late 1954 clearly signaled its intention to develop an operational reconnaissance satellite system.[46]

The command followed up in March 1955 with a formal General Operational Requirement.[47] Now referring to the WS-117L reconnaissance satellite as the Advanced Reconnaissance System (ARS), the requirement prescribed continuous surveillance of "preselected" areas, especially aircraft runways and missile launching sites. In contrast to the Rand study's target resolution parameters, specifications now called for providing visual coverage of objects no larger than 20 feet on a side, and specified electronic and weather coverage capability, too. With an eye to continued technological advances, the scheduled operational date of 1965 seemed achievable. By August, ARDC had named as system project officer Colonel William G. King, Jr. In November he awarded $500,000 contracts to three firms—the Radio Corporation of America, Lockheed, and Glenn L. Martin—for a one-year satellite design competition under the code name "Pied Piper."[48]

Although by late 1955 space advocates might rejoice that at long last a military satellite program seemed underway, a number of long-standing, troublesome issues remained to be solved. One of the most important involved potential competition between satellites and missiles for scarce resources. Trevor Gardner resolved to insure that the Atlas ICBM schedule would not be compromised by satellite requirements. Back in November 1954, he had taken his worries to von Neumann's ICBM Scientific Advisory Group. The members asked General Schriever to assess the challenge of developing simultaneously satellites, Intermediate-Range Ballistic Missiles (IRBMs), and the "high priority" ICBM programs. Meanwhile, in January 1955 von Neumann's group, in an attempt to ease pressure on the ICBM program and accelerate satellite development, recommended that satellite work be confined to the spacecraft and its likely components rather than include booster elements, too. ARDC commander General Thomas S. Power agreed that satellite development not involve booster integration for the present. Nevertheless, the ICBM Scientific Advisory Group continued to worry about potential satellite competition with the ICBM schedule and addressed the issue again in June 1955. It cautioned that conflict could not be avoided because of satellite dependence on components developed through the ICBM program.[49]

General Schriever's analysis of the missile program convinced him that only centralized management of all military satellite and missile programs could minimize the problem of competition for scarce resources and avoid schedule delays. During his investigation, Schriever relied on the advice of Simon Ramo of Ramo-Wooldridge, the Western Development Division's technical consulting firm. Dr. Ramo met with von Neumann's Scientific Advisory Group and the Air Staff's Lieutenant General Donald Putt, Deputy Chief of Staff for Development to warn that the satellite program competed with the ICBM program for the same personnel and launch capabilities. Ramo strongly advised relocating management of the

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By the fall of 1955, with work on the satellite underway at the Western Development Division, General Power agreed to the management transfer, although not until February 1956 would the actual move begin from Wright-Patterson Air Force Base to the suburban Los Angeles facility. That the reassignment took nearly a year and a half to complete from the time Trevor Gardner raised the alarm suggests the reluctance of those concerned. General Schriever, in particular, would have preferred to focus on the ICBM program and not deal with IRBM and satellite competitors, while ARDC understandably preferred to keep the development program at its primary research facility in Ohio. In the long run, centralized management under the Western Development Division seemed the best alternative. At least Schriever's team could provide better management of risk and program scheduling with its "concurrency" approach to systems development and streamlined administrative procedures with higher headquarters. At the same time, satellite development could be expected to benefit from transfer out from under a research facility largely devoted to aeronautics to a "space"-oriented command located in the heart of the missile and satellite environment.

During the course of their deliberations on the ICBM program, Air Force planners and consultants had ample justification for concern over the attention their program would receive from the administration. Not only did they face the challenge of managing their burgeoning satellite and missile programs with limited resources, a new competitor for funds and development priority emerged in the summer of 1955. For over a year, the government had been considering sponsoring a scientific earth satellite to be launched during the International Geophysical Year (IGY), which was scheduled to extend from July 1957 to December 1958. Trevor Gardner and his fellow Air Force advisors kept a wary eye on these discussions of proceeding with an additional satellite program, which certainly contributed to their own concerns about satellite-missile relationships. In July 1955, once the administration formally agreed to sponsor a civilian satellite development program, this potentially high-profile competitor threatened to interfere with Air Force efforts to focus sufficient Defense Department attention and funding on both ICBMs and the Advanced Reconnaissance System.

## The Administration Commits to the First Civilian Satellite

The decision to support a civilian satellite program reflected a genuine interest in promoting science, strong advocacy from certain elements in the scientific community, and the administration's national security concerns—especially the challenge of eliminating the possibility of a surprise nuclear attack on the nation. For most of Eisenhower's advisors, the civilian scientific satellite never represented solely an altruistic, international scientific venture.

By early 1954 Eisenhower expressed grave concerns about inadequate intelligence to the National Security Council (NSC). The President also followed with great interest the work on the country's strategic missile program undertaken by Trevor Gardner and the civilian scientists serving on the Scientific Advisory Committee in the While House Office of Defense Mobilization. In late March he called to the White House a number of prominent scientists, including committee chairman Lee A. DuBridge, president of Cal Tech, and requested their help on the problem of surprise attack. They responded in August by establishing a Technological Capabilities Panel (TCP), chaired by James Killian, president of the Massachusetts Institute of Technology (MIT). After five months of deliberations, in February 1955 it issued a momentous report titled, "Meeting the Threat of Surprise Attack."[51]

The Killian Panel projected changes in the relative posture of American and Soviet strategic forces. Confirming the vital need for pre-hostilities strategic intelligence on Soviet military capabilities, the panel supported development of the Lockheed U-2 high-altitude reconnaissance plane, the solid-fueled Polaris sea-launched ballistic missile, and more rapid construction of the Distant Early Warning (DEW) line across northern Canada. The report advocated an accelerated ICBM program, and rapid development of IRBMs as a stopgap security measure until the ICBM force became operational. The President in September 1955 endorsed their findings together with those of von Neumann's Strategic Missiles Evaluation Committee and assigned to the Atlas and Thor and Jupiter programs the highest possible priority. To the initial consternation of Gardner and Schriever, in December President Eisenhower declared the IRBM programs to be coequal with the ICBM.[52]

As for satellites, the Killian report responded to the growing satellite interest in the scientific community and the panel's strategic intelligence concerns by recommending immediate development of a small scientific satellite that would establish the precedent of "freedom of space" for military satellites to follow. Although government officials and Rand analysts had worried about satellite overflight in international law earlier, here, for the first time, advocates identified the requirement for a "civilian" satellite to establish the overflight precedent. Focused on Project Aquatone, the U-2 project that promised immediate results, the military satellite program received little interest or support from Killian and his experts. At that time, he considered the Air Force's reconnaissance satellite a "peripheral project." This attitude from one so influential helps explain the less than enthusiastic administration support of the Air Force's Advanced Reconnaissance Satellite in the two years preceding Sputnik. Despite the growing need for strategic intelligence and awareness that the U-2 represented a temporary solution, Killian declined to actively support the military satellite until after the launch of the first Sputnik. He believed an American scientific satellite had to precede the launch of a military vehicle to provide the overflight precedent for military satellites to operate with minimum international criticism.[53]

That spring of 1955 Eisenhower and his advisors acted further on the Panel's satellite and overflight recommendations by outlining a policy for outer space analogous to that of the high seas, whereby flight in space would be available to all without legal restriction. At the same time, the President attempted to redefine the legal regime already established for airspace when, on 21 July 1955 at the Geneva summit conference, he called on the Soviet leaders to join him in providing "facilities for aerial photography to the other country" and mutually monitored reconnaissance overflights. Although the Soviets rejected his offer, he continued to advocate his "Open Skies" doctrine, while moving forward to assure the nation of sufficient intelligence to avert surprise attack. The emerging Eisenhower policy on space seemed to accord nicely with the scientists' proposal for launching an experimental scientific satellite during the International Geophysical Year.[54]

Interest in experimental satellite research originated from several sources. Although the satellite studies done by the Navy in 1945 and Rand in 1946 focused more on scientific than military characteristics, only in 1948 did the larger scientific community become aware of this research when portions of the Rand analyses appeared in the so-called "Grimminger Report" in the October issue of the *Journal of Applied Physics*. The report generated widespread interest among various small national rocket societies as well as upper atmosphere research scientists who increasingly worried about continuing their work once the wartime stock of captured V-2 rockets had been used. Another interested group involved space enthusiasts who found a wider audience at proceedings like the Second Congress of the International Astronautical Federation and the First Symposium on Space Flight held in the fall of 1951. By the early 1950s a number of activists offered specific satellite proposals, too. Dr. Fred Singer, University of Maryland physicist, and members of the British Interplanetary Society, for example, proposed the launching of a "Mouse" (Minimum Orbital Unmanned Satellite, Earth) which attracted attention on both sides of the Atlantic. More important for subsequent developments, Wernher von Braun, the chief of the Army's Guided Missile Development Division at the Redstone Arsenal in Huntsville, Alabama, had mounted a campaign in and outside military circles for an experimental satellite using the Army's Redstone rocket as a first-stage booster. He also offered visions of a manned future space station in a series of articles in *Collier's* magazine, which attracted considerable attention. Eventually, interest in von Braun's proposals led the military services to offer their own satellite projects for the International Geophysical Year.[55]

Growing support for launching a scientific satellite led a group of prominent scientists in 1954 to discuss the idea with leading government and congressional leaders. In August of that year, Congress authorized IGY participation by the United States and proposed $10 million to support the American satellite entry. In early 1955, the various scientific satellite proposals arrived at the office of the Assistant Secretary of Defense for Research and Development, Donald Quarles. These included a formal proposal from the United States National Committee for the IGY, appointed by the National Academy of Sciences, along with the Air Force's WS-117L program and the Army's Project Orbiter. Quarles referred all the IGY proposals to his Advisory Group on Special Capabilities for review and recommendations. In early May, the director of the U-2 project, Richard M. Bissell, Jr., met with the director of the Central Intelligence Agency, Alan Dulles, and the director of the National Science Foundation, Alan Waterman, to decide how the scientific satellite initiative could best meet the Killian Report's "freedom of space" objective. Acting on their advice, on 20 May Quarles submitted a draft space policy to the National Security Council for review. The decisions reached at the NSC's 26 May 1955 meeting, issued in the form of NSC Directive 5520, rank among the most important of the early Eisenhower presidency for space policy. Affirming Quarles' recommendations, the NSC declared that an IGY satellite must not interfere with the "high priority" ICBM and IRBM programs then underway, and that the satellite launched for "peaceful purposes" should help establish the "freedom of space" principle and the corresponding right of unimpeded overflight in outer space. The NSC also agreed that the scientific satellites would serve as precursors of later, military satellites. Finally, the NSC showed itself fully aware of the prestige and psychological benefits likely to accrue to the first nation to launch a satellite into orbit. As Nelson Rockefeller, Eisenhower's Special Assistant for Foreign Affairs, noted in a forceful appendix to the directive, "The stake of prestige that is involved makes this a race that we cannot afford to lose."[56]

During the post-Sputnik hysteria, in late 1957, the administration publicly attempted to distinguish between its so-called peaceful satellite project and that of the military-oriented Soviet counterpart by emphasizing the separation of the civilian scientific satellite project from the country's long-range missile program. Yet, the deliberations of the National Security Council clearly show that separation of the satellite and missile program hardly occurred as part of an internationalist, altruistic policy of promoting "pure science." The administration's declaratory policy of "peaceful purposes" purposely obscured its real intentions. When the President publicly announced on 29 July America's participation in the International Geophysical Year effort, he pledged that this scientific venture would remain unconnected to the current military missile development programs. The National Science Foundation would direct the project, with the National Committee for the IGY responsible for the satellite. The Defense Department would furnish the rocket booster and provide logistic and technical support. He gave no hint of the underlying purpose of his emerging space policy for the civilian and military satellite projects then underway. The civilian satellite would serve as a stalking horse to establish the precedent of "freedom of space" for the military satellite, but the administration maintained great secrecy on the latter so that attention would remain focused on the former.[57]

In late 1954, Congressional approval of funding for the IGY project had opened the way for von Braun and others to submit competing satellite proposals. The Defense Department favored a joint service-IGY effort to avoid interservice rivalry, but only the Army's Ordnance Department and the Office of Naval Research could agree to cooperate. Led by the Redstone team of Major General John B. Medaris and von Braun, Project Orbiter envisioned launching a small inert satellite "slug" using a Jupiter IRBM booster with three Loki upper-stage solid-fueled rockets. While the Army developed the booster , the Navy assumed responsibility for satellite, tracking facilities, and data analysis. The Project Orbiter team had vigorously lobbied the Defense Department for their project since early 1955. The Naval Research Laboratory, on the other hand, countered in late spring with Project Vanguard, which specified adapting a Viking sounding rocket as booster for three new upper stages. The Vanguard project included an impressive Minitrack radio-tracking and telemetry system, which would support the scientific focus of the IGY proposal.[58]

The Air Force initially had declined to participate in the IGY competition because it might conflict with its long-range goal of developing heavier, military reconnaissance satellites. However, after Quarles had directed all three services to offer proposals, the Air Force in July submitted its own "World Series" project—an Atlas C booster and a modified Aerobee-Hi space probe rocket. Faced with the dilemma of selecting from among the three rival entries, Quarles appointed an Advisory Group on Special Capabilities in May 1955 under the chairmanship of Homer J. Stewart of JPL. Following a contentious assessment process, the Stewart Committee ultimately selected the Navy's Vanguard proposal, and the Secretary of Defense confirmed this decision on 9 September 1955, just over a month after the White House publicly committed the nation to launching a satellite during the IGY. Although the Air Force entry showed great promise, committee members realized use of the Atlas as booster could conflict with the ICBM schedule. The Army cried foul, claiming that the Vanguard selection represented a major development effort, while von Braun asserted that his Redstone rocket team could launch an 18-pound payload as soon as January 1957, and well under the Vanguard's budget.[59]

Critics of the Vanguard decision argued that the Committee's concern that its choice not "materially delay other major Defense programs" tilted the balance against Project Orbiter.[60] Perhaps so, but the selection issue proved more complex. While the criterion served to rule out the Air Force Atlas ICBM booster, Orbiter's Redstone did not cause similar consternation. Chrysler Corporation was about to begin production of the missile, and the Huntsville group did not receive the Jupiter IRBM assignment until well after the end of the IGY satellite selection process. Von Braun's Redstone team was available. In fact, the non-interference criterion ranked only sixth among nine criteria used by the Stewart Committee. The Committee clearly questioned Orbiter's reliability and its limited potential for future scientific space exploration, and found attractive Vanguard's "maximum scientific utility" and superior tracking system and satellite instrumentation. Rather than merely a question of selecting a "non-military" Navy system over a "military" Army one, the choice reflected efforts to combine the best scientific applications with a launch system that could not avoid military connections in any case.[61]

By the fall of 1955 the administration was supporting two satellite programs, WS-117L and the civilian Project Vanguard. Nevertheless, the Air Force's experience following the decision suggests that the door remained open for a possible military-oriented alternative regardless of the desire to maintain a civilian focus. Problems experienced by Project Vanguard most likely account for the Defense Department's extended review of an alternative Air Force proposal for a scientific satellite.

## The Air Force Reconsiders a "Civilian" Satellite

For well over a year following the Vanguard award, the Air Force became involved with alternative "civilian" satellite proposals while challenged to develop an operational military satellite. The process reveals ambivalent attitudes about accepting a civilian project that threatened to compete for resources not only with the military satellite but the ICBM program as well. Throughout the course of events, Air Force planners seem to have operated without full knowledge of the ground rules, that had effectively eliminated a "military" project from the start, and the degree of seriousness the Defense Department attached to their proposals.[62]

From the start of their involvement in the IGY competition, Air Force planners worried that any Air Force scientific satellite that used an Atlas could interfere directly with the Atlas weapon ICBM schedule, while the competition's ground rules did not seem to exclude military criteria. Although ARDC might have thought the subject closed when the Stewart Committee selected Vanguard in August 1955, on 31 August Air Force headquarters directed ARDC to prepare another proposal that would integrate a scientific satellite with the WS-117L military reconnaissance satellite program. Then, on 14 October, with the new proposal still unfinished, ARDC halted that work, explaining it lacked sufficient funding and, in any case, the decision had been made in Vanguard's favor. In another reversal, the command resumed planning for a scientific satellite on 1 November, and the Western Development Division on 14 January 1956 submitted a scientific satellite variant of WS-117L, a 3500-pound satellite made from ARS components that could be launched by August 1958 atop an Atlas C at a cost of $95.5 million. General Schriever's proposal also specified a number of scientific experiments dealing with atmospheric density, solar radiation, and the upper atmosphere-near space effects on communications. The Schriever proposal reflected a consistent Air Force view that any satellite should serve a specific scientific or military purpose rather than merely serve as a public demonstration of the capability of launching a satellite into orbit.[63]

Most importantly, General Schriever advised that the scientific satellite could be developed without "significant compromise" to the military satellite program—

provided the operation be accorded sufficient funding, personnel, and resolve. He also established criteria to preclude interference with the ICBM program, but warned that any small delay in the Atlas schedule might mean a satellite launch beyond the IGY "window of opportunity." The general's caveats notwithstanding, ARDC forwarded the proposal to the Air Staff in January 1956. After a number of briefings on the subject, the proposal languished at Air Force headquarters in Washington throughout the remainder of 1956. Then, in early 1957, the Air Force notified General Schriever that the Defense Department had decided not to submit it to the Stewart Committee, which continued to oversee Project Vanguard. Evidently, by 1957 the Committee had decided to forego the luxury of a "back-up" satellite for Vanguard.[64]

General Schriever always maintained that his command could have handled development of both missiles and satellites. Yet the Western Development Division, which was redesignated the Air Force Ballistic Missile Division on 1 June 1957, would have required considerable additional resources from a parsimonious Defense Department to support three major long-range missile programs as well as two satellite projects. As for the civilian satellite planning effort during late 1955 and throughout 1956, Schriever and his staff affirmed that it did not significantly interfere with planning or funding for the military satellite. Once again, the so-called "non-military" criterion for an IGY satellite did not seem important enough to rule out lengthy consideration of the most "military" of satellite proposals. If concern for possible delays in the sensitive Atlas ICBM program again proved decisive, the story of the Air Force scientific satellite proposal also suggests that Air Force leaders felt compelled to remain involved in a potential program of questionable value. In view of the Air Force's aspirations to dominate the space mission, it could not remain uninvolved.[65]

## Retrenchment on the Eve of Sputnik

While some Air Force planners labored on proposals for a civilian variant of the WS-117L reconnaissance satellite, work continued on the technical requirements for the military satellite. Following the program's transfer to the Western Development Division in early 1956, General Schriever appointed as project officer Colonel Otto J. Glasser, who directed preparation of a full-scale system development plan based on the winning design entry submitted from the three "Pied Piper" firms. By April the plan had been completed and approved by General Schriever and ARDC commander General Power. In June 1956, the Air Force selected the design from Lockheed's Missile Systems Division and awarded the firm a formal contract in October. The Lockheed choice surprised no one because the company had hired the majority of existing space research specialists, including former Rand analyst James Lipp.[66] Relying on the Atlas C booster, Lockheed proposed building a huge second-stage booster satellite, initially termed Hustler, that could provide high pointing accuracy from its stabilized orbit position. Eventually, this booster satellite would become the workhorse "Agena" that, together with its Atlas booster, would launch the heavier Air Force payloads. Lockheed's winning payload entry also included a unique feature proposed by engineer Joseph J. Knopow for an infrared radiometer and telescope capable of detecting missiles and aircraft from their hot exhaust "signatures." Offering the potential for "real time" data, the infrared system element of the Advanced Reconnaissance System would emerge as a separate missile launch detection alarm system (MIDAS) satellite project designed to provide early warning of missile launches. The Air Force plan predicted an initial orbit date of May 1959, with the complete system, including ground installations, expected to be operational in the summer of 1963.[67]

The advance of technology in 1956 and 1957 served to emphasize the Lockheed proposal's merits. Research on wider and slower reentry vehicles with ablative surfaces offered a solution to the old problem of aerodynamic heating when objects reenter the atmosphere and fostered renewed interest in retrievable reconnaissance systems. Recoverable film containers held the prospect of avoiding image degradation that might occur through TV sensing and transmission through the atmosphere. At the same time, current experiments with panoramic cameras with long-focal-length lenses offered both broad-area coverage and high ground resolution.[68]

Research in new technology promised to be costly, and funding for WS-117L had been a sensitive subject from the start. From General Schriever's perspective, the Advanced Reconnaissance System suffered from guilt by association with the troubled Vanguard project. Although neither satellite program received adequate support, Vanguard's priority status brought it the lion's share of satellite monies. When the civilian satellite experienced management and budgeting problems, the military reconnaissance satellite also encountered difficulties receiving the attention and funding its supporters believed it deserved.

The scientists themselves were largely to blame for Vanguard's problems. Their interest in maximizing the scientific output not only led to additional costly instrumented experiments which Eisenhower criticized as "gold plating," but served to make secondary the essential requirement to establish basic vehicle technology before adding sophisticated payload experiments. The scientists also wanted to increase the number of test launches from six to twelve, which drove up costs and contributed to schedule delays that poor management practices only exacerbated.[69]

In fact, Vanguard had been underfunded from the beginning. Even before the Stewart Committee selected Vanguard, Secretary Quarles indicated his skepticism about the initial Vanguard budget figure of $10 million by raising the satellite budget to $20 million. Even so, the Vanguard budget rose continually from an initial $28.8 million in September 1955 to $110 million by May 1957. Had the Vanguard team payed attention to the early Rand studies, they might have developed a more realistic budget. Nine years earlier, James Lipp proposed a similar satellite at a cost of between $50 million and $150 million, depending on the payload. More attention paid to the Rand reports would also have alerted the scientists to the importance of international prestige associated with the country first into space. Instead, the scientists focused on costly experiments that played havoc with the development schedule. As for the experiments, Vanguard scientists did not clarify for President Eisenhower the important contributions their satellite work offered for ICBM development. Had the scientists done so, they might have been able to convince the administration to elevate satellite priorities in the name of missile progress. Ike, after all, listened to "his" scientists. A higher priority for Project Vanguard might correspondingly have benefited the military satellite and ICBM programs.[70]

Without strong intervention from the scientists, Project Vanguard's managers proved unable to stifle the concerns of an administration that was becoming increasingly exasperated with the spiraling costs. What made matters worse, the administration faced a larger problem brought on by the enormous costs and excessive duplication associated with building two ICBMs and three IRBMs simultaneously. With all five missiles in development in fiscal years 1957 and 1958, the budget for guided missiles reached more than $1.3 billion, an enormous increase from $515 million in fiscal year 1956, $161 million in fiscal year 1955, and only $14 million in fiscal year 1954. In 1957 Eisenhower feared that the spiraling missile costs would force defense spending beyond his fiscal year 1958 ceiling of $38 billion and directed a budget review of all programs. By August Secretary of Defense Wilson had cut the research and development budget by $170 million, reduced overtime work in the Atlas program, accorded Titan a lower priority than Atlas, cut spending on the Navy's Polaris project, and called a temporary halt to Jupiter and Thor production.[71]

The WS-117L satellite program did not prove immune to the budget slasher. Air Force satellite program officers had hoped to obtain $39.1 million of the estimated $114.7 research and development budget for use in fiscal year 1957. In August 1956, however, ARDC received only $3 million to launch the project. On 17 November 1956, General Putt briefed Donald Quarles on the newly approved WS-117L program. If he expected to obtain additional funding from the Secretary, he was disappointed. Secretary Quarles directed the Air Staff's research and development chief to ensure that the Air Force halt its military construction schedule and produce no fabrication mock-up or initial satellite without his express permission.[72] General Schriever felt compelled to vigorously lobby the Air Staff and the Defense Department for an additional $10 million:

> I can recall pounding the halls of the Pentagon in 1957, [he said later,] trying to get $10 million approved for our [USAF] space program. We finally got the $10 million, but it was spelled out that it would be just for component development. No system whatever.[73]

Even so, in July 1957 Secretary Quarles applied additional spending limits to the WS-117L as part of the Defense Department-wide budget slashing exercise that summer. This came after he had received intelligence information that spring predicting that the Soviets would be capable of launching a satellite before the end of the year. Quarles' actions should be viewed in terms of the administration's agenda for military satellites and space operations. The previous year, administration spokesmen had declared that no government officials were to speak publicly about spaceflight. General Schriever found to his dismay that the administration meant business after a February 1957 speech he gave in San Diego, California. Discussing the importance of studying potential military offensive functions in space, he declared the time ripe for the Air Force to "move forward rapidly into space." The following day Secretary of Defense Wilson instructed him to avoid the word "space" in all future speeches.[74]

The administration remained determined that the military satellite would under no circumstances precede the civilian satellite into space. It also opposed any discussion of military space operations that might generate a worldwide debate on the "freedom of space" for military spaceflight. This issue had to be avoided to maintain the declaratory policy of "peaceful purposes" as well as the action policy of having Vanguard provide the precedent for military space operations. As a result, before Sputnik the country supported two modestly funded space programs that did not interfere with ICBMs or other high-technology programs. Neither received the support its advocates sought. Yet neither permanently suffered from the 1957 budget cuts, which proved little more than an embarrassment after the launching of Sputnik on 4 October brought massive increases for satellite and missile programs. Although not of the administration's choosing, *Sputnik I* established the precedent, freedom of space, and underscored the administration's basic space policy.

## Retrospective From the Threshold of Space

On the eve of the Sputnik flights, the Air Force and the nation finally had reached the threshold of space—a full decade after the intrepid Rand analysts first offered the Air Force the prospect of launching an observation earth satellite in five years' time. During the course of the decade Rand produced increasingly convincing analyses of solutions to technical problems, potential military functions, and the important political benefits and prestige that would accrue to the United States. Armed with the Rand studies, Air Force satellite champions repeatedly worked to convince their leaders and the Defense Department and administration officials of the wisdom of their cause. What they confronted, however, proved to be a decade-long pattern of disinterest, inaction, and dogmatic unwillingness to accept change. As a result, the Soviets became the first to launch an orbiting satellite.

President Eisenhower has received considerable criticism for allowing the country to be humiliated and its national security endangered. Yet, if the Eisenhower administration failed to launch the first satellite, Sputnik nonetheless established emerging United States space policy. With unimpeded overflight assured, a clandestine military space program could proceed apace with less scrutiny by domestic and international critics. Civilian spacecraft had set the precedent for the military satellites to follow—even if the pathbreaker proved to be Sputnik and not Vanguard. The failure of the administration's actions lies not in overlooking the importance of prestige, but in assuming that Vanguard, with all its problems, still would be first. Determined that the civilian scientific satellite would take precedence, administration officials remained unwilling to provide the WS-117L program the commitment its supporters desired and expected. Yet the real delay in the reconnaissance satellite program occurred during the Truman years, when the Russians began a serious program and the United States did not. The Soviets had an eight-year start by 1954, when Project Feed Back set the satellite on the sure path to development.

During the Truman era, satellite proposals continually fell victim to the logic of "realism" and higher priorities. The administration argued that national security in the postwar world could be best achieved by strengthening the forces in being, namely strategic bombers and subsonic cruise missiles. Budget austerity after 1946 meant that research programs for forces of the future suffered most. Problematical programs like missiles and satellites faced the severest cuts. The administration's argument received strong support from the scientific community. Experts like Dr. Vannevar Bush dismissed missiles as "fanciful" because they would require a decade of incredible expense to overcome technical problems of guidance, propulsion, and reentry. Bush was far from alone in his pessimism. After all, it would have taken a particularly insightful individual to foresee the incredible advances over the course of the pre-Sputnik decade in chemical fuels, rocket engine combustion technology, instrumentation, and missile frame construction.[75] Postwar America offered too few men of vision like Hap Arnold and Theodore von Kármán in positions of authority. And for all his achievements, von Kármán led the Air Force down the lesser, aerodynamic path of development.

Von Kármán's tenure as chief of the postwar Scientific Advisory Board suggests that Arnold, had he continued to lead the postwar Air Force, would have achieved only modest success against the forces of institutional inertia and intransigence. The newly independent Air Force benefited most from the Truman defense strategy. While General LeMay and others might admit that intercontinental ballistic missiles represented the strategic force of the future, the logic of the present seemed to favor the forces that could best ensure the survival of an independent Air Force and the security of the nation. When those forces happened to be manned aircraft and missiles supporting those aircraft, long-range guided missiles understandably became relegated to the distant future. Satellites suffered a similar fate. Satellites and missiles represented "new" and potentially "revolutionary" change for a service that traditionally viewed itself in terms of the airplane. Even in the early 1950s, when it became clear that technology had made ballistic missiles more feasible and Soviet actions precipitated a major budget increase for research and development, Air Force decision-makers persisted in resisting the acceleration of satellite and missile projects.

On the other hand, the Air Force remained ever vigilant in protecting its authority over satellite and missile development. If it neglected its space programs, it nevertheless kept a wary eye on Army and Naval efforts to weaken the Air Force's claim to exclusive rights to these programs. The fierce contest for control of roles and missions proved to be a running theme throughout the pre-Sputnik decade, and clearly prevented faster progress. While the service squabble centered largely on missiles, on the eve of Sputnik the Air Force faced a competitor for its embryonic satellite program in the guise of Project Vanguard. Although Eisenhower's dual space program remained unclear to many in the mid-1950s, the "civilian" Vanguard satellite represented a future challenge for the Air Force in terms of civil-military roles and missions.

On the eve of Sputnik, the observer of early space events might be tempted to view the previous decade pessimistically as one of frustration and delay. The Air Force experience, however, also suggests a much more positive assessment. Many central characteristics of the future Air Force space mission emerged during the "dawn of the space age." For one, the Air Force made a strong bid for the preeminent military role in space matters, and by 1957 had carved out a relatively strong position with its Atlas and military satellite program. The emphasis on demonstrating military satellite utility served to intensify efforts to define and justify satellite operations in terms of providing better data more effectively than competing systems.

As for research and development, the Arnold and von Kármán legacy appeared far more secure after the downswing in the late 1940s. Reorganization provided research a greater focus, while General Schriever's command arrangements demonstrated impressive flexibility and effective improvisation. To be sure, it took activists like Trevor Gardner and his band of reformers working "outside" the system to facilitate change. Yet, in a sense, the Air Force established a tradition of going outside—to industry, scientists, research laboratories—that von Kármán's *Toward New Horizons* recognized and supported.

Considering both the failures and the achievements, the pre-Sputnik decade is best viewed as the conceptual period of the "New Ocean," during which the new ideas of space had to be tested and inertia and opposition overcome. After all, only a generation separated the upper-atmosphere explorers of the NACA and the early rocketeers from the Atlas and WS-117L teams on the eve of Sputnik. Few are blessed with the vision of Arnold and his disciples or the perceptiveness of the Rand analysts of 1946, who consulted their crystal ball and guessed correctly. Even by late 1957 the path ahead for most seemed unclear. With the nation on the threshold of space, the challenges for the Air Force remained formidable.

# CHAPTER 2

# From Eisenhower to Kennedy:

## The National Space Program and the Air Force's Quest for the Military Space Mission, 1958–1961

The period from late 1957 to the spring of 1961 represents the watershed years of the national space program and the Air Force's place within it. In the wake of the Sputnik crisis, the Eisenhower administration implemented organizational and policy measures that provided the foundation of the nation's space program. Buffeted by pressure and counsel from an alarmed public and congressional and military spokesmen, President Eisenhower found himself fighting a rearguard action to hold to his view of civilian, military, and budget priorities for space activities. His dual military and civilian space program reflected his "space for peace" focus, one that fostered "open skies" for the free passage of future military reconnaissance satellites. Given the sensitivity of overflying the Soviet Union, during the formative years of his administration the civilian space program held center stage, while administration officials consciously downplayed the military space role and service initiatives.

Space advocates in all three military services and their supporters chafed at the government's refusal to sanction a broadly-based military space initiative in response to the Soviet menace. With visions of leading the nation into the space era, Air Force leaders found the situation especially frustrating. Relying on its "aerospace" rationale, they initially argued that the Air Force represented the logical service to head a unified, Defense Department-oriented national space program that would serve both military and civilian requirements. When it became clear that national policy preferred two programs, one a civilian-led effort dependent on military support, the Air Force sought to become the "executive agent" for military space.

The challenge proved formidable. Shortly after Sputnik, concerns with interservice rivalry and duplication, among other reasons, compelled administration officials to create the Advanced Research Projects Agency (ARPA) for all Defense Department space research and development activities. Although the services retained their missile programs, they temporarily lost their independent space programs to the new agency. Moreover, the creation of the National Aeronautics and Space Administration (NASA) in the fall of 1958 further divided the space mission and raised thorny issues of civil-military authority that persisted well beyond Sputnik. Despite repeated government statements to the contrary, for many, a civilian NASA conducted "peaceful" space ventures, while the Defense Department and the military services, by implication, engaged in warlike or non-peaceful space activities. Air Force leaders found "space for peaceful purposes" an albatross that prevented them from pursuing a space program they believed necessary to provide the nation with the security it required. The latter involved not only recognized defense support functions such as satellite communications, reconnaissance, and navigation activities, but potentially offensive functions in space through space-borne antisatellite and antimissile defense measures. The Eisenhower administration believed otherwise, and permitted nothing more than studies of weapons in space.

Constrained by administration policy and the prerogatives of NASA and ARPA, and without a space "mission" to call its own, the Air Force also faced stiff competition from its service counterparts. Indeed, by early 1958, the Army and Navy had far more experience in space than the Air Force. Their success in orbiting the nation's first satellites (Explorer and Vanguard) seemed destined to propel one of them to victory in the quest for future space missions. Yet, by the spring of 1961, NASA had its sights on manned flight to the moon, ARPA had been relegated to obscurity, and the Army and Navy had been removed from any major role in space. The Air Force found itself effectively designated the executive agent for all military space development programs and projects. If Air Force leaders considered the victory incomplete, it nonetheless represented an impressive achievement that established the Air Force as the nation's primary military service for space.

## Sputnik Creates a "National Crisis"

The administration's efforts prior to the Sputnik launches to downplay the American space program through a deliberately-paced civil and military research and development effort came to an abrupt halt following the electrifying news on the morning of 4 October 1957 that the Soviets had launched a 184-pound instrumented satellite into orbit atop a rocket booster weighing nearly 4 tons. By contrast, America's yet-to-be launched Vanguard weighed only 3.5 pounds.[1] *Sputnik I* dramatically demonstrated that the Soviets possessed both a highly advanced satellite program and booster technology sufficient to field an intercontinental ballistic missile force. For the first time, America seemed at risk of an intercontinental attack. Despite warnings of the psychological shock value of satellites repeated through various Rand studies and affirmed by the National Security Council a few years earlier, the administration found itself unprepared for Sputnik's "Pearl Harbor" effect on public opinion.[2]

President Eisenhower sought to reassure the American public and quell the press furor at home and abroad in his first news conference held five days after the Russian launch. On 9 October he downplayed the impact of Sputnik by declaring that, "so far as the satellite itself is concerned, that does not raise my apprehensions, not one iota." People had no reason to panic, and he would not involve the country in a needless space race or accelerate the launch schedule of the civilian Vanguard satellite. But neither the President's soothing words nor unfortunate public comments belittling the importance of the Russian effort by high-ranking administration officials proved able to silence a growing national debate over space and defense policies. They had a national crisis on their hands.[3]

At the same time, Eisenhower and his staff quickly perceived one important benefit from the Sputnik launch. Meeting with the President the day before his post-Sputnik press conference, Deputy Secretary of Defense Donald Quarles observed that "the Russians might have done us a good turn, unintentionally, in establishing the concept of freedom of international space." Eisenhower then requested that his advisors look five years into the future and provide an update of the Air Force's effort to develop a reconnaissance satellite. Clearly, the President intended to continue his public focus on civilian spaceflight and unrestricted satellite overflight to protect the viability of future military satellite operations.[4]

Throughout October administration officials reevaluated the entire missile program and discussed various courses of action. Then, nearly a month later, on 3 November, the Soviets successfully launched the 1,120-pound *Sputnik II* with its passenger, the dog "Laika." Although once again officials tried to calm troubled waters by claiming that the Soviet feat came as "no surprise to the President," the administration rapidly moved to gain control of the debate and reestablish confidence and prestige. On 7 November the President took one of his most important steps, appointing Dr. James R. Killian, his close confidant and chairman of the earlier Killian Committee, as Special Assistant to the President for Science and Technology and Chairman of the President's Science Advisory Committee. He immediately became the administration's "point man" for planning future space organization and policy.[5]

The day after Killian's appointment, the Defense Department authorized the Army Ballistic Missile Agency (ABMA) to proceed with preparations to launch its scientific Explorer satellite during the IGY under Project Orbiter as backup to Project Vanguard. Conveniently, incoming Secretary of Defense Neil McElroy had been visiting the Hunstville, Alabama, complex when the Soviets launched *Sputnik I*. Project director Brigadier General John B. Medaris and Wernher von Braun seized the opportunity to promise a successful Jupiter launch within ninety days. When they received official approval on 8 November, Medaris' team had been hard at work on the Orbiter booster since 5 October. Their hard work would pay off on 31 January 1958, when *Explorer 1* became the first U.S. satellite to achieve successful orbit. Although its miniaturized electronics relayed important scientific data, including discovery of the Van Allen radiation belts surrounding the earth, its 10-pound payload seemed less impressive to the American public than the far larger and heavier, if less scientifically valuable, Sputniks.[6]

Secretary of Defense McElroy followed the Project Orbiter decision by announcing on 20 November his intention to create a new defense agency to control and direct "all our effort in the satellite and space research field." Representing the first step in reorganizing the government for space, Secretary McElroy planned to establish the Advanced Research Projects Agency (ARPA) in early February 1958 at a level above the three military services.[7]

## The Air Force Seizes the Initiative

Meanwhile, the Air Force had been far from idle in the aftermath of *Sputnik I*. While Army and Navy teams continued preparations for Projects Orbiter and Vanguard, respectively, Air Force leaders in late 1957 initiated their own sweeping assessment of the nation's space activities and prospects. They hoped to develop a program of action with the Air Force playing the central role. The wide-ranging post-Sputnik debate on the national space course ahead seemed to present Air Force leaders with a golden opportunity to claim for their service the nation's space mission.

On 21 October 1957, Secretary of the Air Force James H. Douglas convened a committee of distinguished scientists and senior Air Force officers chaired by Dr. Edward Teller to evaluate the nation's missile and space programs. Completed in just two days, the Teller Report chastised the government for administrative and management practices that, it said, prevented either civilian or armed services agencies from achieving a stable and imaginative research and development program. It recommended a unified, closely integrated national space program—under Air Force leadership. A centralized program, the committee argued, would provide focus for an expanded national space program and avoid the divided effort likely to result from a fragmented program. Although the report received attention at high levels of the government, in the unsettled post-Sputnik period it failed to convince government officials to adopt a unified program either under military or civilian direction. Ultimately, the President would commit the nation to a dual program with separate military and civilian elements.[8]

On 7 November 1957, the Air Force's legislative liaison team, alarmed by what seemed to be a preference among congressmen for the Army's space initiatives, described the challenge confronting the Air Force. To avoid defeat in the race for the space mission, the Air Force must base its legitimate case on the position staked out in 1948 by General Vandenberg, that flight in the upper atmosphere and space represent logical extensions of the traditional Air Force realm of operations and the natural evolution of its responsibilities. The officers urged Air Force spokesmen to "emphasize and re-emphasize the logic of this evolution until no doubt exists in the minds of Congress or the public that the Air Force mission lies in space as the mission of the Army is on the ground and the mission of the Navy is on the seas."[9]

On 29 November 1957, Chief of Staff General Thomas D. White made this theme the focus of an important address to the National Press Club. As airpower had provided the means to control operations on land and sea, so in future "whoever has the capability to control space will likewise possess the capability to exert control of the surface of the earth." For the Air Force, he said, "I want to stress that there is no division, per se, between air and space. Air and space are an indivisible field of operations." By implication, an Air Force role in space must embrace offensive operations to provide proper national security. Publicly, Air Force leaders would seldom admit that the atmosphere and space represented fundamentally different mediums. In his talk, General White also addressed another basic institutional theme, affirming the service's traditional research and development focus. The Air Force still depended, he said, on the "skills, talent, ingenuity and cooperativeness of...science and industry to provide us the technological lead we need in the future." This future would be in space.[10]

In public addresses and Congressional testimony, General White and other Air Force spokesmen, including Under Secretary of the Air Force Malcolm A. MacIntyre, Lieutenant General Donald Putt, Deputy Chief of Staff for Development (DCS/D), and Major General Bernard A. Schriever, commander of the Air Force Ballistic Missile Division (AFBMD), would focus on the concept of space as a continuum of the atmosphere, a place for potential military-related operations rather than a function or mission in itself, and the logical arena for Air Force activities. Early in the new year, Air Force leaders coined a new term, "aerospace," to describe their service's legitimate role in space, and the following year "aerospace" officially entered the Air Force lexicon when it appeared in the revised Air Force Manual 1-2, *United States Air Force Basic Doctrine*, issued on 1 December 1959. According to the manual,

> aerospace is an operationally indivisible medium consisting of the total expanse beyond the earth's surface. The forces of the Air Force comprise a family of operating systems—air systems, ballistic missiles, and space vehicle systems. These are the fundamental aerospace forces of the nation.[11]

Along with policy and planning issues, the Air Force also addressed internal organizational concerns for space. To provide better focus for future Air Force space activities, on 10 December General Putt revealed the formation within his office of a Directorate of Astronautics, headed by Brigadier General Homer A. Boushey, whose long career in the "space" field included early rocket-assisted flight experiments with the von Kármán team during World War II. However, having created the new office without consulting Defense Department officials, General Putt and other Air Force leaders were chagrined by the vehement opposition from senior defense officials like William Holaday, newly-appointed Defense Director of Guided Missiles, who accused the Air Force of wanting "to grab the limelight and establish a position." This, of course, is precisely what the Air Force intended to do. When Defense Secretary McElroy objected to the term "astronautics" and criticized the Air Force for seeking public support, Air Force leaders realized they had overstepped military boundaries. The firestorm of protest convinced General Putt to rescind his memorandum three days later. Although Air Staff leaders remained committed to strong centralized headquarters direction of space projects, they continued to face roadblocks from administration officials.[12]

Unable to establish the Air Staff directorate in late 1957, the service's space supporters during the first six months of the new year followed the temporary expedient of coordinating USAF space activities through the Assistant Chief of Staff for Guided Missiles. Only in late July 1958, after the proposed civilian space agency received congressional approval and the National Security Council revised space policy, could the Air Force create a central Air Staff office for space. Even then, the term "astronautics" could not be used, and General Boushey's new office under the DCS/Development became the Directorate of Advanced Technology. Sharing space responsibilities with the Assistant Chief of Staff for Guided Missiles, General Boushey would have to wait another year before his office could be upgraded to assume direction of all headquarters space activities.[13]

In retrospect, given the administration's emphasis on strategic reconnaissance, of which he was well aware, General Putt should have been sensitive to any suggestion of an expanded military role in space. Four days after Sputnik, he and Vice Chief of Staff General Curtis E. LeMay met with Deputy Secretary of Defense Quarles to apprise him of the state of the military reconnaissance program and potential for satellite offensive operations. Quarles readily supported the Advanced Reconnaissance Program, which would become the government's most important space project. Yet, when the two officers advocated an offensive space role to forestall potential Soviet satellite weapon carriers, Quarles in no uncertain terms directed the Air Force not to consider satellites as weapon platforms and to entirely eliminate satellite offensive applications from future Air Force space planning. Air Force leaders would continue to find that the policy of "peaceful uses of outer space" embraced the development of reconnaissance systems but never offensive weapon systems. Weapons in space threatened the reconnaissance assets judged vital to national security.[14]

By the end of the year, the Air Force's initial foray into the space contest had produced mixed results. Its leaders had established the service's policy position for a legitimate space role, yet the lack of a Defense Department response to an Air Force-led space plan for the nation and Air Staff's rebuff suggested the need for a more cautious strategy to achieve Air Force space objectives. In future efforts, the Air Force would develop policy, planning, and organizational proposals as part of a well-organized quest for the military space mission.

## The Government Organizes for Space

Beginning in early 1958, the administration took action to create a national space program. Its focus centered first on organizational measures, then embraced policy issues. By late summer, the National Space Act confirmed a dual civilian-military program designed to pursue a policy of space for peaceful development and exploration. Along the way, the administration and Congress considered various options in their attempt to create the optimum civilian-military balance. Although their decisions would prove enduring, they left unclear the precise relationship between military and civilian space responsibilities.

During the first week of the new year, the Defense Department requested a list of proposed space projects from each of the three services. Air Force leaders viewed this request as an open door for approval of a USAF space program. It had devoted considerable thought to the future space needs of the country ever since the first Sputnik flight and the Teller Committee's deliberations. In early December 1957, the Scientific Advisory Board reported on the subject of space technology. Pointing out that Sputnik and Soviet ICBM capability had produced "a national emergency," the board focused on the rocket field as the area which provided the Air Force the best means of contributing to "a proper national response." Its six-point program also included an accelerated reconnaissance satellite effort and a "vigorous" space initiative with an "immediate goal of landings on the moon." Both military manned spaceflight and the WS-117L Advanced Reconnaissance System would remain centerpieces of future Air Force space proposals, while Air Force leaders would quickly realize that the relationship between missiles and space systems would prove the most effective key to achieving Air Force preeminence in military space.[15]

The result of the Air Force's post-Sputnik deliberations appeared on 24 January 1958, when the Air Staff submitted to the Director of Guided Missiles its "Air Force Astronautics Development Program." It comprised five major space systems: Ballistic Test and Related Systems, a lunar military base system, manned hypersonic (Mach 5 and above) research, the Dyna-Soar orbital glider, and the WS-117L Satellite System. Planners further divided the five proposals into twenty-one major projects that embraced a variety of military missions deemed "essential to the maintenance of our national position and prestige." The planners urged that special emphasis be accorded getting man into space at the earliest time.* Testifying before Congress in early January, Major General Bernard A. Schriever, Commander of the Air Force Ballistic Missile Division, emphasized that the Air Force possessed the means of developing an astronautics program with no detriment to ballistic missile programs. Much to its disappointment, the Air Staff received no reply from Mr. Holaday's office, and Air Force efforts to lead a national space effort proved fruitless. Statements by General Putt and his deputy, General Boushey, advocating missile-firing bases on the moon and eventually militarizing the planets alarmed rather than reassured their audience of civilian leaders in Congress and the Defense Department. By late February, the Air Force initiative had been "overtaken by events," and the Assistant Secretary of Defense assumed responsibility for coordinating military inputs for a national policy on outer space. When the Secretary of Defense created the Advanced Research Projects Agency on 7 February, frustrated Air Force officials realized that the Defense Department's request to the services represented little more than an effort to gain information that would assist the new Defense Department agency in assigning space development responsibilities among the Army, Navy, and Air Force.[16]

The comments by Generals Putt and Boushey reflected the uncertainty of the period and the great unknowns of space in the aftermath of Sputnik. After the demise of the Air Force initiative, Air Force leaders responded to the Defense Department's attempt to coordinate a policy input for the administration by calling for more basic knowledge to determine the potential and limitations of manned and unmanned spaceflight before formulating a national policy covering all available and contemplated space programs. Air Force thinking in the months ahead would be characterized by an emphasis on a "building blocks" approach to space development rather than on advancing fanciful ideas for military bases on the moon and planets from which to attack countries on Earth.[17]

## ARPA Takes Control

ARPA began operations amid a flurry of great expectations from its admirers and dire warnings from its detractors. Secretary of Defense McElroy declared that the new agency would provide a "single control...of our most advanced development projects," while the services would continue with research and development of weapon systems that clearly fell within the "missions of any one of the military departments." ARPA, in fact, gained control over all U.S. space projects, military and civilian, until the National Aeronautics and Space Administration (NASA) commenced operations in the fall of 1958. For another year thereafter, the Defense Department agency retained control, including funding, of all military space

* See Appendix 2-1.

projects. The initial delineation of responsibilities between ARPA and the services proved difficult to maintain. Yet ARPA fulfilled two important administration objectives. For one, it ended the low military priorities heretofore accorded space technology in the absence of clearly defined military applications. For another, it offered the laudable prospect of avoiding interservice rivalry and wasteful duplication by transferring service decision-making power on space projects to the Defense Department agency.[18]

The congressional committees charged with military oversight viewed with suspicion any increase in the powers of the Secretary of Defense at the expense of the military services. In early January 1958, General Schriever and other military spokesmen testified against the creation of any agency with authorization to go beyond policy formulation and program approval to perform development and contractual responsibilities. Research and development, they argued, should be left to the services. Secretary McElroy promised Congress that ARPA's initiative would "be developed in coordination with the military departments to the point of operational use, so that...[weapon systems]...may be phased into the operation of one or more of the military services with a minimum loss of time or interruption of development and production."

The Air Force was not entirely reassured. Roy Johnson, ARPA's aggressive director, seemed too independent of service wishes and possessed too much authority over service space programs. Moreover, the President made ARPA responsible for civilian space projects as well until the proposed civilian space agency became operational. Nevertheless, until ARPA assumed control of most Air Force programs in late June, Air Staff planners, perhaps guilty of wishful thinking, continued to advocate an independent Air Force space program. As the historian of the Air Research and Development Command pointed out, the "classic and foreboding example of things to come...proved to be the reconnaissance satellite program, perhaps the most important single Air Force space program to light upon ARPA." Initially the Air Force applauded ARPA's focus on accelerating the WS-117L program on "the highest national priority basis." In response to Sputnik, by September 1958 ARPA had reprogrammed the Advanced Reconnaissance System into separate component projects with revised designations. The reconnaissance element received the name, Sentry, while MIDAS referred to the infrared sensing system. Under the designation "Discoverer," a cover for the covert CORONA project, ARPA grouped "vehicle tests, biomedical flights, and recovery experiments." In the fall of 1958, ARPA assigned all three projects to different Air Force organizations.

Operating on a project basis, ARPA direction signaled the end of "concurrency," the centralized systems management practice that had proven so successful in the crash ICBM program. In October 1958 ARPA also terminated the Weapon System (WS) designation altogether, declaring that the "system approach employed by the Air Force would be altered in such a way that all other items of the former 117L system would be budgeted as subsystems or components...for reasons of budget justification and program management." Omitting the weapon system designation contributed to the administration's low-profile approach to military space activities.

The other Air Force space programs received similar treatment from ARPA following their transfer in late spring.* The Defense Department agency organized its newly-acquired space activities into four broad programs: Military Reconnaissance Satellites, Missile Defense Against ICBM, Advanced Research for Scientific Purposes, and Developments for Application to Space Technology. Although ARPA redistributed most programs back to the Air Force and the other services, it did so under contract, thereby retaining technical and fiscal control and receiving credit for "its" programs. The Air Staff might set requirements, but ARPA made the decisions, directed the efforts, and dealt directly with other agencies and with private industry.

Air Force leaders also found ARPA's operating procedures highly unsettling. In late March Johnson informed the service secretaries that he intended to "cut red tape" and deal directly with subordinate agencies like the ARDC, AFBMD and other space and missile centers, bypassing established chains of command. At the Air Research and Development Command, for example, ARPA personnel frequently approached individuals and offices directly, which led ARDC commander Lieutenant General Samuel E. Anderson to establish a "focal point" to coordinate ARPA-ARDC activities. Even so, the "focal point" officer and his small staff faced considerable opposition from within the command and criticism from General Boushey's Directorate of Advanced Technology before they succeeded in keeping all parties informed on a consistent basis.

Yet, if the novel Defense Department agency acted high-handedly and pursued management practices that alarmed the services, the intrusion of ARPA could have been far more disruptive. In fact, dire warnings that ARPA might evolve into a "fourth service" proved false. Roy Johnson, much to the dismay of his staff, proved unwilling either to create and operate his own facilities and laboratories or to establish an in-house contracting capability with the armed services functioning as ARPA's contracting agents. In fact, for its expanded space program, ARPA remained dependent on the services for qualified personnel, necessary experience, and resources that included laboratories, launch complexes, rocket boosters, test facilities, and tracking networks. As a result, ARPA designated the military services its executive agents for most projects, with the Air Force receiving the lion's share of eighty percent. Along with the former Advanced Reconnaissance System, these represented the Air Force's most cherished space programs, including lunar probes, the 1.5 million-pound rocket booster, and a variety of measures designed to launch a

* See Appendix 2-2.

military man in space. ARPA, in fact, consistently supported the need for a military manned space mission, and already in late February 1958 had awarded the Air Force development responsibility for military manned spaceflight. Although the Air Force remained unhappy with its subordination to ARPA on space matters, Air Force leaders quickly realized that cooperation with ARPA would prove the best means of gaining development responsibility for space projects and, later, operational responsibility as well.

ARPA's rise to prominence reflected the country's alarm following Sputnik and the need to act rapidly to counter the Soviet advantage. As a result, ARPA became a prime mover for a variety of space projects, some of which, such as the lunar probe program, had no direct military requirement. Specifically authorized by the President, this effort would use available military resources, most notably the Army's Jupiter and the Air Force's Thor IRBMs as boosters. In short, ARPA served as the national space agency through much of 1958. Yet it remained clear from the spring of 1958, when the President submitted his proposal for a National Aeronautics and Space Administration (NASA), that the new civilian space agency would directly challenge ARPA's broad jurisdiction in the space arena and become an additional competitor for traditional Air Force space interests.

## NASA Joins the Competition

Like ARPA, NASA represented an intervening space agency that challenged the Air Force for space responsibilities and program funding. NASA's civilian focus also raised the contentious issue of civilian-military space relationships. Despite the apparent logic in assuming that NASA would be responsible for civilian space activities and the Defense Department would handle military interests, the demarcation line between civilian and military space concerns often proved artificial and unattainable. On the other hand, if the Air Force found NASA an unwanted competitor for the space mission, it quickly perceived the benefits to be gained by cooperating with the civilian agency. For the foreseeable future, NASA would depend heavily on Air Force assistance, while its absorption of Army and Navy space assets would help propel the Air Force toward the military space mission.[19]

The "Sputnik crisis" produced demands by congressmen, scientists, and other civilian leaders for a more sweeping national organizational space effort than ARPA seemed to promise. The hearings begun in late November 1957 by Senate Majority Leader Lyndon Johnson's Preparedness Investigating Subcommittee of the Senate Committee on Armed Services focused on long-term space research and development requirements "from a broad national point of view." This could best be done, the committee's final report suggested, by either improved control and administration within the Defense Department or the establishment of an independent agency.

An independent space agency for long-term research and development outside the Defense Department gained increasing support in early 1958 from scientists concerned that centering space research in the Defense Department would likely alter and reduce the scale of scientific programs. While various individuals and groups proposed organizational alternatives, the National Advisory Committee for Aeronautics (NACA), which had considerably expanded its missile research under Chairman Jimmy H. Doolittle and Director Hugh L. Dryden, took an increasingly active role in the space debate.[20] In late 1957 it convened a special committee on space technology under MIT's D. G. Stever to examine space-age research and development requirements and determine the best role for the NACA to play. On 14 January 1958 the committee's report proposed an interagency cooperative space program that would involve the NACA, the Defense Department and the military services, the National Science Foundation, and the National Academy of Sciences. But just two days later the NACA's main committee passed a strong resolution on spaceflight proposing that fundamental scientific research in the upper atmosphere and space be conducted by the NACA rather than the military.[21]

Meanwhile, in early February 1958, the congressional leadership called for the formation of an independent civilian space agency, and, to address the "national crisis," Congress created two important committees: a Senate Special Committee on Space and Astronautics under Majority Leader Lyndon B. Johnson, and a House Select Committee on Astronautics and Space Exploration chaired by Speaker John W. McCormack. Yet, congressional hearings on the space agency itself began only after the administration submitted its own bill on 2 April 1958. The administration's delay in submitting its proposal is explained by the last ditch disarmament discussions Eisenhower carried out in January and the deliberations over the place of the military in the space program.[22]

In early February, the President charged his science advisor James Killian to proceed with specific recommendations for government organization for space activities. Recalling this early formative period, Killian admitted that he took on the assignment with a clear idea about what should be done.

> From the beginning, it has been my view that the Federal Government had...only two acceptable alternatives in creating its organization for space research, development, and operation. One was to concentrate the entire responsibility, military and nonmilitary, in a single civilian agency. The other was to have dual programs.... A possible third alternative, that of putting our entire space program under the management of the Defense Department always seemed to me to have so many defects as to be practically excluded as a solution.[23]

Because of his concerns for national security, in which strategic reconnaissance loomed large, Eisenhower did not share Killian's views. In fact, shortly after the new year, he thought simply of having the military direct the entire space research and development effort under ARPA's direction. He soon abandoned this idea because of congressional and scientific opposition, and because of the arguments of Killian.[24]

Nevertheless, the President always opposed creating an entirely civilian national space program or of diluting the Defense Department's overall responsibility for space research and applications. During the drafting of the bill, the administration's dilemma involved how much and what kind of military participation to authorize rather than choosing between military and civilian alternatives.

Once the administration accepted a civilian agency based on the NACA, it solicited comments from the Defense Department. Initially, defense officials thought little would change because of the traditionally cooperative military arrangement with the NACA. Commenting on the draft bill prior to its submission to Congress, Deputy Secretary of Defense Quarles reminded budget chief Maurice H. Stans that "it is assumed the operation of the new agency would bear the same relationship to the Defense Department in the field of space and aeronautics as the NACA now does in the aeronautics field." As it was, Quarles objected to a number of passages in the legislation, including one that he perceived as preventing the services from carrying out basic scientific research that had military mission applications. This issue would continue to cause tension long after passage of the Space Act.[25]

The administration's bill, drafted by the NACA general counsel Paul Dembling and sent to Congress on 2 April 1958, proposed that the nation's aeronautical and space science activities be directed by a civilian agency "except insofar as such activities may be peculiar to or primarily associated with weapons systems or military operations, in which case the agency may act in cooperation with, or on behalf of, the Department of Defense." Referred to as the "exception clause," this passage suggested a variety of interpretations. Would the new agency be the prime mover in government space activities, with the military playing a minor role? Did acting on behalf of the Defense Department mean that NASA would undertake military projects? Above all, as Donald Quarles suggested, did the narrowly constructed military mission preclude the Defense Department from performing basic space research closely related to defense missions?[26]

In congressional hearings, witnesses and committee members attempted to determine precise organizational relationships and functions. Defense Department witnesses strongly objected to the "exception clause." ARPA director Roy Johnson also criticized any implication that the law would give NASA veto power over military activities and restrict the Defense Department to operating space systems. His chief scientist, Herbert F. York, agreed and presented the Air Force's argument that space is not a program to be administered by a single civilian agency, but a place of civilian and military applications. From his reading of the bill, it seemed that the Bureau of the Budget and NASA would be responsible for programs either entirely civilian or jointly civilian, leaving the military with only the narrowly defined military agenda. The problem, declared military officials, centered on space requirements that could not be precisely known in advance, but often required identifying and refining during the course of development or research. Therefore, the Defense

Department must be permitted to conduct fundamental exploration of space technology in order to determine if particular defense tasks could be done more effectively in space. The administration's bill pointedly did not provide a clear, fixed division of labor between the military and the new civilian agency. But as an early House of Representatives staff paper concluded, "practically every peaceful use of outer space appears to have a military application."[27] In the bill's final language, Congress approved giving the Defense Department and NASA wide-ranging prerogatives in the space field, yet agreed that the Defense Department had authority both to develop systems and conduct any kind of space research and development "necessary to make effective provision for the defense of the United States." Even so, the gray area would remain.

To overcome the jurisdictional problem and permit basically separate activities without expensive duplication, Congress created two coordinating bodies: a cabinet-level National Aeronautics and Space Council (NASC) and a sub-cabinet-level Civilian-Military Liaison Committee (CMLC). During the remainder of the Eisenhower administration, neither would function effectively. The CMLC met often but had insufficient authority to resolve issues, while the NASC, which possessed the requisite decision-making capability, seldom met. The President refused to be constrained in his management of the space program.[28]

The establishment of NASA reflected the administration's determination to give the space program a civilian focus through a policy of "space for peaceful purposes" that encompassed scientific exploration as well as a less-publicized but far more important national security element. President Eisenhower signed the National Aeronautics and Space Act on 29 July 1958. Along with prescribing organizational and functional responsibilities of the National Aeronautics and Space Administration, the space act addressed policy in unmistakable terms. "The Congress hereby declares that it is the policy of the United States that activities in space should be devoted to peaceful purposes for the benefit of all mankind." [Sec 102(a)] Although the statement reflected Eisenhower's policy statements prior to Sputnik, before inclusion in the space bill James Killian and the Presidential Scientific Advisory Committee (PSAC) conducted a comprehensive examination of broad policy objectives as part of their assessment of organizational requirements.

At the request of the President back in early February 1958, Killian established a panel under the auspices of the PSAC to develop a national space program. Chaired by Nobel laureate Edward Purcell, the panel's deliberations focused on nonmilitary space programs and activities. Arguing that "even the more sober proposals...about space as a future theater of war...do not hold up well on close examination or appear to be achievable at an early date," the Purcell Panel strongly recommended passive military support applications while rejecting any use of military weapons in space. With the President's blessing, Purcell and panel member Herbert York briefed the Cabinet and other groups within the administration, and in late March issued a public version of their report. The brochure, "Introduction to Outer Space," stressed the peaceful, scientific objectives of spaceflight and the administration's cautious approach to the space age. The PSAC report would provide the basic guidelines for the military role in space. Despite strong objections from Air Force officers in the months ahead, the administration would confine offensive military space applications to studies only.[29]

With military satellite launches on the horizon, Eisenhower refined national space policy with two National Security Council directives that closely bracketed the signing of the Space Act. In June NSC Directive 5814, "U.S. Policy on Outer Space," advocated a "political framework which will place the uses of U.S. reconnaissance satellites in a political and psychological context most favorable to the United States." The NSC followed this on 18 August 1958 with a more definitive directive, NSC 5814/1, "Preliminary U.S. Policy on Outer Space," a broad statement which emphasized denying Soviet space superiority. Echoing the early Rand Corporation studies on satellite feasibility, the administration would seek to achieve this objective by "'opening up' the Soviet Bloc through improved intelligence and programs of scientific cooperation." This would be accomplished by the military reconnaissance satellites, whose mission, the directive asserted, fell squarely within the "peaceful purposes" guidelines and represented an asset of "critical importance to U.S. national security."[30] In effect, although NSC 5814/1 advocated an open, cooperative scientific exploration program, it also established the foundation for a national security reconnaissance space capability immune from international inspection or control. The latter received the highest priority from an administration that saw no contradiction in space for peace combined with space for national security.

With the 1958 Space Act, the government formally established a dual space program comprising separate civilian scientific and military applications projects. Both were directed to "peaceful," or scientific, defensive, and nonaggressive purposes. This accorded precisely with Eisenhower's commitment to insure unrestricted overflight in outer space of military reconnaissance satellites that the President so eagerly awaited to replace the increasingly vulnerable U-2 surveillance aircraft that violated national sovereignty in airspace overflight.

Although the framers of the Space Act did not equate "peaceful" with civilian or nonmilitary activities, government officials in the future often found themselves required to explain that both NASA and the Defense Department conducted peaceful space work, one primarily engaged in space exploration and the other in various military support activities devoted to keeping the peace. Air Force space leaders like General Schriever repeatedly criticized this policy which many interpreted as implying that NASA engaged in "peaceful" work while the military, pursued "nonpeaceful" activities. Such inaccuracies, he believed, along with policy restrictions limiting offensive space weapons to the drawing board, prevented the military from providing necessary security through an expanded space program. Air Force advocates of a dynamic, military-oriented national space endeavor remained frustrated by national space policy and organizational constraints that seemed to rule out anything except passive military space applications.[31]

## NASA Takes Shape

With an organization in place by midsummer that provided for dual military and civilian programs, officials turned to the complex mission and project assignments remaining before NASA could commence operations on 1 October 1958. Lines of demarcation remained vague, while competition for prestige and funding promised to be severe. The initial question centered on facilities and infrastructure. During the congressional debate it became clear that the new agency would absorb NACA's existing aeronautical research facilities and personnel. These included nearly 7,000 personnel and the Langley and Ames Aeronautical Laboratories, the Lewis Flight Propulsion Laboratory, the High-Speed Flight Station at Edwards Air Force Base, and the Pilotless Aircraft Research Station at Wallops Island, Virginia.

To achieve space capability quickly, NASA needed an infusion of space programs, facilities, and funding from the military services. In the NASA raid on service assets, the Air Force emerged the clear victor. With little objection from the Navy, NASA received Project Vanguard's personnel and facilities, including its Minitrack satellite tracking network, and more than 400 scientists and engineers from the Naval Research Laboratory. Potential Army losses, however, proved far more sweeping and contentious. Newly-appointed NASA administrator, Keith Glennan, considered the Army space program most important for providing the agency credible space design, engineering, and in-house resources. He initially requested transfer of Cal Tech's contracted Jet Propulsion Laboratory (JPL), whose sympathetic director had visions of turning it into the "national space laboratory," and a portion of the Army Ballistic Missile Agency that included the von Braun team and its giant Saturn booster project. General Medaris, however, strongly objected and waged a public campaign to stall the process and reverse the decision. His effort produced a compromise. The JPL would be transferred to NASA by 3 December 1958, while the Huntsville complex would remain under the Army's jurisdiction and support NASA on a contractual basis. Medaris might postpone but he could not prevent a transfer. A year later the Army would lose to NASA its entire space operation at Huntsville, which would be renamed the Marshall Space Flight Center.[32]

As for the Advanced Research Projects Agency and its Air Force-related programs, the Defense Department agency intended to transfer only elements of its Advanced Research for Scientific Purposes program. In mid-August, however, Eisenhower awarded NASA overall responsibility for human spaceflight. As a result, ARPA relinquished all of its "man in space" projects, which NASA combined under the designation, Project Mercury. ARPA also relinquished its special engine research project, as well as satellite tracking and satellite communications, meteorological, and navigation satellite programs.* Air Force reaction proved mixed. While giving up what amounted to five space probes, three satellite projects, and some propulsion research represented largely scientific projects in early stages of research, the loss of the manned spaceflight mission created apprehension about the future of a military manned role in space. While $800 million for space in the fiscal year 1959 budget represented eight times the space portion before Sputnik, NASA's share outpaced ARPA's by more than $50 million and included $117 million transferred from ARPA. Of the latter, the Air Force gave up $58.8 million. In short order, NASA had acquired the missions of scientific space exploration, including the moon, as well as manned spaceflight and all civil applications satellites. To fund its new programs, NASA received a generous budget, which raised the specter of tough competition between civil and military sectors for space funds in future years.[33]

On the other hand, NASA's absorption of Army and Navy space programs had left the Air Force the front-runner for the military space mission. Air Force leaders quickly perceived the advantage of cooperating with the new agency and making the service indispensable to the national space program. An essential element involved the Air Force's dominance in available space boosters. In a 17 September 1958 memorandum, Under Secretary of the Air Force Malcolm A. MacIntyre offered guidelines for the Secretary of Defense to follow in his discussions with NASA over civil and military program jurisdiction. Under Secretary MacIntyre argued for continuing the Air Force man-in-space program in cooperation with NASA, and reminded the defense secretary that the Air Force possessed the booster engine capability to support manned spaceflight. Responding on 31 October 1958, ARPA Director Roy Johnson noted that the Defense Department and NASA were following the guidelines suggested, and the Space Council would decide jurisdiction in unclear cases. Moreover, he concluded, "the Air Force's foresight in anticipating the requirements of both agencies for booster vehicles is to be commended. The present outlook is that all that have been provided for will be greatly needed and well utilized." In the months ahead the Air Force would continue to work to gain approval of exclusive responsibility for space booster development.[34]

When NASA commenced operations on 1 October 1958, a year after Sputnik initiated the space age, its leaders recognized that it would remain in the Defense Department's shadow for the foreseeable future. The Defense Department continued to focus on system work and big projects. The Air Force, through ARPA, not only pursued space-related missile work on solid propulsion, launch facilities, and test ranges, it also combined space and missile activity through projects like MIDAS, Samos, and antisatellite identification. Its impressive list of projects involved work on a manned orbital glider/bomber, new boosters, a variety of satellites, studies for

* See Appendix 2-3.

developing manned satellites and space stations, and support for Project CORONA, the covert reconnaissance satellite program publicly known as "Discoverer," which planners readied for launch in January 1959. Meanwhile, NASA focused on scientific applications through its existing NACA laboratories, and depended on the Defense Department and the Air Force for assistance with a variety of responsibilities. Of its first eight space probe launches, for example, the Defense Department accepted responsibility for the initial five, with the Air Force launching the first two Pioneer lunar probes.[35]

By the end of 1958 the foundation to support American superiority in space had been laid. Policy prescribed space activities for peaceful, that is nonaggressive, purposes, while organizational arrangements promoted a dual effort with civilian scientific aspects centered in NASA and military research and applications directed by ARPA. Yet much remained unresolved, not only between the Defense Department and NASA, but within the military arena. While the Air Force continued to face challenges with ARPA over program development and operational responsibility, a new Defense Department office appeared in late 1958 to add to the confusion. In August, Congress passed the Defense Reorganization Act which, among other measures to centralize and clarify defense operations, created the office of Director for Defense Research and Engineering (DDR&E), whose chief reported directly to the Secretary of Defense. Subsuming the old responsibilities of the Assistant Secretary of Defense for Research and Development, the new office became the focal point for all defense research and development activities. However, it would be a number of months before the new agency would be able to build its staff, sort out jurisdictional arrangements, and exercise its authority. Meanwhile, ARPA would continue to function as the nation's centralized military space agency. Nevertheless, the fact that the new office received explicit recognition in Public Law, while ARPA had been established only by authority of the Secretary of Defense signaled the ultimate demise of Roy Johnson's space agency. Air Force leaders hoped that the new Defense Department office would allow the service more autonomy in the space arena.[36]

As NASA prepared to begin its operations on 1 October, the Air Force had clearly left the Army and Navy behind in the quest for sole control of the military space mission. Even so, the chief of the Air Force's Legislative Liaison Office perceptively described an Air Staff divided on whether the service should assert itself more directly. Some officers preferred a "wait and see" approach, because the Air Force had received from ARPA a share of the space mission. Others argued for a more active role given the Army's retention of its 1.5 million-pound Saturn booster project as well as signs the Army would be authorized to develop communication satellites and the Navy would proceed with its navigational satellite. By the end of the year, Air Force leaders decided that they could not stand on the sidelines and let events take their course.[37]

## Renewing the Quest for the Military Space Mission

The Air Force decision to promote itself for the military space mission in early 1959 precipitated a wide-ranging review of its current space posture and available courses of action. In early February the Deputy Chief of Staff for Plans described the Air Force's weaknesses in space organization, operations, and research and development that resulted, he said, from its failure to develop a coordinated space program. Rather than formally requesting operating responsibility for space roles and missions, the Air Force should demonstrate successful stewardship, rely on available hardware [boosters], and establish "squatters rights." Despite the presence of ARPA, the Air Force should establish its own integrated space program while working to improve relationships with both ARPA and NASA. The Air Force, he said, "must assume the role of opportunist, aggressively taking advantage of each situation as it arises to assure that the Air Force is always predominate [sic] in any action that has a space connotation."[38]

The Air Force campaign focused on congressional hearings in the winter and spring. Beginning in February 1959 Air Force spokesmen repeatedly elaborated on the Air Force "aerospace" policy that viewed space as...an extension of the medium in which we are now operating in the accomplishment of assigned roles and missions." As General White testified before the House Armed Services Committee, "The missions that we foresee [in space] are largely an extension of the missions that are required in the atmosphere." He went on to argue for funding and program support in terms of three general requirements: first, to improve current forces; second, to develop new systems in areas with recognized military applications; and third, to study and develop systems in areas without clear military applications but with excellent potential for possible future military use. The Air Force's manned space program ranked high among the latter requirements. Unlike NASA, whose mandate encompassed manned spaceflight and exploration of the unknown in outer space, the military would find programs without known applications particularly difficult to justify to congressional budget overseers.[39]

The Air Force's campaign intensified with the convening in late March of Senator Stewart Symington's Subcommittee on Governmental Organization for Space Activities. Scheduled witnesses Under Secretary of the Air Force Malcolm A. MacIntyre and Major General Bernard A. Schriever could expect a sympathetic response to a strong Air Force argument from Senator Symington, who continued to criticize the administration's budgetary frugality in the area of space defense.[40] Aware that the Air Force witnesses appearing before Congress required well-coordinated statements, the Air Staff's Directorate of Technology (DAT) and Schriever's Ballistic Missile Division staff developed position papers that provided a comprehensive assessment of current service strengths and weaknesses as well as a strong case for an increased Air Force role in space.

The Air Staff analysis demonstrated that the Air Force had successfully identified thirteen major military uses of space, nine of which had been included in the important NSC directive, "Preliminary Outer Space Policy."[41] Five of these missions—photographic/visual reconnaissance, electronic reconnaissance, infrared reconnaissance, mapping and charting, and space environmental forecasting and observing—had received approval as Air Force General Operational Requirements (GOR) and represented missions previously identified and analyzed by Rand. At the same time, Air Force headquarters had underway seven important studies with industry or in-house agencies and offices. Moreover, the analysis asserted, Lieutenant General Roscoe C. Wilson's DCS/D had produced an important paper outlining "Priority Listings of Military Space Missions."* In every document cited by the Directorate of Technology's officers, satellites received top billing, with Samos and MIDAS heading the list, followed closely by a variety of manned spaceflight requirements. Despite NASA's human spaceflight mission responsibilities, Air Force space leaders clearly had not relinquished interest in military manned spaceflight.

The Air Staff's analysis focused on constraints that prohibited the Air Force from implementing its aerospace "policy" of performing the space missions formally identified in Air Staff documents and approved as General Operational Requirements. It noted that the Air Force retained authority for planning, budgeting, and development only in non-space areas because NASA's responsibility embraced the scientific space area and ARPA's the military space arena. In effect, the Air Force had no responsibilities for a space program of its own. Echoing long-held criticism of the Defense Department agency, the Air Staff paper faulted ARPA for its practice of assigning system development responsibility to a service on the basis of existing capability but without regard for "existing or likely [space] mission and support roles." ARPA, rather, should focus on policy decisions and forego the "project engineering" detail normally found only at the lowest Air Force operating levels.[42] As for NASA, the Air Staff critique noted that the Air Force, if prohibited from pursuing its own scientific space exploration and research might very well face dependence on the "fall-out and by-products" of the civilian, scientific agency. To avoid this, the Air Force rather than NASA should develop programs of common interest, such as space boosters and satellites, in order to meet the more stringent military specifications and priorities. This would leave NASA to apply its budget to "really scientific projects" like unmanned space probes. Ultimately, concluded the Air Staff directorate, Air Force leaders should lobby Congress for a greater role for the Air Force in space.

General Schriever's staff also agreed that "it is axiomatic that the Air Force has the prime military responsibility for operating in space. Yet the means for developing

* See Appendix 2-4.

this capability are denied by present NASA/ARPA policies and actions."[43] Given the command's responsibilities, the BMD analysts criticized NASA for assuming a major portion of the nation's booster development program, indicating interest in taking over guidance, control, and ground tracking communications programs, and showing signs of building up "a development, production, management and 'operational' capability which will duplicate that presently existing in the AF Ballistic Missile Program." ARPA appeared to acquiesce in NASA's objectives while continuing to pursue its own development activities without regard to the future military operational user. Both agencies appeared oblivious to the "systems" concept of development, leaving the Air Force unable to establish an "integrated Air Force space program with a logical stepwise progression towards stated goals."

General Schriever also found his command becoming overburdened with increased management responsibility for ARPA programs and NASA's requirements for boosters and launch support. In a letter to the chief of staff on 11 February 1959, the general described the critical shortage for the next eighteen months of six Atlas boosters and limited launch pad availability at both Atlantic and Pacific Missile Ranges. Without immediate Air Staff action, he predicted delays in either the ICBM or booster operational schedules. The booster issue proved especially sensitive in view of the new emphasis on using Air Force Thor IRBM and Atlas ICBM requirements as the wedge into an enlarged space arena. As Schriever's staff explained, the close connection between missiles and space vehicles represented the best means of achieving Air Force space objectives because "future booster development as well as subsystem development can be initiated against bona fide ballistic missile requirements." The Air Staff responded by programming for additional boosters and launchers.[44]

In their testimony before the Symington Committee in late April 1959, Under Secretary MacIntyre and General Schriever presented a strong defense of Air Force space projects and the case for a greater Air Force space role. General Schriever, in particular, argued that by 1970 the Air Force's responsibilities for strategic offense and strategic defense would be accomplished by an arsenal of space weapons consisting of "ballistic missiles, satellites and space craft." To help the Air Force move forward with its space missions, he recommended that ARPA be dissolved by 30 June 1959, DDR&E assume the role of providing policy guidance and assigning service operating responsibilities, and space research and development be returned to the military services.[45]

The testimony of General Schriever and other Air Force spokesmen before congressional committees in the spring of 1959 proved especially effective in light of the Air Force's growing involvement in space. They could cite an impressive array of their "own" projects as well as important support the Air Force provided ARPA and NASA on others.[46]

Heading the list of major Air Force projects appeared the three elements of the former WS-117L Advanced Reconnaissance System. Samos, formerly known as Sentry, represented the reconnaissance element. Consisting of the Atlas booster and Lockheed's second-stage spacecraft vehicle Agena, Samos involved collecting photographic and electromagnetic reconnaissance data and transmitting the information by means of a "readout" system or actual "recovery" of data packages by aircraft. In contrast to Project CORONA, which pursued the capsule recovery technique, the Air Force initially had elected the "readout" method, but eventually would attempt both methods of data retrieval. MIDAS (for Missile Defense Alarm System) also relied on the Atlas-Agena booster satellite combination. The MIDAS payload consisted of infrared sensors designed to detect missile exhaust plumes and be able to provide command centers a thirty-minute warning of an ICBM attack.[47]

Both Samos and MIDAS projects experienced technical and management problems not uncommon to projects on the leading edge of technology. For example, civilian and military officials continually differed over technical requirements and capabilities, funding, and operational arrangements. While the Air Force proposed assigning operational control of Samos and MIDAS to the Strategic Air Command (SAC) and the North American Air Defense Command (NORAD), respectively, the Army and Navy argued for a joint command that would operate all military space systems. Air Force officials also favored implementing a systems development approach that would achieve desired performance goals while development and testing proceeded. Solving problems "concurrently," they hoped, would result in achieving early operational capability. The Office of the Secretary of Defense, however, preferred a "fly before buy" arrangement, and focused on component subsystem performance and capabilities. As a result, MIDAS and Samos remained in flux with the Air Staff repeatedly defending and revising development plans, while looking ahead to initial test flights in 1961.

Although publicly Project Discoverer represented a third Air Force project of the former WS-117L program, it actually served as a cover for the covert Project CORONA. After President Eisenhower in February 1958 authorized a secret reconnaissance satellite as a joint CIA-ARPA-Air Force effort, it became known as Project CORONA, an experimental activity within the WS-117L program. However, alarmed by publicity identifying CORONA as a military reconnaissance system, administration officials in the late summer of 1958 decided to sever CORONA's public connection with WS-117L by creating two photo reconnaissance efforts. While the Air Force pursued its Sentry/Samos project using the Atlas booster, CORONA would continue as Project Discoverer and rely on the Thor booster. The Discoverer project embraced tests on satellite stabilization equipment, satellite internal environment, ground support equipment, and biomedical experiments using mice and primates and, most importantly, capsule recovery techniques. Officials had scheduled thirty-two polar orbit launches from Vandenberg Air Force Base using the Thor-Agena combination. Of the four launches attempted by the end of June 1959, the first two achieved orbit for brief periods and passed back useful experimental data despite loss of the capsules. The remaining two failed to achieve orbit.[48] Despite difficulties with the satellite systems during this early developmental phase, the Air Force could claim that it managed or supported the nation's most important satellite programs, and expected to be awarded greater operational responsibility in the near future.

In addition to its own Samos and MIDAS satellite projects, under ARPA's direction the Air Force provided launch support to the Navy's Transit navigational satellite, designed to support Polaris submarines, and the Army's Notus communications satellite effort.[49] The most important, however, proved to be the growing detection, tracking and satellite cataloguing project known as the Space Detection and Tracking System (SPADATS). Begun hurriedly under the name Project Shepherd by ARPA in response to Sputnik, all three services were to participate. The Air Force, under Project Harvest Moon (later Spacetrack), would provide the Interim National Space Surveillance and Control Center (INSSCC) data filtering and cataloguing center at its Cambridge Research Center in Massachusetts. Early efforts brought together radar data from MIT's Lincoln Laboratory's Millstone Hill radar at Westford, the Stanford Research Institute in Palo Alto, California, and an ARDC test radar at Laredo, Texas. Sensors included the new Smithsonian Astrophysical Observatory's Baker-Nunn satellite tracking cameras that it procured for tracking the IGY satellites and available observatory telescopes. The Air Force would also devise the development plan for the future operational system.[50]

ARPA assigned the Navy responsibility for developing and operating its east-west Minitrack radar fence and its data processing facility in Dahlgren, Virginia. Originally designed to support Project Vanguard, the Navy redesignated its sensor and control operation Spasur (Space Surveillance). The Army portion, termed Doploc, envisioned a doppler radar network to augment Spasur and, together, feed data to the INSSCC for cataloguing, trajectory prediction, and dissemination. ARPA and the three services realized the system's limited capability, but agreement on funding necessary improvements proved difficult to achieve. After the Army dropped out of the picture, the Air Force and Navy contested for operational control of the system. The Navy seemed to prefer operating a separate system, while the Air Force wanted its Air Defense Command (ADC) to acquire management responsibility and NORAD to possess operational control. By mid-1959, the controversy had reached the Joint Chiefs of Staff, where it became embroiled in a major roles and missions contest among the services.

As for NASA's requirements, the Air Force agreed to construct infrastructure facilities at Patrick Air Force Base for NASA's space probes and then provide booster support for the Pioneer lunar probes (Thor-Able) and Tiros cloud-cover satellite (Thor-Delta/Able). The Air Force also supported the Centaur high-energy upper stage based on hydrogen and oxygen as fuels, which it hoped to use in support of the Advent communications satellite project. Most importantly, the Air Force supported Project Mercury, NASA's man-in-space project, by furnishing Atlas boosters and launching services, along with considerable technical, biomedical, and personnel assistance. The issue of military manned spaceflight had always been a most sensitive subject for Air Force space enthusiasts. Like their German counterparts, early Air Force space pioneers looked to space as more than an arena for scientific exploration or simply a venue in which to pursue exciting new challenges. They considered a military man in space mission the logical extension and eventual goal of Air Force space operations. Not only did this objective correspond to Air Force thinking on "aerospace," but manned spaceflight seemed the next "logical" step in the chain of operational development from aviation medicine to space medicine. Indeed, by the time of Sputnik, Air Force medical personnel could look back on a wealth of aeromedical experience that put the service at the forefront of knowledge on conditions of flight in the upper atmosphere and near space. Space presented scientists the daunting challenge of mastering the complexity and weight problems in a space environment.[51]

At the close of the Second World War, the Air Force gained the services of a number of German scientists who had performed path-breaking medical research for the Luftwaffe. Although most joined the growing Aeromedical Laboratory at Wright-Patterson Air Force Base in Ohio, six received assignment as research physicians to the Air Force School of Aviation Medicine at Randolph Air Force Base near San Antonio, Texas. In February 1949, the latter established the world's first Department of Space Medicine, under the direction of Dr. Hubertus Strughold, who had coined the term "space medicine" at an important symposium the previous year. In November 1951 the Randolph school held another symposium, entitled "Physics and Medicine of the Upper Atmosphere," to avoid criticism of "Buck Rogers" projects within the Air Force. Nevertheless, at this meeting Strughold advanced the concept of the "aeropause," an area of "space-equivalent conditions" such as anoxia that begins much lower, about 50,000 feet, rather than at the 600-mile barrier normally cited by authorities as the boundary between the atmosphere and space. "What we call upper atmosphere in the physical sense," Strughold said, "must be considered—in terms of biology—as space in its total form." In effect, manned ballistic or orbital flight at the 100-mile altitude would be spaceflight. Strughold would come to be known as the "the father of space medicine" and go on to lead the Air Force's School of Aviation Medicine in exploring the space environment. Together with researchers at the Wright Air Development Center (Aeromedical Laboratory) and the Aeromedical Field Laboratory at Holloman Air Force Base, New Mexico, Air Force space medicine teams from San Antonio pursued a variety of experiments dealing with conditions of "zero g" or weightlessness in space, "g loads," or the effects of heavy acceleration and deceleration primarily through the upper atmosphere rocket plane flights and sounding rockets with animal passengers. Although the crash ICBM program in the 1950s interrupted animal flight research for a six-year period, other human factors experiments continued. By the time of the Sputnik launch, Air Force medical research specialists had accumulated a wealth of data on conditions of manned spaceflight and determined that the basic problems of weightless flight could be solved.[52]

While Air Force medical personnel continued their quest for data on conditions of manned spaceflight, scientists and engineers conducted research and development on space hardware systems that could eventually be powered through the upper atmosphere into earth orbit. Manned space vehicle concepts proceeded along two lines of thought based on the reentry technique used. One involved ballistic reentry using blunt-body capsules, the other aerodynamic reentry with winged vehicles. Although Air Force planners pursued both methods of spaceflight, initial interest centered on the winged suborbital vehicle later known as Dyna-Soar (from dynamic soaring).

Dyna-Soar evolved from the rocket plane studies and experiments of the early 1950s. By May 1955 hypersonic (Mach 5 and above) glide vehicle development had led to three related Air Force projects: Bomi, an acronym for Bomber Missile, but soon redesignated Robo, for Rocket Bomber; Brass Bell, a high altitude reconnaissance system; and Hywards, the actual boost-glide vehicle. Although designed for suborbital flight, the three could be launched into low earth orbits with adequate propulsion. After it became apparent that weapons in space would not proceed, on 30 April 1957 the Air Force merged the three programs under the name Dyna-Soar, and considered it the manned flight successor to turbojet bombers and reconnaissance aircraft. To reflect the requirements of the Air Force's first "aerospace" vehicle, engineers designed the Dyna-Soar as a manned, delta-wing aeronautical vehicle capable of being boosted into orbit while retaining reentry and controlled landing maneuverability. As such it filled a variety of accepted mission functions and could be supported by the vast network of existing ground facilities.

As early as the spring of 1956, the Air Force had discussed with several industrial firms its manned ballistic rocket research program. When the Air Force prepared its ambitious five-year astronautics plan in the heady weeks following the launch of Sputnik, it included projects for a manned capsule test system, manned space stations, and ultimately a manned lunar base. Although critics scoffed at such "fanciful" projects, ARPA director Roy Johnson did not. Shortly after his appointment in February 1958, he declared that "the Air Force has a long term development responsibility for manned space flight." With his blessing, Air Force leaders requested ARPA funds and directed Air Research and Development Command to prepare a development plan, called "Man-in-Space-Soonest" (MISS). It called for a four-phase capsule orbital process, which would first use instruments, to be followed by primates, then a man, with the final objective of landing men on the moon and returning them to earth.

The Army and Navy did not relinquish the field of manned spaceflight to NASA or the Air Force uncontested. In the spring of 1959 the Army unveiled its "Man Very High" proposal, later termed Project Adam, which called for lofting a man in a Jupiter nose-cone capsule on a steep ballistic trajectory that would produce a splashdown about 150 miles downrange from Cape Canaveral, Florida. Project Adam received no support from informed critics like NACA's Hugh Dryden, who explained that "tossing a man up in the air and letting him come back...is about the same technical value as the circus stunt of shooting a young lady from a cannon." The Defense Department rejected the Army plan from the start. The Navy's intriguing alternative, MER I (Manned Earth Reconnaissance), proposed orbiting a cylindrical vehicle with spherical ends. After achieving orbit, the ends would expand laterally to produce a delta-winged inflated glider. Although ARPA conducted studies on the proposal's feasibility, NASA's Project Mercury soon got underway and relegated the Navy plan to an interesting concept too bold for its day.

Although the Air Force MISS proposal came closest to "approval," ARPA balked at the high cost of $1.5 billion and the uncertainties surrounding the future direction of the civilian space agency. When NASA began operations on 1 October 1958, the Air Force had prepared seven Man-In-Space-Soonest development plans, each one dismissed by ARPA for cost, technical, or utility concerns. Fittingly, the last one omitted the word "soonest." When President Eisenhower assigned NASA the human spaceflight mission in August 1958, ARPA transferred its manned space programs and funds to the new civilian agency. Hampered by insufficient funding, the President's "space for peace" policy, and the inability to justify a military man in space, the Air Force had to abandon—at least for the time being—serious plans for a distinct and separate military man-in-space program.

NASA's assumption of the manned space mission left the Air Force with Dyna-Soar, a single-place vehicle, which the Air Force had protected from ARPA's grasp by stressing its suborbital, aeronautics phase of development. Although Dyna-Soar had received approval for development in 1958, by the spring of 1959 the Air Force still had not identified an adequate booster to fulfill the as yet undetermined aeronautical, missile and, especially, space requirements of Dyna-Soar. An initial proposal called for using a cluster of the yet-to-be-developed Minuteman solid-propellant rockets, but the problem of separating the rockets as they would be expended proved too challenging and costly. This opened the door to possible encroachment from the Army and NASA.

The Army's Saturn appeared as a logical candidate, and Wernher von Braun made several attempts to convince the Air Force to accept the Saturn–Dyna-Soar combination. But the Air Force demurred, preferring to continue with its own 1,500,000 lb-thrust engine project it had underway. Given NASA's interest in Saturn, however, the Air Force might very well lose Dyna-Soar to NASA if the civilian space agency acquired the Army's big booster. In the spring of 1959, the Air Force continued to move forward with the Dyna-Soar project and hoped that it could keep alive a military manned spaceflight mission. Meanwhile, it would continue its strong support of NASA's Project Mercury.

By the spring of 1959, the Air Force's expanding role in space led Air Staff leaders on 13 April 1959 to enhance the headquarters focus on space by providing General Boushey's year-old Directorate of Advanced Technology the authority to coordinate within Air Force headquarters all space issues. The new arrangement eliminated the space responsibilities of the Assistant Chief of Staff for Guided Missiles except for coordination activities involving boosters, test facilities, and range and launch complexes. Gone at last was the divided authority within the Air Staff for space requirements.[53]

The Air Force's 1959 campaign for the military space mission did not go unnoticed by the Army and Navy. They closely followed the Air Staff realignment, the growing Air Force responsibilities for space systems, and the coordinated testimony of its spokesmen before Congress. In fact, General Medaris seized his opportunity before Senator Symington's committee to accuse the Air Force of a long history of noncooperation with his Army Ballistic Missile Agency. Although General Schriever provided a lengthy, detailed rebuttal, Medaris refused to withdraw his charges. The dispute only served to reinforce the views of legislators already critical of interservice rivalry.[54]

In a move more threatening to Air Force interests, Admiral Arleigh Burke, Chief of Naval Operations, in late April 1959, made "a bold bid for a major share" of the space mission, by proposing to his Joint Chiefs of Staff colleagues the creation of a joint military space agency. In effect, he advocated a unified command for space based on the "very indivisibility of space," projected large-scale space operations in the near future, and the interests in space of all three services. Army Chief of Staff General Maxwell D. Taylor agreed, arguing that space activities transcended the particular interests of any one service. But Air Force Chief of Staff General White opposed the proposal because, he said, it violated the practice of treating space systems on a functional basis and integrating weapons within unified commands. He argued that space systems represent only a better means of performing existing missions and should be assigned to the appropriate unified or specified command.[55]

The Navy-Army initiative to gain a greater military space role by working through the Joint Chiefs of Staff to realize a joint command compelled General Schriever in mid-May to argue for an Air Force counter-campaign to acquire all or part of the military space mission "as soon as possible." In a letter to Lieutenant General Roscoe C. Wilson, DCS/D, the ARDC commander described his concerns and provided a draft letter for either Air Force Secretary James H. Douglas or Chief of Staff General White to forward to Secretary of Defense McElroy. His suggested letter asserted that "since its inception" the Air Force had been operating in aerospace through the mission areas of strategic attack, defense against attack, and supporting systems that enhanced both the strategic retaliatory and active defense forces. The Air Force had important requirements for earth satellites, which represent aerospace vehicles of the foreseeable future. Characteristically, Schriever criticized existing fragmented satellite program management and advocated a unified, systems development approach that would "achieve the most effective deterrent posture" by coordinating and integrating satellite systems within the broad Air Force strategic and air defense force. Moreover, Army and Navy requirements, the general asserted, would be best achieved by the Air Force acting as "prime operating agency of the military [national] satellite force."[56]

While the services argued over roles and missions, ARPA director Roy Johnson stoked the fire in June by recommending a tri-service Mercury Task Force to support NASA, while Defense Secretary McElroy requested advice from the Joint Chiefs of Staff on assigning the services operational responsibility for several important space projects, including MIDAS and Samos. Service views reflected the division over the unified command issue. While the Navy and Army favored a Mercury Task Force as well as a Defense Astronautical Agency to direct and control all military space systems, the Air Force opposed both for the reasons General White explained earlier in response to Admiral Burke's proposal.[57]

With no resolution of the differences by the fall of 1959, Secretary McElroy in September made three decisions that propelled the Air Force further forward in its quest for exclusive responsibility for military space activities. Differing with Admiral Burke's prediction, DDR&E director Herbert York had argued that the country could expect relatively few satellites in orbit in the foreseeable future, and thus the nation did not need a unified space command. The Secretary of Defense agreed, and sided with the Air Force position by declaring that "establishment of a joint military organization with control over operational space systems does not appear desirable at this time." He too disapproved both the proposed Defense Astronautical Agency and Mercury Task Force. In place of the latter, he designated Air Force Major General Donald N. Yates, Atlantic Missile Range commander, to "direct military support" for NASA's manned space project. Most significantly for the Air Force, the Defense Secretary assigned to it responsibility for "the development, production and launching of space boosters" as well as payload integration. Satellite operational responsibility, however, would continue to be assigned to the services on a case-by-case basis. Initially, the Air Force would receive Samos and MIDAS (in November) and, in a separate action, Discoverer (in December). The Navy acquired the Transit navigation satellite, and the Army four Notus communications satellites. In short, Secretary McElroy agreed with Dr. York and Air Force critics by reversing his established policy that favored ARPA and reassigning space projects among the three services. The Air Force received the major share. Admiral Burke's proposal for a unified command for space would prove twenty-five years too early.[58]

Secretary McElroy's directive in September represented the first fruits of the Air Force campaign of 1959 for the military space mission. Legitimately hailed as a landmark decision on the Air Force's road to space, it nevertheless provided the Air Force an incomplete victory over its protagonists. Pessimists pointed out that civilian control over development of military space systems remained unchanged at the secretarial level, and ARPA retained its authority to conduct project engineering supervision. Moreover, the Air Force received responsibility for space boosters but not for all space satellite systems. On the other hand, the Air Force had warded off a joint operational agency for space and received designation as the nation's "military space booster service"—a major objective of the spring campaign, and a further blow to the Army's space fortunes. The Air Force now found itself poised to assume command and control of operational space systems, while receiving operational control of Samos, MIDAS, and Dyna-Soar—all space systems with growth potential.

On balance, in the fall of 1959, Air Force leaders could express optimism about the space future, fully aware that much needed to be done to consolidate the September gains. At the Air Force major commanders' conference on 1 October 1959, the audience heard that "the Army and Navy can be expected to continue their efforts to neutralize this interim Air Force victory" by showing that missile range and tracking facilities as well as satellite payloads deserved unified command direction and control. Now that the Air Force had gained its first chance to issue plans for development and operation of particular space systems, it would need to make good use of this opportunity. "Future steps toward gaining the assignment of space responsibilities will be determined...by the manner in which the Air Force handles the responsibilities it has just been assigned."[59]

Before it discharged any of these responsibilities, however, the Air Force began lobbying for the Army's Saturn heavy-lift booster project. As Vice Chief of Staff General Curtis E. LeMay explained to the Secretary of the Air Force on 29 September 1959, "in view of this directive [18 September 1959] it appears that the in-house capability of the Department of the Army for the development of space boosters and systems, which is represented by the Army Ballistic Missile Agency at Huntsville, Alabama may now be available for transfer to the Air Force."[60] But Saturn was not a weapon system, and NASA, with funds available and manned spaceflight on the horizon, could make a far better case for the big booster than could the Defense Department. Try as they did, Air Force planners could not specifically justify the need for a 1,500,000-pound thrust engine. Apparently, Secretary of Defense McElroy offered the Saturn to NASA's Director Glennan, who contacted General Medaris. After DDR&E York publicly confirmed that the Air Force would develop all space boosters needed by the Defense Department, integrate space payloads and launch the combination, Medaris preferred to transfer to NASA the entire von Braun team and missile operation, rather than have the Redstone complex and personnel separated and parceled out to various agencies. Despite objections from the Joint Chiefs of Staff, President Eisenhower approved Saturn's transfer to NASA on 2 November 1959. The Air Force would have to await more favorable circumstances to gain authority to develop military superboosters.[61]

With the President's decision underscoring NASA's claim to human spaceflight, Air Force leaders realized that the Dyna-Soar project had become endangered. At the end of October 1959, General Boushey, chief of the Directorate of Advanced Technology, declared that the Saturn decision suggested that "the loss of the Dyna-Soar project to NASA appears imminent." He predicted such an action would effectively remove the Air Force from super booster development and nullify the 18 September 1959 memorandum assigning the Air Force space booster responsibilities. Events proved General Boushey's pessimism misplaced. York reaffirmed the Air Force's Dyna-Soar project and the service selected Boeing as contractor in November 1959. Yet Air Force leaders remained aware of the fragile state of the project's future.[62]

The end of the year also brought the official demise of ARPA as the central Defense Department agency for space activities. Following the transfer of most of its space projects to the services in the fall, a 30 December 1959 directive from Secretary McElroy designated ARPA as "an operating research and development agency of the DoD under the direction and supervision of the DDR&E." In the future, ARPA would manage only a limited number of advanced research programs. General Schriever and other Air Force leaders rejoiced at ARPA's demise and the return of development responsibilities to the user agencies. Yet it meant removing a high profile centralized space management agency close to the Defense Secretary. With the military spotlight on space now reduced, space projects faced competition from other worthy service requirements in the battle for funding, while greater service rivalry over space systems without clear service roles became a distinct possibility.[63]

DDR&E now became the dominant Defense Department reviewing office with far more authority over Air Force research and development proposals than ARPA possessed. In late 1959 Lieutenant General Roscoe C. Wilson, Deputy Chief of Staff for Development, expressed his concerns about the civilian technical influence that resulted in considerable wasted time and effort before decisions from "on-high" reached the Air Force. He also complained about civil-military relationships within the Air Force community. One involved Secretary of the Air Force James H. Douglas' initiative, in October 1959, to have all space decisions taken by the civilian-led Air Force Ballistic Missile Committee in the Office of the Secretary of the Air Force without significant Air Staff participation. Although Douglas' successor, Dudley C. Sharp, agreed to allow prior review of space issues by the Air Staff and increase its role in space development planning, the final decisions remained with his Ballistic Missile Committee.[64]

Despite these concerns, by end of the year the Air Force clearly had become recognized as the dominant military service in space. Lacking the boosters, facilities, and space experience of the Air Force, the Army and Navy found themselves on the periphery of the space picture, while ARPA had been reduced to insignificance. The changes in late 1959 affected the "space budget," too. The Air Force benefited the most from ARPA's loss of 80% of its funding. While NASA succeeded in nearly doubling its fiscal year 1961 budget from $535,6000,000 to $915,000,000, Air Force funding multiplied by nearly 120 times, from $2,200,000 to $249,700,000. Air Force leaders now could argue that the service had regained control of much of its "own" space program. Moreover, NASA remained dependent on the Air Force for launch boosters and range support and, Project Mercury notwithstanding, the Air Force's Dyna-Soar manned space program continued on the drawing board. If the Air Force had not achieved the complete victory sought by its leaders, it nonetheless seemed well on its way to gaining management responsibility for all service requirements as the Defense Department's executive agent for space.[65]

## The Air Force Seeks to Consolidate Its Position

As the Eisenhower administration entered its final year, the President could take pride in the country's space program. In the spring of 1960, the number of American scientific and space probe launches totaled 24, of which 14 had achieved successful orbit. The Soviets had succeeded only in launching three such spacecraft, although they continued to garner world prestige from their spectacular "feat" of hitting the moon and photographing its far side. On the international front, the United Nations (UN) prepared to establish a permanent 24-nation Committee on Peaceful Uses of Outer Space, ten European nations discussed formation of a joint agency for scientific space exploration, and the administration continued its nuclear test ban and disarmament efforts by offering the Soviets use of America's global tracking network for its manned space experiments.[66]

Nevertheless, Air Force leaders continued to chafe at what they considered a policy that produced too modest a defense-support space program and prevented offensive space weapon system development altogether. They centered their criticism on the administration's National Security Council 18 August 1958 national space policy, "Preliminary Policy on Outer Space." If this directive represented a preliminary statement of policy, hopefully a more conclusive formulation of policy would provide specific recognition of military requirements. Back on 30 June 1959, President Eisenhower had charged the National Aeronautics and Space Council to review the preliminary policy. It took the group a full six months to prepare their report. Approved by the NSC as Directive 5918 on 26 January 1960, the "U.S. Policy on Outer Space" continued to emphasize a policy of civilian "peaceful" scientific exploration and development activity. It lauded the UN's approval of the "launching and flight of space vehicles...regardless of what territory they passed over"—as long as they involved the "peaceful uses of outer space." Although the directive accorded the military mission better recognition, it restricted military space functions to defense support and, once again, specifically limited offensive space weapon systems to study only.[67]

Although the revised space policy disappointed military leaders, Eisenhower's attempt to have the Space Act amended in early 1960 provided another opportunity to promote greater recognition of the military space role. The President believed that the single national space program implied in the act was impractical and undesirable. Dual civilian and military programs represented reality and should be formally recognized. Because NASA and the Defense Department cooperated effectively without what he considered inappropriate congressional mandates, the National Aeronautics and Space Council and Civilian-Military Liaison Committee should be abolished. Furthermore, he desired presidential relief from direct program planning responsibility but, to avoid duplication, sought specific authority to "assign responsibility for the development of each new launch vehicle, regardless of its intended use, to either NASA or the Department of Defense."[68]

The President sent his proposed amendments to Congress on 14 January 1960, where they received considerable scrutiny in hearings that winter and spring. Not only did many legislators remain unhappy that the country seemed to trail the Soviet Union in space progress despite administration statements to the contrary, the fact that 1960 was a presidential election year assured a lively and contentious debate on space in the months ahead. Overton Brooks, Democrat from Louisiana and Chairman of the House Committee on Science and Astronautics, predicted as much in late fall of 1959 when he warned that Congress early in the new year would "probe every facet of the [space] program." Brooks, in fact, had been trying since the spring of 1958 to convince the administration that the country should have an integrated space program.[69]

Representative Brooks and other congressional leaders convened a number of committees to examine the President's request and review the merits of whether the country had or should have one or two space programs under civilian and/or military control. Since there appeared no ready solution to the issue, the Eisenhower administration's preference of separate programs continued. As for the President's recommendations, the House agreed to eliminate both oversight bodies, but in so doing convinced the administration to accept a substitute, the Aeronautics and Astronautics Coordinating Board (AACB). Cochaired by the Defense Department's Director for Defense Research and Engineering (DDR&E) and NASA's Deputy Administrator, the new coordinating body, unlike the CMLC, possessed the authority to make binding decisions. The Senate, however, chose not to act on the President's request until a new administration could review the issue. As a result, the NASC and CMLC continued in law yet ceased to function, while the AACB began operating in September 1960.[70]

The hearings provided an opportunity for Defense Department witnesses to lobby for a wider military role in space. At the same time, pointed questions about space planning revealed the weaker side of the Defense Department and Air Force approach to space. NASA impressed committee members by presenting a "10-year plan" with funding milestones for research, development and exploratory space activities in pursuit of peaceful objectives. The NASA initiative placed the Defense Department on the defensive. The Defense Department had no such plan and, as DDR&E Herbert York explained, it saw no reason to prepare one. Testifying on 30 March 1960 before Senator John Stennis' NASA Authorization Subcommittee of the Committee on Aeronautical and Space Sciences, his argument reflected the logic of the Air Force concept for space planning and operations. Unlike NASA, he said, the Defense Department did not view space as a mission, with spaceflight and exploration as ends in themselves, but rather as a means for achieving better military space applications to improve existing terrestrial military mission capabilities. "Considering the nature of our space objectives, it is not logical to formulate a long-range military space program which is separate and distinct from the overall defense program." The Defense Department's planning process also served the administration's political agenda by highlighting the civilian program rather than the military.[71]

While DDR&E York presented a sound argument, the Defense Department's unwillingness to produce a space plan left it open to criticism from a Congress sensitive to duplication and effective development and coordination between NASA and the Defense Department. While NASA seemed to know where it wanted its space program to go in future, the Defense Department appeared less certain. Especially in the field of space exploration, which demanded initial funding for programs without definite military mission applications, the military found it difficult to convince Congress without benefit of an effective long-range plan. For the Air Force, this meant that its budget reflected space not as a program in itself, but as part of traditional mission areas. The Samos reconnaissance satellite, for example, appeared under strategic elements, while the MIDAS early warning system supported air defense mission requirements. Even after ARPA had transferred space projects to the Air Force, the scattering of space projects throughout the budget prevented a strong focus for advocacy of a military space program during the budget process.

At the same time, Air Force planners encountered difficulty in development and operational planning for space systems. While the so-called indivisibility of "aerospace" provided a conceptual approach to space that supported the service's quest for military space missions, it did not contribute effectively to a planning process that required consideration of space as a separate medium. Not only did space systems, in fact, involve different technical challenges, determined by orbital dynamics in a hard vacuum, but the lack of basic knowledge about many aspects of space contributed to the complexities of the planning process.

Nevertheless, the Defense Department's lack of a space plan per se did not mean that the Air Force conducted no long-range space planning. Air Staff planners had attempted since early 1958 to develop conceptual plans for space by means of an Air Force Objective Series (AFOS) paper. An agreed-upon AFOS paper would be complemented by a Required Operational Capability (ROC) document, which would identify the forces necessary to achieve the objectives (AFOS). Only by September 1959 could planners agree on a ten-year plan for peacetime and wartime operations that seemed to meet Air Force requirements without conflicting with national policy. Yet critics claimed that the draft document treated space as a separate "entity" in violation of the "aerospace" concept, and subsequent AFOS drafts failed to gain approval throughout the spring. Meanwhile, Air Staff officers working on the ROC also encountered roadblocks when they presented their "revolutionary" developmental program. Looking ahead to an operational date of 1975, they proposed a high-profile program with major funding increases to achieve innovations in propulsion, structural materials, and guidance, as well as "human factors development" as part of a future military man-in-space program. The ROC clearly treated space as a mission by calling for development of space weapons regardless of whether earth-based aeronautical systems might provide a more efficient and cost-effective alternative. Air Staff critics dismissed the plan as too "utopian" and risky. Without approval of these two planning documents, the 120-page qualitative force structure analysis that would logically follow in the form of a Research and Development Objectives (RDO) paper, remained a "dead letter."[72]

Not until the fall of 1960 could Air Force planners agree among themselves and gain the necessary approval for their ROC and RDO proposals. Another nine months would pass before the Air Force issued its first Objective Series statement depicting long-range concepts and its vision of military space activities. By then, Air Force leaders dealt with another administration that appeared to be far more sympathetic to their objectives. Much of the planning dilemma resulted from the unwillingness of General White and other Air Force leaders to issue official guidance for meeting national space policy and engage in an Air Force-wide educational campaign on space. The administration's "space for peace" policy tended to inhibit independent, high-profile Air Force military initiatives, while any official Air Force statement on space would prove of marginal value as long as space remained the preserve of ARPA or NASA for funding, management, and overall technical direction.[73]

While Air Force leaders might very well ballyhoo the concept of "aerospace" in public forums and argue that "aerospace power is peace power," current political and organizational constraints called for a more cautious approach to Air Force pronouncements on space. Back in July 1959 Air Staff planners initiated a formal space policy study, which received greater attention following ARPA's demise in the fall. By the end of the year, the Chief of Staff's "policy book" contained a number of statements for use in the 1960 congressional hearings. General White, however, desired a comprehensive space policy statement he could issue officially. After numerous reviewers on the Air Staff and in the Office of the Secretary of the Air

Force had their say, a final version seemed ready for publication in mid-March. Yet General White considered the timing "inappropriate." As the Air Force headquarters historian concluded, the chief of staff worried that "publication of an official [space] policy statement at a time when so many facets of the space program were still undecided would have unfavorable reverberations in Congress, the Office of the Secretary of Defense, and the other military services."[74]

General White's caution was not misplaced. In early May 1960, shortly after the Air Force had submitted its operational plans for MIDAS and Samos, Admiral Arleigh Burke, the Chief of Naval Operations, again challenged the Air Force position on space operations. He reaffirmed the need for a joint [unified] military space agency based on major technological developments of the last year and a half that propelled several systems to the "operational threshold." He also referred to the substantial interservice support for NASA's Project Mercury, and the joint agencies soon to be established for command, control, and communications functions. After dividing along the lines of the previous summer, the Joint Chiefs of Staff forwarded its divergent views to Secretary of Defense Thomas Gates, who had held the post since December 1959. On 16 June 1960, he reaffirmed the decision earlier taken by his predecessor on 18 September 1959.[75]

For a second time, the Air Force had deflected an Army-Navy challenge to its growing military role in space. Its prudent, cautious approach to asserting its prominence in the military space picture seemed vindicated. By late summer, however, the Air Force would lose control of one of its largest and most important space missions.

The downing of the U-2 reconnaissance aircraft piloted by Francis Gary Powers on 1 May 1960 destroyed plans for an East-West Summit Conference and limited reconnaissance flights exclusively to the periphery of the USSR. It also brought the troubled Samos and MIDAS satellite programs more funding from the administration and Congress, while compelling officials to reassess the reconnaissance satellite program at the highest government levels.[76]

Eisenhower's "peaceful purposes" space policy covered CIA as well as military involvement in a reconnaissance satellite program. Back in February 1958 the President authorized the CIA to develop a reconnaissance satellite, assisted by elements of the Air Force, after being told by his Board of Consultants on Foreign Intelligence Activities that Samos could not meet near-term requirements, because it used film readout and relied on the Atlas booster. While the Atlas would not be operational for several years, by using the Thor IRBM, the CIA might have a film recovery satellite launched by the spring of 1959. Using as a cover the Air Force's Discoverer project, the CIA designated its highly sensitive operation Project CORONA.

Satellites had to fill the intelligence gap created by the loss of the U-2. On 10 June 1960 Eisenhower directed Secretary of Defense Gates to reassess intelligence requirements and the prospects for fulfilling them using the Air Force Samos readout system. In turn, he appointed a three-man panel made up of the President's science advisor, George B. Kistiakowsky, John H. Rubel, Deputy Director for Defense Research and Engineering, and Joseph V. Charyk, Under Secretary of the Air Force. Apparently, over the summer Kistiakowski and the President's Scientific Advisory Committee performed most of the work, assisted by Richard Bissell and his CIA science advisory panel. CORONA, meanwhile, achieved its second success in fourteen attempts on 20 August, recovering the first film capsule. Kistiakowsky presented the Samos findings to the President in a NSC meeting on 25 August. The report concluded that the Samos satellites, like CORONA and the U-2, represented a national asset. As such, the project should not be directed by a military service, but by a civilian agency in the Defense Department. The President agreed and authorized an accelerated program directed by the Secretary of the Air Force and reporting to the Secretary of Defense.[77]

The new program arrangements took shape quickly. On 31 August Secretary of the Air Force Dudley Sharp created within his department the Office of Missile and Satellite Systems under the Assistant Secretary of the Air Force, who would be responsible for coordinating Air Force, CIA, and later Navy and National Security Agency (NSA) intelligence reconnaissance activities. Secretary Sharp named Brigadier General Robert E. Greer director of the Samos west coast development field office. At the same time, the Secretary established two advisory bodies: a Satellite Reconnaissance Advisory Group made up of four civilian technical specialists, and a Satellite Reconnaissance Advisory Council. Chaired by the Under Secretary of the Air Force, the council included General Greer, the three Air Force assistant secretaries, the vice chief of staff of the Air Force and two senior Air Staff officers. Within months, the Office of Missile and Satellite Systems became the secret National Reconnaissance Office (NRO), directed by the Under Secretary of the Air Force, and responsible for all reconnaissance satellite projects, including CORONA. The Samos effort disappeared from public view as surely as it did from Air Force control.[78]

Although the new reconnaissance satellite offices remained within the Office of the Secretary of the Air Force and employed serving Air Force officers, Air Force headquarters was essentially excluded from the operations of this highly sensitive national project. As a result, the military satellite reconnaissance program would operate outside the Air Force area of responsibility. Moreover, when continued funding and technical problems led to cancellation of Samos in the early 1960s, only the equally troubled MIDAS missile early warning satellite and the Vela nuclear detection spacecraft remained in the Air Force satellite inventory.

While the Air Force lost control of the Samos satellite program, it took action to create The Aerospace Corporation to insure that it would have the technical competence to meet current and future space age challenges.[79] Although the new systems approach had proven successful during the crash missile program, the systems engineering role played by Ramo-Wooldridge Corporation generated criticism from aerospace firms and Congress about its privileged position. When, on 31 October 1958, it merged with Thompson Products, Inc., to become Thompson-Ramo-Wooldridge (TRW), Inc., its Space Technology Laboratory (STL) became an "independent" subsidiary of TRW. Nevertheless, conflict-of-interest charges and congressional scrutiny compelled General Schriever to seek an alternative based on a nonprofit, noncompetitive arrangement.

Secretary of the Air Force James H. Douglas and other Air Force leaders agreed. A special committee confirmed the nonprofit corporation approach, and in the spring of 1960 General Schriever and Under Secretary Joseph Charyk worked with an organizing committee to form a new corporation. By 3 June they had established The Aerospace Corporation on El Segundo Boulevard in Inglewood, California, adjacent to the Ballistic Missile Division headquarters. At a news conference on 25 June, Chairman of the Board Roswell L. Gilpatrick declared that his organization represented "a new approach on the part of the Air Force in the management of its missile and space programs." By the end of the year, the new corporation had acquired more than 1700 employees and responsibility for twelve major Air Force programs. Eventually, the Aerospace Corporation would provide general systems engineering and technical direction (GSE/TD) for every missile and space program undertaken by the Air Force.

Air Force leaders had good reason for optimism in the fall of 1960. They had beaten back space challenges from the Navy and Army and had created the Aerospace Corporation. Despite losing control of the Samos program, the Air Force continued to expand its space role in the Space Detection and Tracking System, in booster development, and in development of infrastructure to support national space policy. The Air Staff's Brigadier General Homer A. Boushey forecast in the fall of 1960: "We can go into space with our feet firmly planted on the ground." Yet, Air Force leaders soon threw caution to the wind. With the prospect of a new and potentially more space-oriented administration on the horizon after the November 1960 election, Air Force leaders decided to embark on a campaign to influence the thinking of the new administration on space issues.[80]

## The Military Space Mission Goes to the Air Force

Senator John F. Kennedy made space an issue in the 1960 presidential election campaign. Referring to Soviet "firsts," he cautioned that "if the Soviets control space they can control the earth, as in past centuries the nation that controlled the seas dominated the continents....We cannot run second in this vital race. To insure peace and freedom, we must be first." He called for an accelerated space program.[81]

Shortly after his narrow victory over Vice President Richard M. Nixon, Kennedy appointed a committee to review the country's space program. Chaired by MIT's Jerome B. Wiesner, the "Wiesner Committee" included among its nine distinguished members Trevor Gardner, prime mover of the Air Force Atlas missile program. While serving on the Wiesner Committee, Gardner also accepted an invitation from General Schriever to chair a committee that would examine the status of Air Force space activities. Schriever hoped that Gardner would be able to produce a von Neumann Committee type of report that would lead to a "comprehensive, dynamic Air Force space development program" along the lines of the crash ICBM program.[82]

The Wiesner Report appeared on 10 January 1961.[83] It began by severely criticizing the organization and management of NASA and what it termed a "fractionated military space program." It recommended that one agency or military service be made responsible for all military space development and cited the Air Force as the logical choice. Already providing ninety percent of the support and resources for other military agencies, the Air Force, said the report's authors, represented the nation's "principal resource for the development and operation of future space systems, except those of a purely scientific nature assigned by law to NASA." Their recommendations also included more emphasis on booster development, manned space activities, and military applications in space. The Air Force could not have been happier with the Committee's report.

Meanwhile, early in 1961 the Air Force had to confront the unwanted fruits of its assertive late-fall campaign for a greater space role. Back in late November 1960, the Air Staff's Deputy Director of War Plans, Brigadier General J. D. Page, prepared a paper describing the Air Force position on space for use in briefing the new administration's officials. The paper restated the Air Force view of "aerospace," stressed the importance of space applications, and described seven such projects: Samos, MIDAS, a space-based antisatellite or missile system, a satellite inspector known as Saint, the Space Detection and Tracking System (SPADATS), the Advent communications satellite, and the Transit navigation satellite. Additionally, four more projects, Discoverer, Dyna-Soar, the Aerospace Plane, and HETS, a so-called hyper-environmental test system, were identified as "learning type" projects designed to determine the feasibility of new technology for space. General Page's rationale also assessed relations with NASA, suggesting that the Air Force work to have the Space Act be amended to provide clear recognition of the military's role in developing space systems.[84]

The Page paper seemed at variance with General White's efforts to promote a good relationship with the civilian space agency. In late 1959, for example, the chief of staff had circulated a letter to the Air Staff directing the fullest possible cooperation with NASA and had continued to foster good relations between NASA and the Air Force. General Page's paper of late 1960, however, suggested that less harmony existed between the two organizations than publicly admitted, and a more forceful effort might be needed to right the balance. The Page paper coincided with an intense public and internal information campaign to express Air Force views on space to congressmen, journalists, businessmen, and other influential people. The self-promotion effort immediately raised a storm of protest in the press over what it termed the Air Force's "political offensive to bring about changes in national space policy and law." Critics predicted an approaching contest with NASA for the country's major role in space.[85]

The outcry came to the attention of Congressman Overton Brooks, whose House Committee on Science and Astronautics planned to meet in February 1961 to examine the possible Air Force-industry "plot to undercut the space agency." Brooks' intentions prompted General White to write the congressman a letter, in which the Chief of Staff declared that "any action or statements by any Air Force individual or groups which tend to create such impressions [of unhealthy competition between the service and NASA] are in direct contradiction to the established beliefs and policies of the Air Force." General White requested Congressman Brooks to identify the "'pressure groups within the USAF'...and the specific actions taken by these groups toward 'degrading the position of NASA.'" Despite General White's assurances, the chairman reiterated his concerns in a 14 February 1961 letter to NASA director Glennan, who passed the letter on to General White. The Air Force Chief of Staff responded by assuring Glennan he was sending his key officers to meet with the new NASA leadership to determine how they could lay to rest the "ghost of this alleged NASA-Air Force dissension and duplication" once and for all.[86]

General White also appeared before the Brooks Committee in March to deny that his service had a plan "to take over NASA." During the congressional hearings in 1960, he had reassured his questioners that all was well between the two agencies, and that Air Force support to NASA had been extensive. This included providing the space agency sixteen Atlas D boosters modified for Mercury capsules and adapters, launch facilities at the Atlantic Missile Range Complex 14, and one-half of Hanger J with adaptations to accommodate telemetry, communications, and data transfer equipment. Along with normal base support and office space and equipment, Air Force infrastructure support also encompassed guidance sites and computers used for the Atlas, along with more than 400 Air Force military and civilian personnel. General White specifically referred to good working relations in evaluating requirements and preparing schedules, reaching agreements to share facilities on a priority basis, and cooperating on a demarcation of missions. As for the latter, he declared that the Air Force had no conflict with NASA handling space exploration and civilian uses and the Defense Department pursuing military applications. General White did, however, suggest the need for a single point of contact for Defense Department-NASA affairs and argued that the Air Force represented the logical Defense Department representative.[87]

Nevertheless, Congressman Brooks in March 1961 called on the new president to clarify the civilian and military roles and explain what seemed to be a tilt toward the military by the Wiesner Committee. In reply, President Kennedy declared that, while he never intended NASA to be subordinated to the Defense Department, there remained "legitimate missions in space for which the military services should assume responsibility."[88]

In fact, the President had already agreed to a new military directive that assigned remaining military space efforts and effectively awarded the Air Force the bulk of the military space "mission." Shortly after taking office, Secretary of Defense Robert McNamara directed his staff to review the military space program in light of the Wiesner Report's criticism of the "fractionated military space program." After studying the issue and soliciting comments from important Defense Department officials, the Defense Secretary decided to centralize space system development within the Defense Department by assigning the Air Force responsibility for "research, development, test, and engineering of Department of Defense space development programs or projects." Air Force enthusiasm remained tempered by other parts of the directive which authorized each service to conduct preliminary research and asserted that operational assignment of space systems would be done on a project-by-project basis. Nevertheless, by effectively making the service the executive agent for military space development projects and, thereby, the lead military service in space, the directive represented a major step in the Air Force's quest for the military space mission.[89]

On 17 March General White announced a major reorganization to better manage the missile and space programs. Although the timing suggests that the Defense Department directive precipitated the Air Force action, actually the reverse describes the course of events more accurately.[90] Apparently in early January 1961, Roswell Gilpatrick, the new Deputy Secretary of Defense, bolstered by the Wiesner Report's findings, contacted General White and offered the Air Force major responsibility for the military space mission if it "put its house in order." Gilpatrick and General Schriever had discussed the fragmented state of Air Force research and development activities when they worked together in forming the Aerospace Corporation the previous year. At that time, the main split in weapons systems responsibilities was between research and development, and procurement, the former function being assigned to Air Research and Development Command and the latter to Air Materiel Command. General Schriever had argued that the Air Force could not handle the military space mission unless one Air Force command held responsibility for research and development, system testing, and acquisition of space systems. The ARDC commander had advocated such a reorganization for a number of years. The problem had become more urgent by 1960. While the ARDC's Ballistic Missile Division in Los Angeles had retained research and development responsibility for space projects, its most important mission in 1960 involved close coordination with Air Materiel Command's collocated Ballistic Missiles Center to activate the new intercontinental ballistic missile (ICBM) force. As a result, two major, national programs—missiles and space—competed for resources and management focus within a single research and development organization. General Schriever expressed his concerns to General White in September 1960 and received authority to begin dividing the west coast space and missile functions by moving the latter to Norton Air Force Base, California, and retaining all space responsibilities at the Los Angeles site. Yet the ARDC commander remained convinced that the Air Force required more sweeping organizational reform. Deputy Secretary of Defense Gilpatrick agreed.

Following Gilpatrick's offer, General White asked Schriever to form a small task force to prepare an acceptable plan for centralizing weapon system development and procurement. Only Secretary of the Air Force Eugene Zuckert, Under Secretary of the Air Force Joseph V. Charyk, and Generals White and Schriever had been informed of Gilpatrick's offer, and General White preferred to keep the knowledge to a minimum. Although the Air Staff's Major General Howell M. Estes, Jr., Assistant Deputy Chief of Staff for Operations, chaired the small group, Schriever's chief appointee, Colonel Otto Glasser, actually formulated the plan and briefed it to senior officers and officials in the Air Force and to Defense Secretary McNamara. Afterward, General White informed the Air Council of what had transpired.

The centerpiece of the Air Force reorganization of the spring of 1961 involved creation of the Air Force Systems Command (AFSC), with responsibility for all research, development and acquisition of aerospace and missile systems. With the inactivation of the Air Materiel Command, a new Logistics Command was established to handle maintenance and supply only. To carry out this challenging assignment, AFSC received four subordinate divisions: an Electronics Division, an Aeronautical Systems Division, a Ballistic Missile Division, and a Space Systems Division. The new arrangement reflected the separation of missile and space management functions that General Schriever had favored for the past two years. The new Space Systems Division would be formed at the Los Angeles site from elements of ARDC's Ballistic Missile Division and AMC's Ballistic Missiles Center. The Ballistic Missile Division, also comprised of elements from ARDC's Ballistic Missile Division and AMC's Ballistic Missiles Center as well as the Army Corps of Engineers' Ballistic Missile Construction Office, would relocate to Norton Air Force Base. An additional measure involved establishment of an Office of Aerospace Research (OAR) on the Air Staff for basic research elements.

The Air Force reorganization represented a fitting complement to the Defense Department's directive assigning to the service future military space development responsibilities. With its own house in order, space activities promised to receive the management and research and development they would need in the years ahead. Fittingly, General Schriever received a promotion to four-star rank and became the first commander of Air Force Systems Command.

The Defense Department directive awarding the military space development mission to the Air Force could not be expected to please Army and Navy leaders.

Their grumblings reached the ears of Congressman Brooks, who held hearings beginning on 17 March, the day the Air Force announced its organizational changes. Before the committee, however, Army General Lyman Lemnitzer, Chairman of the Joint Chiefs of Staff, and other Army and Navy representatives denied opposing the directive. At the same time, Deputy Defense Secretary Gilpatrick assured the committee that centralization of space research and development would prevent duplication and prevent "misuse of resources," while General White declared that the Air Force would "bend over backward to meet the requirements of the Army and Navy as prescribed by the directive." The Chief of Staff also stressed that the new arrangement would improve cooperative relationships with NASA. The committee took no action, but promised "continuing close scrutiny" of the new directive's implementation.

Meanwhile, on 20 March 1961, three days after the public announcement of the Air Force reorganization, Trevor Gardner submitted his committee's report to General Schriever.[91] Although General Schriever had hoped to have Gardner's report by mid-January 1961, the former Assistant Secretary of the Air Force found it necessary to establish two study groups to provide the managerial and technical information needed. The report's conclusions proved alarming. The United States, it claimed, could not overtake the Soviet Union in space achievements for another three to five years without a major increase in the Defense Department's space effort. The report reserved particular criticism for the Eisenhower administration's emphasis on separate "military" and "peaceful" space programs, which had relegated the military program to a "stepchild" status with little participation in the scientific exploration of space, which was reserved to NASA. Above all, the report recommended that planners avoid prescribing detailed space requirements and operational systems in favor of first developing a firm technological basis, with the Defense Department and NASA focusing on fundamentals or "building blocks." Finally, like the Wiesner Report, the Gardner Report called for military participation in a comprehensive, lunar landing program that would land and return astronauts sometime between 1967 and 1970. The broad technological capabilities resulting from such a major national effort, the report predicted, would provide important "fallout" for both military and civilian space purposes.

While the Gardner Report underwent high-level review, on 12 April 1961, Soviet cosmonaut Yuri Gagarin became the first man to orbit the earth. Motivated in part by this Soviet space "spectacular," Secretary of Defense McNamara directed Herbert York, DDR&E, and Secretary of the Air Force Zuckert to assess the national space program in terms of defense interests and the Gardner Report's conclusions. The Defense Secretary's initiative led to an intense two-week study effort that centered on a special task force at the Space Systems Division led by Major General Joseph R. Holzapple, Air Force Systems Command's Assistant Deputy Commander for Aerospace Systems. On 1 May 1961, in forwarding the report to Secretary McNamara, Secretary Zuckert reiterated the Air Force's concerns about "the inadequacy of our current National Space Program." Not surprisingly, the Air Force's "Holzapple Report" confirmed the conclusions reached by Trevor Gardner's committee. Following an analysis of military space objectives and current development efforts designed to meet them, the report focused on the large booster program as the most pressing problem and reason for Soviet supremacy. Like the Gardner Report, the Air Force proposal also called for a national lunar landing initiative, whose framework would provide an urgently needed comprehensive research and development "effort." Although the Air Force recognized that NASA would head the expedition, it looked forward to a close, cooperative effort that would enable it to reenter the field of superbooster research that had been a NASA preserve since it acquired the Army rocket team in October 1959.[92]

The Air Force recommendations ultimately were incorporated into the National Space Program announced by President Kennedy in May. Shortly after receiving the Air Force proposal, Secretary McNamara and newly-appointed NASA Administrator James E. Webb met to propose major initiatives and budget increases necessary "to establish and to direct an 'Integrated National Space Program.'" Although the lunar landing objective topped their list of essential projects, they also called for developing global space communications and meteorological networks and large boosters for both civilian and military programs.

After receiving public and congressional support for an expanded space program, President Kennedy on 25 May 1961 appeared before a joint session of Congress to challenge the nation to overtake the Russians in space.

> If we are to win the battle that is now going on around the world between freedom and tyranny, the dramatic achievements in space... should have made clear to us all...the impact of this adventure on the minds of men everywhere who are attempting to make a determination of which road they should take....It is time to take larger strides—time for a great new American enterprise—time for this nation to take a clearly leading role in space achievements."[93]

Echoing the agreement between McNamara and Webb on the nation's future course, the President listed the moon expedition as the first space goal, followed by development of nuclear rockets [big boosters] for interplanetary space exploration, and creation of global communication and meteorological satellite systems as soon as possible. Congress had already raised the funding of the Defense Department's large solid-fuel booster project from $3 to $15 million. As a result of the Kennedy-proposed space program, the Air Force, as the "space booster service," would receive $77 million to begin development of both an upper stage and a large solid-fuel booster to compete with NASA's liquid-fueled Nova engine.[94]

By May of 1961 President Kennedy realized the importance to national security of reconnaissance satellites. Although he did not alter the Eisenhower policy of "space for peaceful purposes," he clearly believed that the nation found itself in a race for space supremacy with the Soviets and should accept the challenge. The Air Force fully expected to play a central part in the ambitious space program that lay ahead and to benefit from the technological achievements along the way.

## The Air Force Rise to Military Space Preeminence

The Eisenhower administration's space policy never wavered from its central objective of permitting the launch and operation of military reconnaissance satellites. The "spy satellites" would enable the country to guard against the President's old nemesis of surprise attack, while reinforcing the moral high ground of "space for peace" by providing the means to verify future arms agreements and nuclear test ban treaties. Relying on the "Sputnik precedent," he preferred to avoid direct confrontation with the Soviets by stressing civilian spaceflight and limiting military operations to defense support activities. This would best insure the success of clandestine satellite operations for the nation's defense.

Throughout the late 1950s Air Force leaders often failed to appreciate the subtleties of the Eisenhower space policy. For them, the policy of "space for peaceful purposes" served only to restrict military space activities to modest defense support projects and no offensive initiatives beyond the study phase. As military planners, they preferred defense preparations to combat potential enemy capabilities rather than prepare for operating in an "outer space sanctuary." Given their focus on space as the ultimate "high ground" and the extension of traditional Air Force operations, Air Force leaders believed that the country should achieve space "supremacy" in order to deny offensive space operations to the enemy. Because such activity might jeopardize space reconnaissance assets, the Eisenhower regime categorically refused to permit it.

Given these circumstances, the Air Force remained unable to conduct an independent space program. Prevented after Sputnik from leading a nationwide space effort to overtake the Soviets, it found itself forced to respond to ARPA's direction, then compete with NASA for funds and projects. Only with the demise of ARPA in late 1959 did the Air Force regain control of its "own" space program. Even then the future course with NASA and DDR&E seemed unclear, while key projects continued to experience growing pains. Moreover, much of the Air Force space responsibility involved supporting other agencies with booster and infrastructure assistance. Operational direction remained the responsibility of other services and agencies. This did not always seem to reflect the aspirations of the service that had been assigned, in the words of General Schriever, the "prime responsibility for the military space mission."

By the end of the Eisenhower presidency, the program had in place the five functional areas of defense support operations that would characterize Air Force space operations from then until the Reagan administration reopened the issue of weapons in space.* Of the five areas, only the missile detection and space defense functions remained largely under Air Force control. At this time, the Air Force supported others who had responsibility for communications, navigational, and meteorological satellites, while "observation of the earth" now encompassed highly sensitive "black" systems outside the Air Force's control. Looking back on the McNamara directive's impact on the Air Force following loss of reconnaissance assets to the National Reconnaissance Office, Air Force Secretary Eugene Zuckert declared, "It was like getting a franchise to run a bus line in the Sahara Desert." Yet, Secretary Zuckert's comment did not express pique at the service not getting what it wanted. The March 1961 decision, he explained, was jurisdictional and provided the Air Force all the jurisdiction it needed in the space field. How much support the service would get remained in doubt. In effect, the Air Force received the research and development franchise for space systems, including offensive space-based systems, but it awaited customers and support from the Defense Department in the course ahead.[95]

If the Air Force did not acquire all the military space missions it desired, it had much to celebrate in the spring of 1961. Of its space programs, the MIDAS early warning infrared satellite remained a high national priority, and the Air Force continued to develop its Samos reconnaissance project. At the same time, it provided important launch and infrastructure support to the national reconnaissance effort under Project Discoverer. By the end of June 1961, the Air Force had launched twenty-six Discoverer satellites in support of various projected space systems, and the program had been expanded from an original thirty-five planned vehicles to forty-four. The Air Force also played a major part in the Space Detection and Tracking System with overall planning responsibilities and development of its Spacetrack network, and it moved forward with an elaborate air and missile defense system that would provide collateral support for Spacetrack. Already, the Air Force programmed the Thor and Atlas boosters as standard launch vehicles of the future, with an improved Titan to follow. Boosters had been the foundation of the Air Force dominance in space and represented the best means to perpetuate that dominance. At the same time, despite Project Mercury, in these, the Dyna-Soar years, the military man-in-space mission remained a viable option.

In the aftermath of Secretary McNamara's directive and President Kennedy's lunar challenge, the Air Force could look back on the years since Sputnik with satisfaction. Its cautious, well-orchestrated, opportunistic three-and-one-half-year quest for the military space mission had succeeded. The losses of rival service assets to NASA had resulted in Air Force gains, and efforts to create a unified space command for space had been successfully thwarted. Along the way the Air Force

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* See Appendix 2-5.

prepared itself for the space mission by demonstrating the flexibility to establish its own in-house technical expertise with the Aerospace Corporation, and implement a major reorganization to better handle the research and development challenges ahead. The Air Force had staked its claim to space through the "logic of aerospace," and it had been accepted. Most importantly, despite the difficulties with space program advocacy this often presented, Air Force leaders remained convinced that space must be approached in terms of its utility for traditional operations. This would be an important legacy for the future. In the years to come, space would become an increasingly important medium in support of both strategic and tactical military operations. That, in turn, would serve to institutionalize space within the Air Force. In 1961, the Air Force had garnered the bulk of the military space "mission." The challenge now would be to strengthen its position by developing a military space program vital to the nation's defense.